

CS 441 Discrete Mathematics for CS

Discrete Mathematics for Computer Science

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5329 Sennott Square

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Course administrivia

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Course web page:

<http://www.cs.pitt.edu/~milos/courses/cs441/>

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Course administrivia

Lectures:

- Tuesdays, Thursdays: 11:00 AM - 12:15 PM
- 205 LAWRN

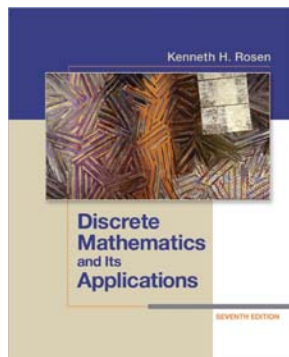
Recitations:

- held in 5313 SENSQ
 - **Section 1: Thursdays** 4:00 – 4:50 PM
 - **Section 2:** Fridays: 11:00 – 11:50 AM

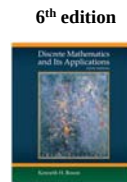
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Textbook:

- Kenneth H. Rosen. *Discrete Mathematics and Its Applications*, 7th Edition, McGraw Hill, 2012.



Exercises from the book will be given for homework assignments



Course administrivia

Grading policy

- Exams: (50%)
- Homework assignments: 40%
- Lectures/recitations: 10%

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Weekly homework assignments

- Assigned in class and posted on the course web page
- Due one week later at the beginning of the lecture
- No extension policy

Collaboration policy:

- You may discuss the material covered in the course with your fellow students in order to understand it better
- However, homework assignments should be worked on and written up **individually**

Course administria

Course policies:

- Any un-intellectual behavior and cheating on exams, homework assignments, quizzes will be dealt with severely
- If you feel you may have violated the rules speak to us as soon as possible.
- Please make sure you read, understand and abide by the Academic Integrity Code for the Faculty and College of Arts and Sciences.

Course syllabus

Tentative topics:

- Logic and proofs
- Sets
- Functions
- Integers and modular arithmetic
- Sequences and summations
- Counting
- Probability
- Relations
- Graphs

Course administrivia

Questions



Discrete mathematics

- **Discrete mathematics**
 - study of mathematical structures and objects that are fundamentally **discrete** rather than **continuous**.
- **Examples of objects** with discrete values are
 - **integers, graphs, or statements in logic.**
- Discrete mathematics and **computer science**.
 - Concepts from discrete mathematics are useful for describing **objects and problems in computer algorithms and programming languages**. These have applications in cryptography, automated theorem proving, and software development.

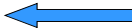
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Course syllabus

Tentative topics:

- **Logic and proofs** 
- Sets
- Functions
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Logic

Logic:

- defines a formal language for representing knowledge and for making logical inferences
- It helps us to understand how to construct a valid argument

Logic defines:

- Syntax of statements
- The meaning of statements
- The rules of logical inference (manipulation)

Propositional logic

- The simplest logic
- Definition:
 - A **proposition** is a statement that is either true or false.
- Examples:
 - Pitt is located in the Oakland section of Pittsburgh.
 - (T)
 - $5 + 2 = 8$.
 - (F)
 - It is raining today.
 - (either T or F)

Propositional logic

- **Examples (cont.):**
 - How are you?
 - a question is not a proposition
 - $x + 5 = 3$
 - since x is not specified, neither true nor false
 - 2 is a prime number.
 - (T)
 - She is very talented.
 - since she is not specified, neither true nor false
 - There are other life forms on other planets in the universe.
 - either T or F

Composite statements

- More complex propositional statements can be build from elementary statements using **logical connectives**.

Example:

- Proposition A: It rains outside
- Proposition B: We will see a movie
- A new (combined) proposition:
 - If it rains outside then we will see a movie

Composite statements

- More complex propositional statements can be build from elementary statements using **logical connectives**.
- Logical connectives:
 - Negation
 - Conjunction
 - Disjunction
 - Exclusive or
 - Implication
 - Biconditional

Negation

Definition: Let p be a proposition. The statement "It is not the case that p ." is another proposition, called the **negation of p** . The negation of p is denoted by $\neg p$ and read as "not p ."

Example:

- Pitt is located in the Oakland section of Pittsburgh.
 $\rightarrow$
- It is **not the case** that Pitt is located in the Oakland section of Pittsburgh.

Other examples:

- $5 + 2 \neq 8$.
- 10 is **not** a prime number.
- It is **not** the case that buses stop running at 9:00pm.

Negation

- **Negate the following propositions:**

- It is raining today.
 - It is **not** raining today.
- 2 is a prime number.
 - 2 is **not** a prime number
- There are other life forms on other planets in the universe.
 - It is **not the case** that there are other life forms on other planets in the universe.

Negation

- A **truth table** displays **the relationships between truth values** (T or F) of different propositions.

p	$\neg p$
T	F
F	T

Rows: all possible values of elementary propositions:

Conjunction

- **Definition:** Let p and q be propositions. The proposition "**p and q**" denoted by $p \wedge q$, is true when both p and q are true and is false otherwise. The proposition $p \wedge q$ is called the **conjunction** of p and q .
- **Examples:**
 - Pitt is located in the Oakland section of Pittsburgh **and** $5 + 2 = 8$
 - It is raining today **and** 2 is a prime number.
 - 2 is a prime number **and** $5 + 2 \neq 8$.
 - 13 is a perfect square **and** 9 is a prime.

Disjunction

- **Definition:** Let p and q be propositions. The proposition "**p or q**" denoted by $p \vee q$, is false when both p and q are false and is true otherwise. The proposition $p \vee q$ is called the **disjunction** of p and q .
- **Examples:**
 - Pitt is located in the Oakland section of Pittsburgh **or** $5 + 2 = 8$.
 - It is raining today **or** 2 is a prime number.
 - 2 is a prime number **or** $5 + 2 \neq 8$.
 - 13 is a perfect square **or** 9 is a prime.

Truth tables

- **Conjunction and disjunction**
- Four different combinations of values for p and q

p	q	$p \wedge q$	$p \vee q$
T	T		
T	F		
F	T		
F	F		

Rows: all possible combinations of values for elementary propositions: 2^n values

Truth tables

- **Conjunction and disjunction**
- Four different combinations of values for p and q

p	q	$p \wedge q$	$p \vee q$
T	T	T	
T	F	F	
F	T	F	
F	F	F	

- NB: $p \vee q$ (the or is used inclusively, i.e., $p \vee q$ is true when either p or q or both are true).

Truth tables

- **Conjunction and disjunction**
- Four different combinations of values for p and q

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

- NB: $p \vee q$ (the or is used inclusively, i.e., $p \vee q$ is true when either p or q or both are true).

Exclusive or

- **Definition:** Let p and q be propositions. The proposition "**p exclusive or q**" denoted by $p \oplus q$, is true when exactly one of p and q is true and it is false otherwise.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Implication

- **Definition:** Let p and q be propositions. The proposition " **p implies q** " denoted by $p \rightarrow q$ is called **implication**. It is false when p is true and q is false and is true otherwise.
- In $p \rightarrow q$, p is called the **hypothesis** and q is called the **conclusion**.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Implication

- $p \rightarrow q$ is read in a variety of equivalent ways:
 - if p then q
 - p only if q
 - p is sufficient for q
 - q whenever p
- **Examples:**
 - if Steelers win the Super Bowl in 2013 then 2 is a prime.
 - If F then T ?

Implication

- $p \rightarrow q$ is read in a variety of equivalent ways:
 - if p then q
 - p only if q
 - p is sufficient for q
 - q whenever p
- **Examples:**
 - if Steelers win the Super Bowl in 2013 then 2 is a prime.
 - T
 - if today is Tuesday then $2 * 3 = 8$.
 - What is the truth value ?

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Implication

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 - if p then q
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 - if Steelers win the Super Bowl in 2013 then 2 is a prime.
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 - if today is Tuesday then $2 * 3 = 8$.
 - If T then F

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 - if Steelers win the Super Bowl in 2013 then 2 is a prime.
 - T
 - if today is Tuesday then $2 * 3 = 8$.
 - F

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Implication

- The **converse** of $p \rightarrow q$ is $q \rightarrow p$.
- The **contrapositive** of $p \rightarrow q$ is $\neg q \rightarrow \neg p$
- The **inverse** of $p \rightarrow q$ is $\neg p \rightarrow \neg q$
- **Examples:**
 - If it snows, the traffic moves slowly.
 - p : it snows q : traffic moves slowly.
 - $p \rightarrow q$
 - **The converse:**
 - If the traffic moves slowly then it snows.
 - $q \rightarrow p$

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Implication

- The **contrapositive** of $p \rightarrow q$ is $\neg q \rightarrow \neg p$
- The **inverse** of $p \rightarrow q$ is $\neg p \rightarrow \neg q$
- **Examples:**
 - If it snows, the traffic moves slowly.
 - **The contrapositive:**
 - If the traffic does not move slowly then it does not snow.
 - $\neg q \rightarrow \neg p$
 - **The inverse:**
 - If it does not snow the traffic moves quickly.
 - $\neg p \rightarrow \neg q$

Biconditional

- **Definition:** Let p and q be propositions. The **biconditional** $p \leftrightarrow q$ (**read p if and only if q**), is true when p and q have the same truth values and is false otherwise.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- **Note:** two truth values always agree.

Constructing the truth table

- **Example: Construct a truth table for**
 $(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
- Simpler if we decompose the sentence to elementary and intermediate propositions

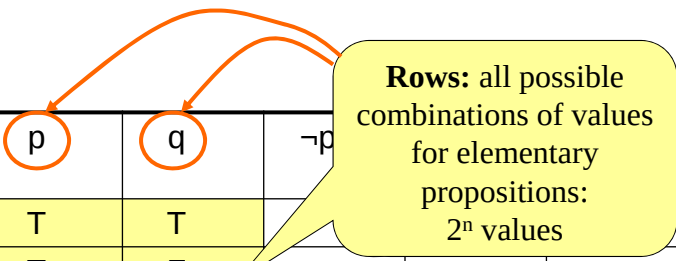
p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
T	T				
T	F				
F	T				
F	F				

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Constructing the truth table

- **Example: Construct the truth table for**
 $(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$



p	q	$\neg p$				$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
T	T					
T	F					
F	T					
F	F					

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Constructing the truth table

- Example: Construct the truth table for

$$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$$

Typically the target
(unknown) compound
proposition and its
values

p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
T	T				
T	F				
F	T				
F	F				

Auxiliary compound
propositions and their
values

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Constructing the truth table

- Examples: Construct a truth table for

$$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$$

p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
T	T	F			
T	F	F			
F	T	T			
F	F	T			

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Constructing the truth table

- Examples: Construct a truth table for
 $(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$

p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
T	T	F	T		
T	F	F	F		
F	T	T	T		
F	F	T	T		

Constructing the truth table

- Examples: Construct a truth table for
 $(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$

p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
T	T	F	T	F	
T	F	F	F	T	
F	T	T	T	T	
F	F	T	T	F	

Constructing the truth table

- **Examples: Construct a truth table for**

$$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$$

Simpler if we decompose the sentence to elementary and intermediate propositions

p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
T	T	F	T	F	F
T	F	F	F	T	F
F	T	T	T	T	T
F	F	T	T	F	F

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Lecture 2

Propositional logic

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Course administration

Homework 1

- First homework assignment is out today will be posted on the course web page
- Due next week on Thursday

Recitations:

- **today at 4:00pm SENSQ 5313**
- **tomorrow at 11:00 SENSQ 5313**

Course web page:

<http://www.cs.pitt.edu/~milos/courses/cs441/>

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Propositional logic: review

- **Propositional logic:** a formal language for representing knowledge and for making logical inferences
- A **proposition** is a statement that is either true or false.
- A **compound proposition** can be created from other propositions using logical connectives
- **The truth of a compound proposition** is defined by truth values of elementary propositions and the meaning of connectives.
- **The truth table for a compound proposition:** table with entries (rows) for all possible combinations of truth values of elementary propositions.

Compound propositions

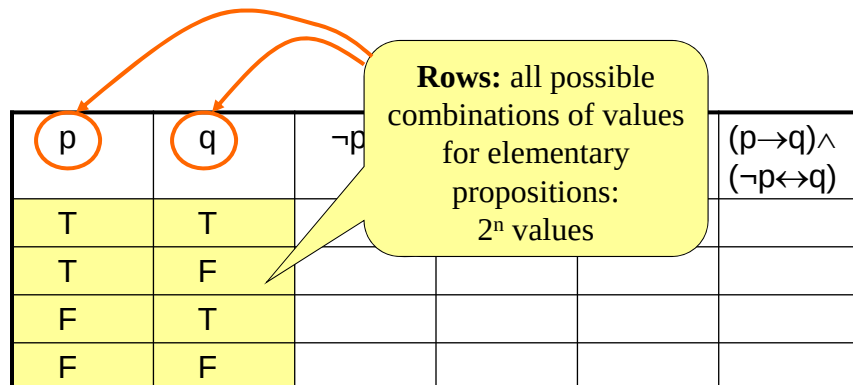
- Let p : 2 is a prime **T**
 q : 6 is a prime **F**
- Determine **the truth value** of the following statements:
 - $\neg p$: **F**
 - $p \wedge q$: **F**
 - $p \wedge \neg q$: **T**
 - $p \vee q$: **T**
 - $p \oplus q$: **T**
 - $p \rightarrow q$: **F**
 - $q \rightarrow p$: **T**

Constructing the truth table

- **Example:** Construct the truth table for
 $(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$

Constructing the truth table

- Example: Construct the truth table for
 $(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$



p	q	$\neg p$	$(p \rightarrow q)$	$(\neg p \leftrightarrow q)$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
T	T	F	T	F	F
T	F	F	F	T	F
F	T	T	T	T	T
F	F	T	T	F	F

Constructing the truth table

- Example: Construct the truth table for

$$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$$

Typically the target
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T	T				
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Constructing the truth table

- Examples: Construct a truth table for

$$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$$

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T	F	F	F	T	F
F	T	T	T	T	T
F	F	T	T	F	F

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Computer representation of True and False

We need to encode two values **True and False**:

- Computers represents data and programs using 0s and 1s
- Logical truth values – True and False
- A bit is sufficient to represent two possible values:
 - 0 (False) or 1(True)
- A variable that takes on values 0 or 1 is called a **Boolean variable**.
- **Definition:** A **bit string** is a sequence of zero or more bits. The **length** of this string is the number of bits in the string.

Bitwise operations

- T and F replaced with 1 and 0

p	q	$p \vee q$	$p \wedge q$
1	1	1	1
1	0	1	0
0	1	1	0
0	0	0	0

p	$\neg p$
1	0
0	1

Bitwise operations

- **Examples:**

$$\begin{array}{r} 1011\ 0011 \\ \vee \underline{0110\ 1010} \\ 1111\ 1011 \end{array}$$

$$\begin{array}{r} 1011\ 0011 \\ \wedge \underline{0110\ 1010} \\ 0010\ 0010 \end{array}$$

$$\begin{array}{r} 1011\ 0011 \\ \oplus \underline{0110\ 1010} \\ 1101\ 1001 \end{array}$$

Applications of propositional logic

- **Translation of English sentences**
- **Inference and reasoning:**
 - new true propositions are inferred from existing ones
 - Used in Artificial Intelligence:
 - Rule based (expert) systems
 - Automatic theorem provers
- **Design of logic circuit**

Translation

Assume a sentence:

If you are older than 13 or you are with your parents then you can attend a PG-13 movie.

Parse:

- If (you are older than 13 or you are with your parents) then (you can attend a PG-13 movie)

Atomic (elementary) propositions:

- A= you are older than 13
- B= you are with your parents
- C=you can attend a PG-13 movie

- Translation: $A \vee B \rightarrow C$

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Translation

- **General rule for translation.**
- Look for patterns corresponding to logical connectives in the sentence and use them to define elementary propositions.

- **Example:**

You can have free coffee if you are senior citizen and it is a Tuesday

Step 1 find logical connectives

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Translation

- **General rule for translation.**
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Step 1 find logical connectives

Translation

- **General rule for translation.**
- Look for patterns corresponding to logical connectives in the sentence and use them to define elementary propositions.
- **Example:**

You can have free coffee if you are senior citizen and it is a Tuesday

Step 2 break the sentence into elementary propositions

Translation

- **General rule for translation.**
- Look for patterns corresponding to logical connectives in the sentence and use them to define elementary propositions.
- **Example:**

You can have free coffee if you are senior citizen and it is a Tuesday

a

b

c

Step 2 break the sentence into elementary propositions

Translation

- **General rule for translation .**
- Look for patterns corresponding to logical connectives in the sentence and use them to define elementary propositions.
- **Example:**

You can have free coffee if you are senior citizen and it is a Tuesday

a

b

c

Step 3 rewrite the sentence in propositional logic

$$\mathbf{b \wedge c \rightarrow a}$$

Translation

- Assume two elementary statements:
 - **p: you drive over 65 mph ; q: you get a speeding ticket**
- **Translate each of these sentences to logic**
 - you do not drive over 65 mph. ($\neg p$)
 - you drive over 65 mph, but you don't get a speeding ticket. ($p \wedge \neg q$)
 - you will get a speeding ticket if you drive over 65 mph. ($p \rightarrow q$)
 - if you do not drive over 65 mph then you will not get a speeding ticket. ($\neg p \rightarrow \neg q$)
 - driving over 65 mph is sufficient for getting a speeding ticket. ($p \rightarrow q$)
 - you get a speeding ticket, but you do not drive over 65 mph. ($q \wedge \neg p$)

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Application: inference

Assume we know the following sentences are true:

If you are older than 13 or you are with your parents then you can attend a PG-13 movie. You are older than 13.

Translation:

- **If (you are older than 13 or you are with your parents) then (you can attend a PG-13 movie) . (You are older than 13).**
 - A= you are older than 13
 - B= you are with your parents
 - C=you can attend a PG-13 movie
- **(A $\vee$ B $\rightarrow$ C), A**
- **(A $\vee$ B $\rightarrow$ C) $\wedge$ A is true**
- **With the help of the logic we can infer the following statement (proposition):**
 - **You can attend a PG-13 movie or C is True**

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Application: inference

The field of Artificial Intelligence:

- Builds programs that act intelligently
- Programs often rely on symbolic manipulations

Expert systems:

- Encode knowledge about the world in logic
- Support inferences where new facts are inferred from existing facts following the semantics of logic

Theorem provers:

- Encode existing knowledge (e.g. about math) using logic
- Show that some hypothesis is true

Example: MYCIN

- MYCIN: an expert system for diagnosis of bacterial infections
- **It** represents
 - Facts about a specific patient case
 - Rules describing relations between entities in the bacterial infection domain

If	1. The stain of the organism is gram-positive, and 2. The morphology of the organism is coccus, and 3. The growth conformation of the organism is chains
Then	the identity of the organism is streptococcus

- **Inferences:**

- manipulates the facts and known relations to answer diagnostic queries (consistent with findings and rules)

Tautology and Contradiction

- Some propositions are interesting since their values in the truth table are always the same

Definitions:

- A compound proposition that is always true for all possible truth values of the propositions is called a **tautology**.
- A compound proposition that is always false is called a **contradiction**.
- A proposition that is neither a tautology nor contradiction is called a **contingency**.

Example: $p \vee \neg p$ is a **tautology**.

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Tautology and Contradiction

- Some propositions are interesting since their values in the truth table are always the same

Definitions:

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Example: $p \wedge \neg p$ is a **contradiction**.

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Equivalence

- We have seen that some of the propositions are equivalent. Their truth values in the truth table are the same.
- Example: $p \rightarrow q$ is equivalent to $\neg q \rightarrow \neg p$ (**contrapositive**)

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

- Equivalent statements are important for **logical reasoning** since they can be substituted and can help us to:
(1) make a logical argument and (2) infer new propositions

Logical equivalence

Definition: The propositions p and q are called **logically equivalent** if $p \leftrightarrow q$ is a tautology (alternately, if they have the same truth table). The notation $p \Leftrightarrow q$ denotes p and q are logically equivalent.

Example of important equivalences

- **DeMorgan's Laws:**

- 1) $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- 2) $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

Example: Negate "The summer in Mexico is cold and sunny"
with DeMorgan's Laws

Solution: ?

Equivalence

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Example: Negate "The summer in Mexico is cold and sunny"
with DeMorgan's Laws

Solution: "The summer in Mexico is not cold or not sunny."

Equivalence

Example of important equivalences

- **DeMorgan's Laws:**

- 1) $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- 2) $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

**To convince us that two propositions are logically equivalent
use the truth table**

p	q	$\neg p$	$\neg q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F		
T	F	F	T		
F	T	T	F		
F	F	T	T		

Equivalence

Example of important equivalences

- **DeMorgan's Laws:**

- 1) $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- 2) $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

To convince us that two propositions are logically equivalent use the truth table

p	q	$\neg p$	$\neg q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	F	
T	F	F	T	F	
F	T	T	F	F	
F	F	T	T	T	

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Equivalence

Example of important equivalences

- **DeMorgan's Laws:**

- 1) $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- 2) $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

To convince us that two propositions are logically equivalent use the truth table

p	q	$\neg p$	$\neg q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	F	F
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

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Equivalence

Example of important equivalences

- **DeMorgan's Laws:**

- 1) $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- 2) $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

To convince us that two propositions are logically equivalent use the truth table

p	q	$\neg p$	$\neg q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	F	F
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

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Important logical equivalences

- **Identity**

- $p \wedge T \Leftrightarrow p$
- $p \vee F \Leftrightarrow p$

- **Domination**

- $p \vee T \Leftrightarrow T$
- $p \wedge F \Leftrightarrow F$

- **Idempotent**

- $p \vee p \Leftrightarrow p$
- $p \wedge p \Leftrightarrow p$

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Important logical equivalences

- **Double negation**

- $\neg(\neg p) \Leftrightarrow p$

- **Commutative**

- $p \vee q \Leftrightarrow q \vee p$

- $p \wedge q \Leftrightarrow q \wedge p$

- **Associative**

- $(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$

- $(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$

Important logical equivalences

- **Distributive**

- $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$

- $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

- **De Morgan**

- $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$

- $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

- **Other useful equivalences**

- $p \vee \neg p \Leftrightarrow T$

- $p \wedge \neg p \Leftrightarrow F$

- $p \rightarrow q \Leftrightarrow (\neg p \vee q)$

Using logical equivalences

- Equivalences can be used in proofs. A proposition or its part can be transformed using equivalences and some conclusion can be reached.
- **Example:** Show $(p \wedge q) \rightarrow p$ is a tautology.
- **Proof:** (we must show $(p \wedge q) \rightarrow p \Leftrightarrow T$)
 - $(p \wedge q) \rightarrow p \Leftrightarrow \neg(p \wedge q) \vee p$ Useful
 - $\Leftrightarrow [\neg p \vee \neg q] \vee p$ DeMorgan
 - $\Leftrightarrow [\neg q \vee \neg p] \vee p$ Commutative
 - $\Leftrightarrow \neg q \vee [\neg p \vee p]$ Associative
 - $\Leftrightarrow \neg q \vee [T]$ Useful
 - $\Leftrightarrow T$ Domination

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Using logical equivalences

- Equivalences can be used in proofs. A proposition or its part can be transformed using equivalences and some conclusion can be reached.
- **Example:** Show $(p \wedge q) \rightarrow p$ is a tautology.
- Alternate proof:

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

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Using logical equivalences

- Equivalences can be used in proofs. A proposition or its part can be transformed using equivalences and some conclusion can be reached.

- **Example 2:** Show $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$

Proof:

- $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$
- $\Leftrightarrow ?$

Using logical equivalences

- Equivalences can be used in proofs. A proposition or its part can be transformed using equivalences and some conclusion can be reached.

- **Example 2:** Show $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$

Proof:

- $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$
- $\Leftrightarrow \neg(\neg q) \vee (\neg p)$ Useful
- $\Leftrightarrow ?$

Using logical equivalences

- Equivalences can be used in proofs. A proposition or its part can be transformed using equivalences and some conclusion can be reached.

- **Example 2:** Show $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$

Proof:

- $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$
- $\Leftrightarrow \neg(\neg q) \vee (\neg p)$ Useful
- $\Leftrightarrow q \vee (\neg p)$ Double negation
- $\Leftrightarrow ?$

Using logical equivalences

- Equivalences can be used in proofs. A proposition or its part can be transformed using equivalences and some conclusion can be reached.

- **Example 2:** Show $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$

Proof:

- $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$
- $\Leftrightarrow \neg(\neg q) \vee (\neg p)$ Useful
- $\Leftrightarrow q \vee (\neg p)$ Double negation
- $\Leftrightarrow \neg p \vee q$ Commutative
- $\Leftrightarrow ?$

Using logical equivalences

- Equivalences can be used in proofs. A proposition or its part can be transformed using equivalences and some conclusion can be reached.

- **Example 2:** Show $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$

Proof:

- $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$
- $\Leftrightarrow \neg(\neg q) \vee (\neg p)$ Useful
- $\Leftrightarrow q \vee (\neg p)$ Double negation
- $\Leftrightarrow \neg p \vee q$ Commutative
- $\Leftrightarrow p \rightarrow q$ Useful

End of proof

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Lecture 3

Predicate logic

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Propositional logic: review

- **Propositional logic:** a formal language for making logical inferences
- A **proposition** is a statement that is either true or false.
- A **compound proposition** can be created from other propositions using logical connectives
- **The truth of a compound proposition** is defined by truth values of elementary propositions and the meaning of connectives.
- **The truth table for a compound proposition:** table with entries (rows) for all possible combinations of truth values of elementary propositions.

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Tautology and Contradiction

What is a tautology?

- A compound proposition that is always true for all possible truth values of the propositions is called a **tautology**.
-

Example: $p \vee \neg p$ is a **tautology**.

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

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Tautology and Contradiction

What is a contradiction?

- A compound proposition that is always false is called a **contradiction**.

Example: $p \wedge \neg p$ is a **contradiction**.

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

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Equivalence

- How do we determine that two propositions are equivalent?
Their truth values in the truth table are the same.
- Example: $p \rightarrow q$ is equivalent to $\neg q \rightarrow \neg p$ (**contrapositive**)

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

- Equivalent statements** are important for logical reasoning since they can be substituted and can help us to make a logical argument.

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Logical equivalence

- Definition:** The propositions p and q are called **logically equivalent** if $p \leftrightarrow q$ is a tautology (alternately, if they have the same truth table). The notation $p \Leftrightarrow q$ denotes p and q are logically equivalent.

a	b	$a \rightarrow b$	$\neg a \rightarrow \neg b$	$(a \rightarrow b) \leftrightarrow (\neg a \rightarrow \neg b)$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

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Important logical equivalences

- **Identity**

- $p \wedge T \Leftrightarrow p$
- $p \vee F \Leftrightarrow p$

- **Domination**

- $p \vee T \Leftrightarrow T$
- $p \wedge F \Leftrightarrow F$

- **Idempotent**

- $p \vee p \Leftrightarrow p$
- $p \wedge p \Leftrightarrow p$

Important logical equivalences

- **Double negation**

- $\neg(\neg p) \Leftrightarrow p$

- **Commutative**

- $p \vee q \Leftrightarrow q \vee p$
- $p \wedge q \Leftrightarrow q \wedge p$

- **Associative**

- $(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$
- $(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$

Important logical equivalences

- **Distributive**

- $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
- $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

- **De Morgan**

- $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

- **Other useful equivalences**

- $p \vee \neg p \Leftrightarrow T$
- $p \wedge \neg p \Leftrightarrow F$
- $p \rightarrow q \Leftrightarrow (\neg p \vee q)$

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Showing logical equivalence

Example: Show $(p \wedge q) \rightarrow p$ is a tautology

In other words $((p \wedge q) \rightarrow p \Leftrightarrow T)$

Proof via truth table:

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

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Showing logical equivalences

- **Equivalences can be used in proofs.** A proposition or its part can be transformed using equivalences and some conclusion can be reached.

Example: Show that $(p \wedge q) \rightarrow p$ is a tautology.

- Proof: (we must show $(p \wedge q) \rightarrow p \Leftrightarrow T$)
$$\begin{aligned}(p \wedge q) \rightarrow p &\Leftrightarrow \neg(p \wedge q) \vee p && \text{Useful} \\ &\Leftrightarrow [\neg p \vee \neg q] \vee p && \text{DeMorgan} \\ &\Leftrightarrow [\neg q \vee \neg p] \vee p && \text{Commutative} \\ &\Leftrightarrow \neg q \vee [\neg p \vee p] && \text{Associative} \\ &\Leftrightarrow \neg q \vee [T] && \text{Useful} \\ &\Leftrightarrow T && \text{Domination}\end{aligned}$$

Propositional logic

- **Definition:**
 - A **proposition** is a statement that is either true or false.
- **Examples:**
 - Pitt is located in the Oakland section of Pittsburgh.
 - $5 + 2 = 8$.
 - It is raining today
 - 2 is a prime number
 - If (you do not drive over 65 mph) then (you will not get a speeding ticket).
- **Not a proposition:**
 - How are you?
 - $x + 5 = 3$

Limitations of the propositional logic

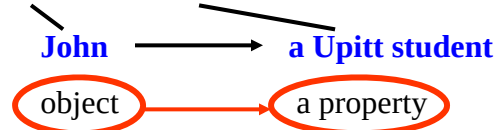
Propositional logic: the world is described in terms of elementary propositions and their logical combinations

Elementary statements:

- Typically refer to objects, their properties and relations.
But these are not explicitly represented in the propositional logic

– **Example:**

- “John is a UPitt student.”



- Objects and properties are hidden in the statement, it is not possible to reason about them

Limitations of the propositional logic

(1) Statements that must be repeated for many objects

- In propositional logic these must be exhaustively enumerated

• **Example:**

- If John is a CS UPitt graduate then John has passed cs441

Translation:

- John is a CS UPitt graduate $\rightarrow$ John has passed cs441

Similar statements can be written for other Upitt graduates:

- Ann is a CS Upitt graduate $\rightarrow$ Ann has passed cs441

- Ken is a CS Upitt graduate $\rightarrow$ Ken has passed cs441

– ...

- **What is a more natural solution to express the above knowledge?**

Limitations of the propositional logic

(1) Statements that must be repeated for many objects

- **Example:**

- If John is a CS UPitt graduate then John has passed cs441

Translation:

- John is a CS UPitt graduate $\rightarrow$ John has passed cs441

Similar statements can be written for other Upitt graduates:

- Ann is a CS Upitt graduate $\rightarrow$ Ann has passed cs441
- Ken is a CS Upitt graduate $\rightarrow$ Ken has passed cs441
- ...

- **Solution:** make statements with **variables**

- If x is a CS Upitt graduate then x has passed cs441
- x is a CS UPitt graduate $\rightarrow$ x has passed cs441

Limitations of the propositional logic

(2) Statements that define the property of the group of objects

- **Example:**

- All new cars must be registered.
- Some of the CS graduates graduate with honors.

- **Solution:** make statements with **quantifiers**

- **Universal quantifier** – the property is satisfied by all members of the group
- **Existential quantifier** – at least one member of the group satisfy the property

Predicate logic

Remedies the limitations of the propositional logic

- Explicitly models objects and their properties
- Allows to make statements with variables and quantify them

Basic building blocks of the predicate logic:

- **Constant** –models a specific object
Examples: “John”, “France”, “7”
- **Variable** – represents object of specific type (**defined by the universe of discourse**)
Examples: x, y
(universe of discourse can be people, students, numbers)
- **Predicate** - over one, two or many variables or constants.
 - Represents properties or relations among objects**Examples:** Red(car23), student(x), married(John,Ann)

Predicates

Predicates represent properties or relations among objects

A predicate $P(x)$ assigns a value **true or false** to each x depending on whether the property holds or not for x .

- The assignment is best viewed as a big table with the variable x substituted for objects from **the universe of discourse**

Example:

- Assume **Student(x)** where the universe of discourse are people
- Student(John) T (if John is a student)
- Student(Ann) T (if Ann is a student)
- Student(Jane) F (if Jane is not a student)
- ...

Predicates

Assume a predicate $P(x)$ that represents the statement:

- x is a prime number

What are the truth values of:

- | | |
|----------|---|
| • $P(2)$ | T |
| • $P(3)$ | T |
| • $P(4)$ | F |
| • $P(5)$ | T |
| • $P(6)$ | F |
| • $P(7)$ | T |

All statements $P(2)$, $P(3)$, $P(4)$, $P(5)$, $P(6)$, $P(7)$ are propositions

Predicates

Assume a predicate $P(x)$ that represents the statement:

- x is a prime number

What are the truth values of:

- | | |
|----------|---|
| • $P(2)$ | T |
| • $P(3)$ | T |
| • $P(4)$ | F |
| • $P(5)$ | T |
| • $P(6)$ | F |
| • $P(7)$ | T |

Is $P(x)$ a proposition? No. Many possible substitutions are possible.

Predicates

- Predicates can have **more arguments** which represent the **relations between objects**

Example:

- $\text{Older}(\text{John}, \text{Peter})$ denotes 'John is older than Peter'
 - this is a proposition because it is either true or false
- $\text{Older}(x, y)$ - 'x is older than y'
 - not a proposition, but after the substitution it becomes one

Predicates

- Predicates can have **more arguments** which represent the **relations between objects**

Example:

- Let $Q(x, y)$ denote ' $x+5 > y$ '
 - Is $Q(x, y)$ a proposition?

Predicates

- Predicates can have **more arguments** which represent the **relations between objects**

Example:

- Let $Q(x,y)$ denote ' $x+5 > y$ '
 - Is $Q(x,y)$ a proposition? **No!**
 - Is $Q(3,7)$ a proposition? **Yes.** It is true.
 - What is the truth value of:
 - $Q(3,7)$ T
 - $Q(1,6)$ F
 - $Q(2,2)$ T
 - Is $Q(3,y)$ a proposition? **No!** We cannot say if it is true or false.

Compound statements in predicate logic

Compound statements are obtained via logical connectives

Examples:

$\text{Student}(\text{Ann}) \wedge \text{Student}(\text{Jane})$

- **Translation:** "Both Ann and Jane are students"
- **Proposition:** yes.

$\text{Country}(\text{Sienna}) \vee \text{River}(\text{Sienna})$

- **Translation:** "Sienna is a country or a river"
- **Proposition:** yes.

$\text{CS-major}(x) \rightarrow \text{Student}(x)$

- **Translation:** "if x is a CS-major then x is a student"
- **Proposition:** no.

Predicates

Important:

- statement $P(x)$ is **not a proposition** since there are more objects it can be applied to

This is the same as in propositional logic ...

... But the difference is:

- predicate logic allows us to explicitly manipulate and substitute for the objects
- Predicate logic permits quantified sentences where variables are substituted for statements about the group of objects

Quantified statements

Predicate logic lets us to make statements about groups of objects

- To do this we use special quantified expressions

Two types of quantified statements:

- **universal**

Example: ‘all CS Upitt graduates have to pass cs441’

– the statement is true for all graduates

- **existential**

Example: ‘Some CS Upitt students graduate with honor.’

– the statement is true for some people

Universal quantifier

Defn: The universal quantification of $P(x)$ is the proposition:

" $P(x)$ is true for all values of x in the domain of discourse." The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$, and is expressed as **for every x , $P(x)$** .

Example:

- Let $P(x)$ denote $x > x - 1$.
- What is the truth value of $\forall x P(x)$?
- Assume the universe of discourse of x is all real numbers.
- **Answer:** Since every number x is greater than itself minus 1. Therefore, **$\forall x P(x)$ is true.**

Universal quantifier

Quantification converts a propositional function into a **proposition** by binding a variable to a set of values from the universe of discourse.

Example:

- Let $P(x)$ denote $x > x - 1$.
- Is $P(x)$ a proposition? **No.** Many possible substitutions.
- Is $\forall x P(x)$ a proposition? **Yes.** True if for all x from the universe of discourse $P(x)$ is true.

Universally quantified statements

Predicate logic lets us make statements about groups of objects

Universally quantified statement

- $\text{CS-major}(x) \rightarrow \text{Student}(x)$
 - **Translation:** “if x is a CS-major then x is a student”
 - **Proposition:** **no.**
- $\forall x \text{CS-major}(x) \rightarrow \text{Student}(x)$
 - **Translation:** “(For all people it holds that) if a person is a CS-major then she is a student.”
 - **Proposition:** **yes.**

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Existential quantifier

Definition: The **existential quantification** of $P(x)$ is the proposition “*There exists an element in the domain (universe) of discourse such that $P(x)$ is true.*” The notation $\exists x P(x)$ denotes the existential quantification of $P(x)$, and is expressed as **there is an x such that $P(x)$ is true.**

Example 1:

- Let $T(x)$ denote $x > 5$ and x is from Real numbers.
- What is the truth value of $\exists x T(x)$?
- **Answer:**
- Since $10 > 5$ is true. Therefore, it is **true that $\exists x T(x)$.**

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Existential quantifier

Definition: The **existential quantification** of $P(x)$ is the proposition "*There exists an element in the domain (universe) of discourse such that $P(x)$ is true.*" The notation $\exists x P(x)$ denotes the existential quantification of $P(x)$, and is expressed as **there is an x such that $P(x)$ is true.**

Example 2:

- Let $Q(x)$ denote $x = x + 2$ where x is real numbers
- What is the truth value of $\exists x Q(x)$?
- **Answer:** Since no real number is 2 larger than itself, the truth value of $\exists x Q(x)$ is false.

Existentially quantified statements

Statements about groups of objects

Example:

- $\text{CS-UPitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “ x is a CS-UPitt-graduate and x is an honor student”
 - **Proposition:** ?

Quantified statements

Statements about groups of objects

Example:

- $\text{CS-Upitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “x is a CS-Upitt-graduate and x is an honor student”
 - **Proposition:** **no.**
- $\exists x \text{CS-Upitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “There is a person who is a CS-Upitt-graduate and who is also an honor student.”
 - **Proposition:** ?

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Quantified statements

Statements about groups of objects

Example:

- $\text{CS-Upitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “x is a CS-Upitt-graduate and x is an honor student”
 - **Proposition:** **no.**
- $\exists x \text{CS-Upitt-graduate}(x) \wedge \text{Honor-student}(x)$
 - **Translation:** “There is a person who is a CS-Upitt-graduate and who is also an honor student.”
 - **Proposition:** **yes.**

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Summary of quantified statements

- When $\forall x P(x)$ and $\exists x P(x)$ are true and false?

Statement	When true?	When false?
$\forall x P(x)$	$P(x)$ true for all x	There is an x where $P(x)$ is false.
$\exists x P(x)$	There is some x for which $P(x)$ is true.	$P(x)$ is false for all x .

Suppose the elements in the universe of discourse can be enumerated as $x_1, x_2, \dots, x_N$ then:

- $\forall x P(x)$ is true whenever $P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_N)$ is true
- $\exists x P(x)$ is true whenever $P(x_1) \vee P(x_2) \vee \dots \vee P(x_N)$ is true.

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Lecture 4

Predicate logic

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Announcements

- **Homework assignment 1 due today**
- **Homework assignment 2:**
 - **posted on the course web page**
 - **Due on Thursday January 23, 2013**
- **Recitations today and tomorrow:**
 - **Practice problems related to assignment 2**

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Propositional logic: limitations

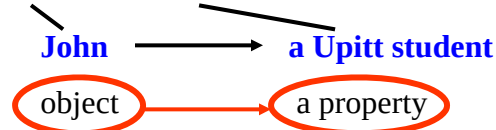
Propositional logic: the world is described in terms of elementary propositions and their logical combinations

Elementary statements/propositions:

- Typically refer to objects, their properties and relations.
But these are not explicitly represented in the propositional logic

– **Example:**

- “John is a UPitt student.”



- Objects and properties are hidden in the statement, it is not possible to reason about them

Propositional logic: limitations

(1) Statements that hold for many objects must be enumerated

• **Example:**

- John is a CS UPitt graduate $\rightarrow$ John has passed cs441
- Ann is a CS UPitt graduate $\rightarrow$ Ann has passed cs441
- Ken is a CS UPitt graduate $\rightarrow$ Ken has passed cs441
- ...

• **Solution:** make statements with **variables**

- x is a CS UPitt graduate $\rightarrow x$ has passed cs441

Propositional logic: limitations

(2) Statements that define the property of the group of objects

- **Example:**
 - All new cars must be registered.
 - Some of the CS graduates graduate with honor.
- **Solution:** make statements with **quantifiers**
 - **Universal quantifier** – the property is satisfied by all members of the group
 - **Existential quantifier** – at least one member of the group satisfy the property

Predicate logic

Remedies the limitations of the propositional logic

- Explicitly models objects and their properties
- Allows to make statements with variables and quantify them

Predicate logic:

- **Constant** – models a specific object
Examples: “John”, “France”, “7”
- **Variable** – represents object of specific type (**defined by the universe of discourse**)
Examples: x, y
(universe of discourse can be people, students, numbers)
- **Predicate** – over one, two or many variables or constants.
 - Represents properties or relations among objects**Examples:** $\text{Red}(\text{car23})$, $\text{student}(x)$, $\text{married}(\text{John}, \text{Ann})$

Predicates

Predicates represent properties or relations among objects

- A predicate $P(x)$ assigns a value **true or false** to each x depending on whether the property holds or not for x .
- The assignment is best viewed as a big table with the variable x substituted for objects from *the universe of discourse*

Example:

- Assume **Student(x)** where the universe of discourse are people
- Student(John) T (if John is a student)
- Student(Ann) T (if Ann is a student)
- Student(Jane) F (if Jane is not a student)
- ...

Predicates

Assume a predicate $P(x)$ that represents the statement:

- **x is a prime number**

Truth values for different x :

- | | |
|----------|---|
| • $P(2)$ | T |
| • $P(3)$ | T |
| • $P(4)$ | F |
| • $P(5)$ | T |
| • $P(6)$ | F |

All statements $P(2)$, $P(3)$, $P(4)$, $P(5)$, $P(6)$ are propositions

...

But $P(x)$ with variable x is not a proposition

Quantified statements

Predicate logic lets us to make statements about groups of objects

- To do this we use **special quantified expressions**

Two types of quantified statements:

- **universal**

Example: ‘all CS Upitt graduates have to pass cs441’

- the statement is true for all graduates

- **existential**

Example: ‘Some CS Upitt students graduate with honor.’

- the statement is true for some people

Universal quantifier

Quantification converts a propositional function into a **proposition** by binding a variable to a set of values from the universe of discourse.

Example:

- Let $P(x)$ denote $x > x - 1$. Assume x are real numbers.
- Is $P(x)$ a proposition? **No**. Many possible substitutions.
- Is $\forall x P(x)$ a proposition? **Yes**.
- What is the truth value for $\forall x P(x)$?
 - **True**, since $P(x)$ holds for all x .

Existential quantifier

Quantification converts a propositional function into a **proposition** by binding a variable to a set of values from the universe of discourse.

Example:

- Let $T(x)$ denote $x > 5$ and x is from Real numbers.
- Is $T(x)$ a proposition? **No.**
- Is $\exists x T(x)$ a proposition? **Yes.**
- What is the truth value for $\exists x T(x)$?
 - Since $10 > 5$ is true. Therefore, $\exists x T(x)$ is **true**.

Summary of quantified statements

- When $\forall x P(x)$ and $\exists x P(x)$ are true and false?

Statement	When true?	When false?
$\forall x P(x)$	$P(x)$ true for all x	There is an x where $P(x)$ is false.
$\exists x P(x)$	There is some x for which $P(x)$ is true.	$P(x)$ is false for all x .

Suppose the elements in the universe of discourse can be enumerated as $x_1, x_2, \dots, x_N$ then:

- $\forall x P(x)$ is true whenever $P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_N)$ is true
- $\exists x P(x)$ is true whenever $P(x_1) \vee P(x_2) \vee \dots \vee P(x_N)$ is true.

Translation with quantifiers

Sentence:

- All Upitt students are smart.
- **Assume:** the domain of discourse of x are Upitt students
- **Translation:**
- $\forall x \text{ Smart}(x)$
- **Assume:** the universe of discourse are students (all students):
- $\forall x \text{ at}(x, \text{Upitt}) \rightarrow \text{Smart}(x)$
- **Assume:** the universe of discourse are people:
- $\forall x \text{ student}(x) \wedge \text{at}(x, \text{Upitt}) \rightarrow \text{Smart}(x)$

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Translation with quantifiers

Sentence:

- Someone at CMU is smart.
- **Assume:** the domain of discourse are all CMU affiliates
- **Translation:**
- $\exists x \text{ Smart}(x)$
- **Assume:** the universe of discourse are people:
- $\exists x \text{ at}(x, \text{CMU}) \wedge \text{Smart}(x)$

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Translation with quantifiers

- Assume two predicates $S(x)$ and $P(x)$

Universal statements typically tie with implications

- All $S(x)$ is $P(x)$
 - $\forall x (S(x) \rightarrow P(x))$
- No $S(x)$ is $P(x)$
 - $\forall x (S(x) \rightarrow \neg P(x))$

Existential statements typically tie with conjunctions

- Some $S(x)$ is $P(x)$
 - $\exists x (S(x) \wedge P(x))$
- Some $S(x)$ is not $P(x)$
 - $\exists x (S(x) \wedge \neg P(x))$

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Nested quantifiers

- More than one quantifier may be necessary to capture the meaning of a statement in the predicate logic.

Example:

- Every real number has its corresponding negative.
- **Translation:**
 - Assume:
 - a real number is denoted as x and its negative as y
 - A predicate $P(x,y)$ denotes: “ $x + y = 0$ ”
- Then we can write:
 - $\forall x \exists y P(x,y)$

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Nested quantifiers

- More than one quantifier may be necessary to capture the meaning of a statement in the predicate logic.

Example:

- There is a person who loves everybody.
- **Translation:**
 - Assume:
 - Variables x and y denote people
 - A predicate $L(x,y)$ denotes: “ x loves y ”
- Then we can write in the predicate logic:
 $\exists x \forall y L(x,y)$

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Order of quantifiers

The order of nested quantifiers matters if quantifiers are of different type

- $\forall x \exists y L(x,y)$ is not the same as $\exists y \forall x L(x,y)$

Example:

- Assume $L(x,y)$ denotes “ x loves y ”
- Then: $\forall x \exists y L(x,y)$
- Translates to: Everybody loves somebody.
- And: $\exists y \forall x L(x,y)$
- Translates to: There is someone who is loved by everyone.

The meaning of the two is different.

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Order of quantifiers

The order of nested quantifiers does not matter if quantifiers are of the same type

Example:

- For all x and y , if x is a parent of y then y is a child of x
- **Assume:**
 - $\text{Parent}(x,y)$ denotes “ x is a parent of y ”
 - $\text{Child}(x,y)$ denotes “ x is a child of y ”
- Two equivalent ways to represent the statement:
 - $\forall x \forall y \text{Parent}(x,y) \rightarrow \text{Child}(y,x)$
 - $\forall y \forall x \text{Parent}(x,y) \rightarrow \text{Child}(y,x)$

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Translation exercise

Suppose:

- Variables x,y denote people
- $L(x,y)$ denotes “ x loves y ”.

Translate:

- Everybody loves Raymond. $\forall x L(x, \text{Raymond})$
- Everybody loves somebody. $\forall x \exists y L(x,y)$
- There is somebody whom everybody loves. $\exists y \forall x L(x,y)$
- There is somebody who Raymond doesn't love.
 $\exists y \neg L(\text{Raymond}, y)$
- There is somebody whom no one loves.
 $\exists y \forall x \neg L(x,y)$

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Lecture 5

Predicate logic

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Negation of quantifiers

English statement:

- Nothing is perfect.
- **Translation:** $\neg \exists x \text{ Perfect}(x)$

Another way to express the same meaning:

- **Everything ...**

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Negation of quantifiers

English statement:

- Nothing is perfect.
- **Translation:** $\neg \exists x \text{ Perfect}(x)$

Another way to express the same meaning:

- **Everything is imperfect.**
- **Translation:** $\forall x \neg \text{Perfect}(x)$

Conclusion: $\neg \exists x P(x)$ is equivalent to $\forall x \neg P(x)$

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Negation of quantifiers

English statement:

- It is not the case that all dogs are fleabags.
- **Translation:** $\neg \forall x \text{ Dog}(x) \rightarrow \text{Fleabag}(x)$

Another way to express the same meaning:

- There is a dog that ...

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Negation of quantifiers

English statement:

- It is not the case that all dogs are fleabags.
- **Translation:** $\neg \forall x \text{ Dog}(x) \rightarrow \text{Fleabag}(x)$

Another way to express the same meaning:

- There is a dog that is not a fleabag.
- **Translation:** $\exists x \text{ Dog}(x) \wedge \neg \text{Fleabag}(x)$
- Logically equivalent to:
 - $\exists x \neg (\text{Dog}(x) \rightarrow \text{Fleabag}(x))$

Conclusion: $\neg \forall x P(x)$ is equivalent to $\exists x \neg P(x)$

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Negation of quantified statements (aka DeMorgan Laws for quantifiers)

Negation	Equivalent
$\neg \exists x P(x)$	$\forall x \neg P(x)$
$\neg \forall x P(x)$	$\exists x \neg P(x)$

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Formal and informal proofs

Theorems and proofs

- The truth value of some statement about the world is obvious and easy to assign
- The truth of other statements may not be obvious, ...
.... But it may still follow (be derived) from known facts about the world

To show the truth value of such a statement following from other statements we need to provide **a correct supporting argument**

- **a proof**

Important questions:

- When is the argument correct?
- How to construct a correct argument, what method to use?

Theorems and proofs

- **Theorem:** a statement that can be shown to be true.

– Typically the theorem looks like this:

$$(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n) \rightarrow q$$



- **Example:**

Fermat's Little theorem:

- If p is a prime and a is an integer not divisible by p ,
then: $a^{p-1} \equiv 1 \pmod{p}$

Theorems and proofs

- **Theorem:** a statement that can be shown to be true.

– Typically the theorem looks like this:

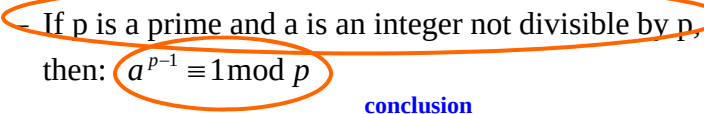
$$(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n) \rightarrow q$$



- **Example:**

Premises (hypotheses)

Fermat's Little theorem:

- If p is a prime and a is an integer not divisible by p ,
then: $a^{p-1} \equiv 1 \pmod{p}$
- 

Formal proofs

Proof:

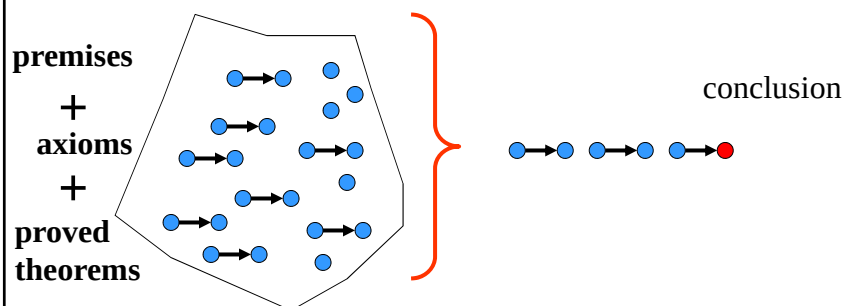
- Provides an argument supporting the validity of the statement
- Proof of the theorem:
 - shows that the conclusion follows from premises
 - May use:
 - Premises
 - Axioms
 - Results of other theorems

Formal proofs:

- steps of the proofs **follow logically** from the set of premises and axioms

Formal proofs

- Formal proofs:
 - show that **steps of the proofs follow logically** from the set of hypotheses and axioms



In this class we assume formal proofs in the **propositional logic**

Using logical equivalence rules

Proofs based on logical equivalences. A proposition or its part can be transformed using a sequence of equivalence rewrites till some conclusion can be reached.

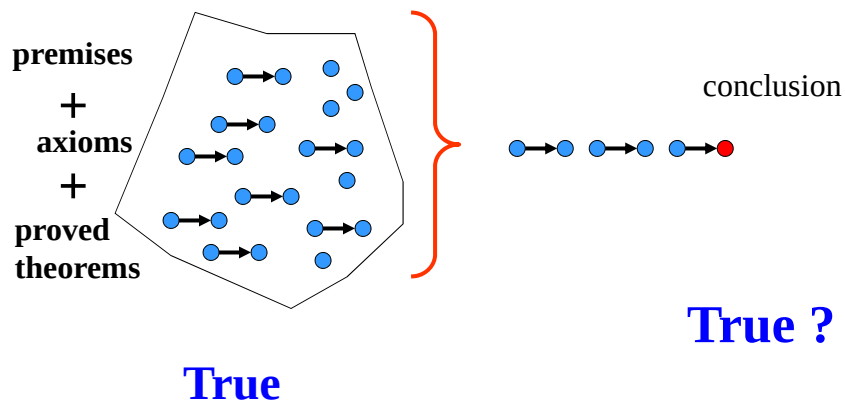
Example: Show that $(p \wedge q) \rightarrow p$ is a tautology.

- Proof: (we must show $(p \wedge q) \rightarrow p \Leftrightarrow T$)

$$\begin{aligned}
 (p \wedge q) \rightarrow p &\Leftrightarrow \neg(p \wedge q) \vee p && \text{Useful} \\
 &\Leftrightarrow [\neg p \vee \neg q] \vee p && \text{DeMorgan} \\
 &\Leftrightarrow [\neg q \vee \neg p] \vee p && \text{Commutative} \\
 &\Leftrightarrow \neg q \vee [\neg p \vee p] && \text{Associative} \\
 &\Leftrightarrow \neg q \vee [T] && \text{Useful}
 \end{aligned}$$

Formal proofs

Allow us to infer new True statements from known True statements



Rules of inference

Rules of inference:

- Allow us to infer new True statements from existing True statements
- Represent logically valid inference patterns

Example:

- **Modus Ponens**, or the Law of Detachment
- Rule of inference

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

- Given p is true and the implication $p \rightarrow q$ is true then q is true.

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Rules of inference

Rules of inference: logically valid inference patterns

Example;

- **Modus Ponens**, or the Law of Detachment
- Rule of inference

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

- Given p is true and the implication $p \rightarrow q$ is true then q is true.

p	q	$p \rightarrow q$
<i>False</i>	<i>False</i>	<i>True</i>
<i>False</i>	<i>True</i>	<i>True</i>
<i>True</i>	<i>False</i>	<i>False</i>
<i>True</i>	<i>True</i>	<i>True</i>

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Rules of inference

Rules of inference: logically valid inference patterns

Example;

- **Modus Ponens**, or the Law of Detachment
- Rule of inference

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$
- Given p is true and the implication $p \rightarrow q$ is true then q is true.

p	q	$p \rightarrow q$
False	False	True
False	True	True
True	False	False
True	True	True

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Rules of inference

Rules of inference: logically valid inference patterns

Example:

- **Modus Ponens**, or the Law of Detachment
- Rules of inference

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$
- Given p is true and the implication $p \rightarrow q$ is true then q is true.

- **Tautology Form:** $(p \wedge (p \rightarrow q)) \rightarrow q$

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Rules of inference

- **Addition**

$$p \rightarrow (p \vee q) \qquad \frac{p}{\therefore p \vee q}$$

- **Example:** It is below freezing now. Therefore, it is below freezing or raining snow.

- **Simplification**

$$(p \wedge q) \rightarrow p \qquad \frac{p \wedge q}{\therefore p}$$

- **Example:** It is below freezing and snowing. Therefore it is below freezing.

Rules of inference

- **Modus Tollens (modus ponens for the contrapositive)**

$$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p \qquad \frac{\neg q \quad p \rightarrow q}{\therefore \neg p}$$

- **Hypothetical Syllogism**

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r) \qquad \frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$$

- **Disjunctive Syllogism**

$$[(p \vee q) \wedge \neg p] \rightarrow q \qquad \frac{p \vee q \quad \neg p}{\therefore q}$$

Rules of inference

- Logical equivalences (discussed earlier)

$$A \Leftrightarrow B$$

$A \rightarrow B$ is a tautology

Example: De Morgan Law

$$\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$$

$\neg(p \vee q) \rightarrow \neg p \wedge \neg q$ is a tautology

Rules of inference

- A **valid argument** is one built using the rules of inference from premises (hypotheses). When all premises are true the argument should lead us to the correct conclusion.
- $(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n) \rightarrow q$
- **How to use the rules of inference?**

Applying rules of inference

Assume the following statements (hypotheses):

- It is not sunny this afternoon and it is colder than yesterday.
- We will go swimming only if it is sunny.
- If we do not go swimming then we will take a canoe trip.
- If we take a canoe trip, then we will be home by sunset.

Show that all these lead to a conclusion:

- We will be home by sunset.

Applying rules of inference

Text:

- (1) It is not sunny this afternoon and it is colder than yesterday.
- (2) We will go swimming only if it is sunny.
- (3) If we do not go swimming then we will take a canoe trip.
- (4) If we take a canoe trip, then we will be home by sunset.

Propositions:

- p = It is sunny this afternoon, q = it is colder than yesterday,
 r = We will go swimming, s = we will take a canoe trip
- t = We will be home by sunset

Translation:

- **Assumptions:** (1) $\neg p \wedge q$, (2) ?
- **We want to show:** t

Applying rules of inference

- **Approach:**

- p = It is sunny this afternoon, q = it is colder than yesterday,
 r = We will go swimming, s = we will take a canoe trip
- t = We will be home by sunset

Translation: “We will go swimming only if it is sunny”.

- Ambiguity: $r \rightarrow p$ or $p \rightarrow r$?
- Sunny is a must before we go swimming
- Thus, if we indeed go swimming it must be sunny,
therefore $r \rightarrow p$

Applying rules of inference

Text:

- (1) It is not sunny this afternoon and it is colder than yesterday.
- (2) We will go swimming only if it is sunny.
- (3) If we do not go swimming then we will take a canoe trip.
- (4) If we take a canoe trip, then we will be home by sunset.

Propositions:

- p = It is sunny this afternoon, q = it is colder than yesterday,
 r = We will go swimming, s = we will take a canoe trip
- t = We will be home by sunset

Translation:

- **Assumptions:** (1) $\neg p \wedge q$, (2) $r \rightarrow p$, (3) $\neg r \rightarrow s$, (4) $s \rightarrow t$
- **We want to show:** t

Proofs using rules of inference

Translations:

- **Assumptions:** $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, $s \rightarrow t$
- **We want to show:** t

Proof:

- 1. $\neg p \wedge q$ Hypothesis
- 2. $\neg p$ Simplification
- 3. $r \rightarrow p$ Hypothesis
- 4. $\neg r$ Modus tollens (step 2 and 3)
- 5. $\neg r \rightarrow s$ Hypothesis
- 6. s Modus ponens (steps 4 and 5)
- 7. $s \rightarrow t$ Hypothesis
- 8. t Modus ponens (steps 6 and 7)
- **end of proof**

Informal proofs

Proving theorems in practice:

- The steps of the proofs are not expressed in any formal language as e.g. propositional logic
- Steps are argued less formally using English, mathematical formulas and so on
- One must always watch the consistency of the argument made, logic and its rules can often help us to decide the soundness of the argument if it is in question
- **We use (informal) proofs to illustrate different methods of proving theorems**

Methods of proving theorems

Basic methods to prove the theorems:

- **Direct proof**
 - $p \rightarrow q$ is proved by showing that if p is true then q follows
- **Indirect proof**
 - Show the contrapositive $\neg q \rightarrow \neg p$. If $\neg q$ holds then $\neg p$ follows
- **Proof by contradiction**
 - Show that $(p \wedge \neg q)$ contradicts the assumptions
- **Proof by cases**
- **Proofs of equivalence**
 - $p \leftrightarrow q$ is replaced with $(p \rightarrow q) \wedge (q \rightarrow p)$

Sometimes one method of proof does not go through as nicely as the other method. Hint: try more than one approach.

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Direct proof

- $p \rightarrow q$ is proved by showing that if p is true then q follows
- **Example:** Prove that “If n is odd, then n^2 is odd.”

Proof:

- Assume the hypothesis is true, i.e. suppose n is odd.
- Then $n = 2k + 1$, where k is an integer.

$$\begin{aligned}n^2 &= (2k + 1)^2 \\&= 4k^2 + 4k + 1 \\&= 2(2k^2 + 2k) + 1\end{aligned}$$

- Therefore, n^2 is odd. $\square$

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Indirect proof

- To show $p \rightarrow q$ prove its contrapositive $\neg q \rightarrow \neg p$
- Why? **$p \rightarrow q$ and $\neg q \rightarrow \neg p$ are equivalent !!!**
- Assume $\neg q$ is true, show that $\neg p$ is true.

Example: Prove If **$3n + 2$ is odd** then n is odd.

Proof:

- Assume n is even, that is $n = 2k$, where k is an integer.
- Then:
$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k+1) \end{aligned}$$
- Therefore **$3n + 2$ is even.**

Indirect proof

- To show $p \rightarrow q$ prove its contrapositive $\neg q \rightarrow \neg p$
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- Then:
$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k+1) \end{aligned}$$
- Therefore $3n + 2$ is even.
- We proved **$\neg$ “ n is odd” $\rightarrow$ $\neg$ “ $3n + 2$ is odd”**. This is equivalent to **“ $3n + 2$ is odd” $\rightarrow$ “ n is odd”**. $\square$

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Lecture 6

Informal proofs

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Proofs

- The truth value of some statements about the world are obvious and easy to assess
- The truth of other statements may not be obvious, ...
.... But it may still follow (be derived) from known facts about the world

Proof: shows that the truth value of such a statement follows from (or can be inferred) from the truth value of other statements

Important questions:

- When is the argument correct?
- How to construct a correct argument, what method to use?

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Theorems

- **Theorem:** a statement that can be shown to be true.

– Typically the theorem looks like this:

$$\underbrace{(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n)}_{\text{Premises}} \rightarrow \underbrace{q}_{\text{conclusion}}$$

- **Example:**

Premises (hypotheses)

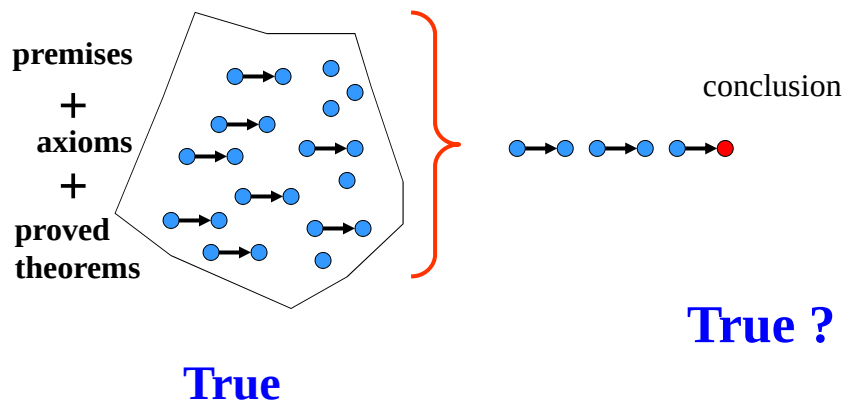
Fermat's Little theorem:

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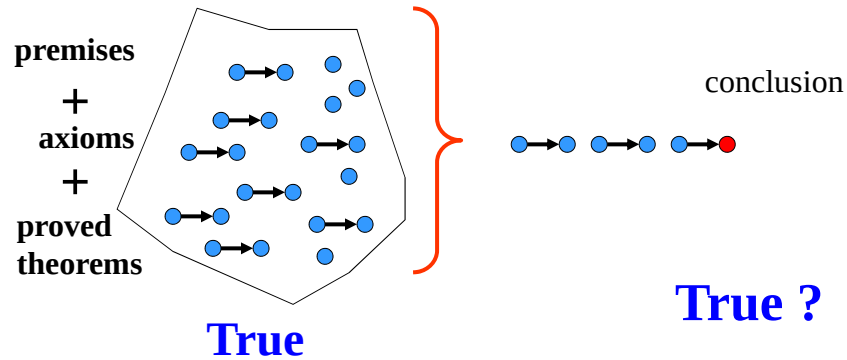
conclusion

Formal proofs

Allow us to infer from **new True statements** from **known True statements**



Formal proofs



Steps of the proof for statements in the propositional logic are argued using:

- Equivalence rules
- Rules of inference (e.g. modus ponens)

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Proofs using rules of inference

Translations:

- **Assumptions:** $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, $s \rightarrow t$
- **We want to show:** t

Proof:

- 1. $\neg p \wedge q$ Hypothesis
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- **end of proof**

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Informal proofs

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Direct proof

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Proof:

- Assume the hypothesis is true, i.e. suppose n is odd.
- Then $n = 2k + 1$, where k is an integer.

$$\begin{aligned}n^2 &= (2k + 1)^2 \\&= 4k^2 + 4k + 1 \\&= 2(2k^2 + 2k) + 1\end{aligned}$$

- Therefore, n^2 is odd. $\square$

Indirect proof

- To show $p \rightarrow q$ prove its contrapositive $\neg q \rightarrow \neg p$
- Why? **$p \rightarrow q$ and $\neg q \rightarrow \neg p$ are equivalent !!!**
- Assume $\neg q$ is true, show that $\neg p$ is true.

Example: Prove If $3n + 2$ is odd then n is odd.

Proof:

- Assume n is even, that is $n = 2k$, where k is an integer.

$$\begin{aligned}\text{Then: } 3n + 2 &= 3(2k) + 2 \\&= 6k + 2 \\&= 2(3k+1)\end{aligned}$$

- Therefore $3n + 2$ is even.
- We proved $\neg$ “ n is odd” $\rightarrow$ $\neg$ “ $3n + 2$ is odd”. This is equivalent to “ $3n + 2$ is odd” $\rightarrow$ “ n is odd”. $\square$

Proof by contradiction

- We want to prove $p \rightarrow q$
- The only way to reject (or disprove) $p \rightarrow q$ is to show that $(p \wedge \neg q)$ can be true
- However, if we manage to prove that either q or $\neg p$ is True then we contradict $(p \wedge \neg q)$
 - and subsequently $p \rightarrow q$ must be true
- Proof by contradiction. Show that the assumption $(p \wedge \neg q)$ leads either to q or $\neg p$ which generates a contradiction.

Proof by contradiction

- We want to prove $p \rightarrow q$
- To reject $p \rightarrow q$ show that $(p \wedge \neg q)$ can be true
- To reject $(p \wedge \neg q)$ show that either q or $\neg p$ is True

Example: Prove If $3n + 2$ is odd then n is odd.

Proof:

- Assume $3n + 2$ is odd and n is even, that is $n = 2k$, where k an integer.

Proof by contradiction

- We want to prove $p \rightarrow q$
- To reject $p \rightarrow q$ show that $(p \wedge \neg q)$ can be true
- To reject $(p \wedge \neg q)$ show that either q or $\neg p$ is True

Example: Prove If $3n + 2$ is odd then n is odd.

Proof:

- Assume $3n + 2$ is odd and n is even, that is $n = 2k$, where k an integer.
- Then:
$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k + 1) \end{aligned}$$
- Thus $3n + 2$ is even. This is a contradiction with the assumption that $3n + 2$ is odd. Therefore n is odd. $\square$

Vacuous proof

We want to show $p \rightarrow q$

- Suppose p (the hypothesis) is always false
- Then $p \rightarrow q$ is always true.

Reason:

- $F \rightarrow q$ is always T, whether q is True or False

Example:

- Let $P(n)$ denotes “if $n > 1$ then $n^2 > n$ ” is TRUE.
- Show that $P(0)$.

Proof:

- For $n=0$ the premise is False. Thus $P(0)$ is always true.

Trivial proofs

We want to show $p \rightarrow q$

- Suppose the conclusion q is always true
- Then the implication $p \rightarrow q$ is trivially true.
- **Reason:**
- $p \rightarrow T$ is always T , whether p is True or False

Example:

- Let $P(n)$ is “if $a \geq b$ then $a^n \geq b^n$ ”
- Show that $P(0)$

Proof:

$a^0 \geq b^0$ is $1=1$ trivially true.

Proof by cases

- We want to show $p_1 \vee p_2 \vee \dots \vee p_n \rightarrow q$
- Note that this is equivalent to
 - $(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)$
- **Why?**
- $p_1 \vee p_2 \vee \dots \vee p_n \rightarrow q \iff$ (useful)
- $\neg (p_1 \vee p_2 \vee \dots \vee p_n) \vee q \iff$ (De Morgan)
- $(\neg p_1 \wedge \neg p_2 \wedge \dots \wedge \neg p_n) \vee q \iff$ (distributive)
- $(\neg p_1 \vee q) \wedge (\neg p_2 \vee q) \wedge \dots \wedge (\neg p_n \vee q) \iff$ (useful)
- $(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)$

Proof by cases

We want to show $p_1 \vee p_2 \vee \dots \vee p_n \rightarrow q$

- Equivalent to $(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)$

Prove individual cases as before. All of them must be true.

Example: Show that $|x||y|=|xy|$.

Proof:

- 4 cases:
- $x \geq 0, y \geq 0$
- $x \geq 0, y < 0$
- $x < 0, y \geq 0$
- $x < 0, y < 0$

Proof by cases

We want to show $p_1 \vee p_2 \vee \dots \vee p_n \rightarrow q$

- Equivalent to $(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)$

Prove individual cases as before. All of them must be true.

Example: Show that $|x||y|=|xy|$.

Proof:

- 4 cases:
- $x \geq 0, y \geq 0$ $xy \geq 0$ and $|xy|=xy=|x||y|$
- $x \geq 0, y < 0$ $xy < 0$ and $|xy|=-xy=x(-y)=|x||y|$
- $x < 0, y \geq 0$ $xy < 0$ and $|xy|=-xy=(-x)y=|x||y|$
- $x < 0, y < 0$ $xy > 0$ and $|xy|=(-x)(-y)=|x||y|$
- All cases proved.

Proof of equivalences

We want to prove $p \leftrightarrow q$

- Statements: p if and only if q .
- Note that $p \leftrightarrow q$ is equivalent to $[(p \rightarrow q) \wedge (q \rightarrow p)]$
- Both implications must hold.

Example:

- Integer is odd if and only if n^2 is odd.

Proof of $(p \rightarrow q)$:

- **$(p \rightarrow q)$** If n is odd then n^2 is odd
- we use a direct proof
- Suppose n is odd. Then $n = 2k + 1$, where k is an integer.
- $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$
- Therefore, n^2 is odd.

Proof of equivalences

We want to prove $p \leftrightarrow q$

- Note that $p \leftrightarrow q$ is equivalent to $[(p \rightarrow q) \wedge (q \rightarrow p)]$
- Both implications must hold.

- Integer is odd if and only if n^2 is odd.

Proof of $(q \rightarrow p)$:

- **$(q \rightarrow p)$:** if n^2 is odd then n is odd
- we use an indirect proof $(\neg p \rightarrow \neg q)$ is a contrapositive
- n is even that is $n = 2k$,
- then $n^2 = 4k^2 = 2(2k^2)$
- Therefore n^2 is even. Done proving the contrapositive.

Since both $(p \rightarrow q)$ and $(q \rightarrow p)$ are true the equivalence is true

Proofs with quantifiers

- **Existence proof** – sentences expressed with an existential quantifiers
 - **Constructive**
 - Find an example (through search) that shows the statement holds.
 - **Nonconstructive**
 - Show the statement holds for one example but we do not have the witness example. Typically relies on the proof by contradiction – negate the existentially quantified statement and show that it implies a contradiction.

Proofs with quantifiers

- **Universally quantified statements**
 - **Prove the property holds for all examples**
 - can be tricky
 - **proof by cases** to divide the proof to the different subgroups may help
 - **Counterexamples:**
 - use to disprove universal statements
 - Similar to constructive proofs for existentially

CS 441 Discrete Mathematics for CS
Lecture 7

Sets and set operations

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Basic discrete structures

- **Discrete math** =
 - study of the discrete structures used to represent discrete objects
- Many discrete structures are built using [sets](#)
 - **Sets = collection of objects**

Examples of discrete structures built with the help of sets:

- **Combinations**
- **Relations**
- **Graphs**

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Set

- **Definition:** A **set** is a (unordered) collection of objects. These objects are sometimes called **elements** or **members** of the set. (Cantor's naive definition)
- **Examples:**
 - **Vowels in the English alphabet**
 $V = \{ a, e, i, o, u \}$
 - **First seven prime numbers.**
 $X = \{ 2, 3, 5, 7, 11, 13, 17 \}$

Representing sets

Representing a set by:

- 1) Listing (enumerating) the members of the set.
- 2) Definition by property, using the set builder notation
 $\{x \mid x \text{ has property } P\}.$

Example:

- Even integers between 50 and 63.
 - 1) $E = \{50, 52, 54, 56, 58, 60, 62\}$
 - 2) $E = \{x \mid 50 \leq x < 63, x \text{ is an even integer}\}$

If enumeration of the members is hard we often use ellipses.

Example: a set of integers between 1 and 100

- $A = \{1, 2, 3, \dots, 100\}$

Important sets in discrete math

- **Natural numbers:**
 - $\mathbf{N} = \{0, 1, 2, 3, \dots\}$
- **Integers**
 - $\mathbf{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- **Positive integers**
 - $\mathbf{Z}^+ = \{1, 2, 3, \dots\}$
- **Rational numbers**
 - $\mathbf{Q} = \{p/q \mid p \in \mathbf{Z}, q \in \mathbf{Z}, q \neq 0\}$
- **Real numbers**
 - $\mathbf{R}$

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Russell's paradox

Cantor's naive definition of sets leads to Russell's paradox:

- **Let $S = \{x \mid x \notin x\}$,**
is a set of sets that are not members of themselves.
- **Question:** Where does the set S belong to?
 - Is $S \in S$ or $S \notin S$?
- **Cases**
 - **$S \in S$?:** S does not satisfy the condition so it must hold that $S \notin S$ (or $S \in S$ does not hold)
 - **$S \notin S$?:** S is included in the set S and hence $S \notin S$ does not hold
- **A paradox:** we cannot decide if S belongs to S or not
- **Russell's answer:** theory of types – used for sets of sets

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Equality

Definition: Two sets are equal if and only if they have the same elements.

Example:

- $\{1,2,3\} = \{3,1,2\} = \{1,2,1,3,2\}$

Note: Duplicates don't contribute anything new to a set, so remove them. The order of the elements in a set doesn't contribute anything new.

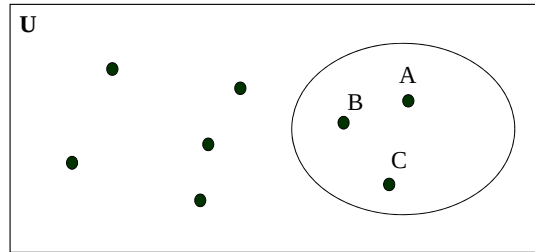
Example: Are $\{1,2,3,4\}$ and $\{1,2,2,4\}$ equal?
No!

Special sets

- **Special sets:**
 - The universal set is denoted by U : the set of all objects under the consideration.
 - The empty set is denoted as $\emptyset$ or $\{ \}$.

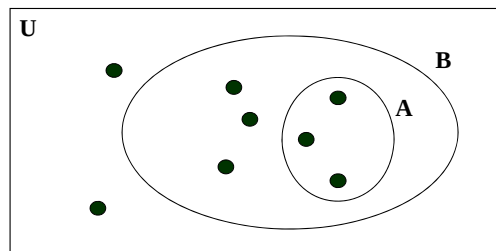
Venn diagrams

- A set can be visualized using **Venn Diagrams**:
 - $V = \{ A, B, C \}$



A Subset

- **Definition:** A set A is said to be a **subset** of B **if and only if** every element of A is also an element of B . We use $A \subseteq B$ to indicate **A is a subset of B** .



- Alternate way to define A is a subset of B :
$$\forall x (x \in A) \rightarrow (x \in B)$$

Empty set/Subset properties

Theorem $\emptyset \subseteq S$

- Empty set is a subset of any set.

Proof:

- Recall the definition of a subset: all elements of a set A must be also elements of B: $\forall x (x \in A \rightarrow x \in B)$.
- We must show the following implication holds for any S
 $\forall x (x \in \emptyset \rightarrow x \in S)$
- Since the empty set does not contain any element, $x \in \emptyset$ is **always False**
- Then the implication is **always True**.

End of proof

Subset properties

Theorem: $S \subseteq S$

- Any set S is a subset of itself

Proof:

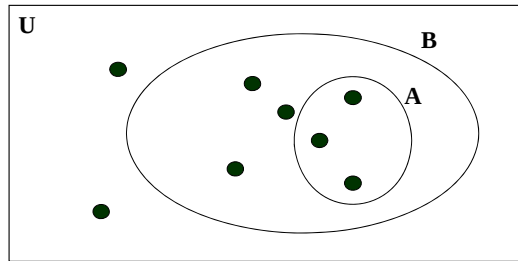
- the definition of a subset says: all elements of a set A must be also elements of B: $\forall x (x \in A \rightarrow x \in B)$.
- Applying this to S we get:
- $\forall x (x \in S \rightarrow x \in S)$ which is trivially **True**
- End of proof

Note on equivalence:

- Two sets are equal if each is a subset of the other set.

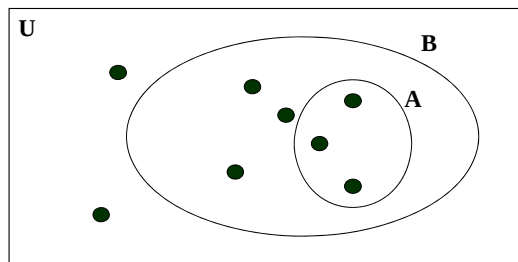
A proper subset

Definition: A set A is said to be a **proper subset** of B if and only if $A \subseteq B$ and $A \neq B$. We denote that A is a proper subset of B with the notation $A \subset B$.



A proper subset

Definition: A set A is said to be a **proper subset** of B if and only if $A \subseteq B$ and $A \neq B$. We denote that A is a proper subset of B with the notation $A \subset B$.



Example: $A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4, 5\}$

Is: $A \subset B$? Yes.

Cardinality

Definition: Let S be a set. If there are exactly n distinct elements in S , where n is a nonnegative integer, we say S is a finite set and that n is the **cardinality of S** . The cardinality of S is denoted by $|S|$.

Examples:

- $V = \{1, 2, 3, 4, 5\}$
 $|V| = 5$
- $A = \{1, 2, 3, 4, \dots, 20\}$
 $|A| = 20$
- $|\emptyset| = 0$

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Infinite set

Definition: A set is **infinite** if it is not finite.

Examples:

- The set of natural numbers is an infinite set.
- $N = \{1, 2, 3, \dots\}$
- The set of reals is an infinite set.

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Power set

Definition: Given a set S , the **power set** of S is the set of all subsets of S . The power set is denoted by $P(S)$.

Examples:

- Assume an empty set $\emptyset$
- What is the power set of $\emptyset$? $P(\emptyset) = \{ \emptyset \}$
- What is the cardinality of $P(\emptyset)$? $|P(\emptyset)| = 1$.

- Assume set $\{1\}$
- $P(\{1\}) = \{ \emptyset, \{1\} \}$
- $|P(\{1\})| = 2$

Power set

- $P(\{1\}) = \{ \emptyset, \{1\} \}$
- $|P(\{1\})| = 2$

- Assume $\{1,2\}$
- $P(\{1,2\}) = \{ \emptyset, \{1\}, \{2\}, \{1,2\} \}$
- $|P(\{1,2\})| = 4$

- Assume $\{1,2,3\}$
- $P(\{1,2,3\}) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\} \}$
- $|P(\{1,2,3\})| = 8$

- **If S is a set with $|S| = n$ then $|P(S)| = ?$**

Power set

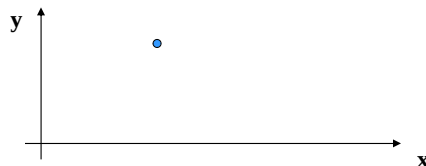
- $P(\{1\}) = \{\emptyset, \{1\}\}$
- $|P(\{1\})| = 2$
- Assume $\{1,2\}$
- $P(\{1,2\}) = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$
- $|P(\{1,2\})| = 4$
- Assume $\{1,2,3\}$
- $P(\{1,2,3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$
- $|P(\{1,2,3\})| = 8$
- **If S is a set with $|S| = n$ then $|P(S)| = 2^n$**

N-tuple

- Sets are used to represent unordered collections.
- **Ordered-n tuples** are used to represent an ordered collection.

Definition: An **ordered n-tuple** $(x_1, x_2, \dots, x_N)$ is the ordered collection that has x_1 as its first element, x_2 as its second element, ..., and x_N as its N -th element, $N \geq 2$.

Example:



- Coordinates of a point in the 2-D plane $(12, 16)$

Cartesian product

Definition: Let S and T be sets. The **Cartesian product of S and T** , denoted by $S \times T$, is the set of all ordered pairs (s,t) , where $s \in S$ and $t \in T$. Hence,

- $$S \times T = \{ (s,t) \mid s \in S \wedge t \in T \}.$$

Examples:

- $S = \{1,2\}$ and $T = \{a,b,c\}$
- $S \times T = \{ (1,a), (1,b), (1,c), (2,a), (2,b), (2,c) \}$
- $T \times S = \{ (a,1), (a,2), (b,1), (b,2), (c,1), (c,2) \}$
- Note: $S \times T \neq T \times S$!!!!

Cardinality of the Cartesian product

- $|S \times T| = |S| * |T|.$

Example:

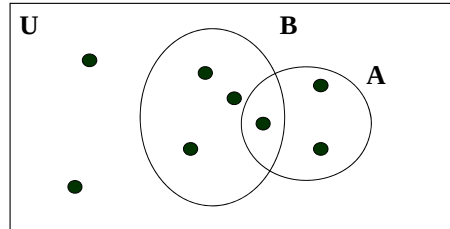
- $A = \{ \text{John, Peter, Mike} \}$
- $B = \{ \text{Jane, Ann, Laura} \}$
- $A \times B = \{ (\text{John, Jane}), (\text{John, Ann}), (\text{John, Laura}), (\text{Peter, Jane}), (\text{Peter, Ann}), (\text{Peter, Laura}), (\text{Mike, Jane}), (\text{Mike, Ann}), (\text{Mike, Laura}) \}$
- $|A \times B| = 9$
- $|A|=3, |B|=3 \rightarrow |A| |B|= 9$

Definition: A subset of the Cartesian product $A \times B$ is called a relation from the set A to the set B .

Set operations

Definition: Let A and B be sets. The **union of A and B**, denoted by $A \cup B$, is the set that contains those elements that are either in A or in B, or in both.

- Alternate: $A \cup B = \{ x \mid x \in A \vee x \in B \}$.



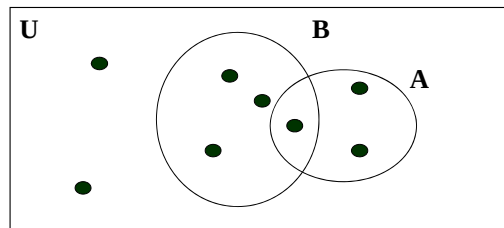
- **Example:**

- $A = \{1, 2, 3, 6\}$ $B = \{2, 4, 6, 9\}$
- $A \cup B = \{1, 2, 3, 4, 6, 9\}$

Set operations

Definition: Let A and B be sets. The **intersection of A and B**, denoted by $A \cap B$, is the set that contains those elements that are in both A and B.

- Alternate: $A \cap B = \{ x \mid x \in A \wedge x \in B \}$.



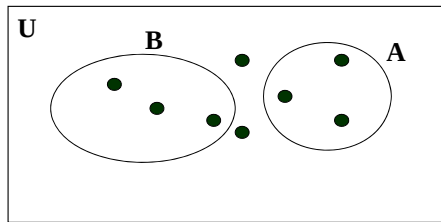
Example:

- $A = \{1, 2, 3, 6\}$ $B = \{2, 4, 6, 9\}$
- $A \cap B = \{2, 6\}$

Disjoint sets

Definition: Two sets are called **disjoint** if their intersection is empty.

- Alternate: A and B are disjoint **if and only if** $A \cap B = \emptyset$.



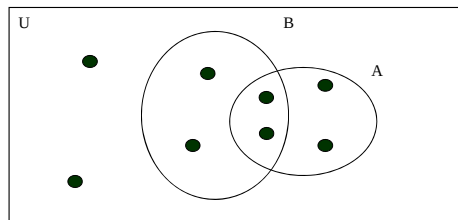
Example:

- $A = \{1, 2, 3, 6\}$ $B = \{4, 7, 8\}$ Are these disjoint?
- Yes.
- $A \cap B = \emptyset$

Cardinality of the set union

Cardinality of the set union.

- $|A \cup B| = |A| + |B| - |A \cap B|$

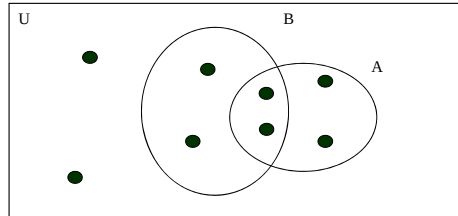


- Why this formula?

Cardinality of the set union

Cardinality of the set union.

- $|A \cup B| = |A| + |B| - |A \cap B|$

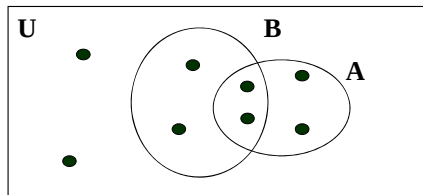


- Why this formula? Correct for an over-count.
- More general rule:
 - **The principle of inclusion and exclusion.**

Set difference

Definition: Let A and B be sets. The **difference of A and B**, denoted by **$A - B$** , is the set containing those elements that are in A but not in B. The difference of A and B is also called the complement of B with respect to A.

- Alternate: $A - B = \{ x \mid x \in A \wedge x \notin B \}$.



Example: $A = \{1, 2, 3, 5, 7\}$ $B = \{1, 5, 6, 8\}$

- $A - B = \{2, 3, 7\}$

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Lecture 8

Sets and set operations: cont. Functions.

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Set

- **Definition:** A **set** is a (unordered) collection of objects. These objects are sometimes called **elements** or **members** of the set. (Cantor's naive definition)
- **Examples:**
 - **Vowels in the English alphabet**
 $V = \{ a, e, i, o, u \}$
 - **First seven prime numbers.**
 $X = \{ 2, 3, 5, 7, 11, 13, 17 \}$

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Sets - review

- **A subset of B:**
 - A is a subset of B if all elements in A are also in B.
- **Proper subset:**
 - A is a proper subset of B, if A is a subset of B and $A \neq B$
- **A power set:**
 - The power set of A is a set of all subsets of A

Sets - review

- **Cardinality of a set A:**
 - The number of elements of in the set
- **An n-tuple**
 - An ordered collection of n elements
- **Cartesian product of A and B**
 - A set of all pairs such that the first element is in A and the second in B

Set operations

Set union:

- $A = \{1,2,3,6\}$ $B = \{2,4,6,9\}$
- $A \cup B = \{1,2,3,4,6,9\}$

Set intersection:

- $A = \{1,2,3,6\}$ $B = \{2,4,6,9\}$
- $A \cap B = \{2,6\}$

Set difference:

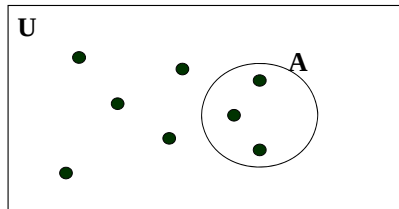
- $A = \{1,2,3,6\}$ $B = \{2,4,6,9\}$
- $A - B = \{1,3\}$
- $B - A = \{4,9\}$

Complement of a set

Definition: Let U be the **universal set**: the set of all objects under the consideration.

Definition: The **complement of the set A** , denoted by $\bar{A}$, is the complement of A with respect to U .

- Alternate: $\bar{A} = \{x \mid x \notin A\}$



Example: $U = \{1,2,3,4,5,6,7,8\}$ $A = \{1,3,5,7\}$

- $\bar{A} = \{2,4,6,8\}$

Set identities

Set Identities (analogous to logical equivalences)

- **Identity**

- $A \cup \emptyset = A$
- $A \cap U = A$

- **Domination**

- $A \cup U = U$
- $A \cap \emptyset = \emptyset$

- **Idempotent**

- $A \cup A = A$
- $A \cap A = A$

Set identities

- **Double complement**

- $\overline{\overline{A}} = A$

- **Commutative**

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

- **Associative**

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

- **Distributive**

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Set identities

- **DeMorgan**

$$\begin{aligned} - \overline{(A \cap B)} &= \overline{A} \cup \overline{B} \\ - \overline{(A \cup B)} &= \overline{A} \cap \overline{B} \end{aligned}$$

- **Absorbion Laws**

$$\begin{aligned} - A \cup (A \cap B) &= A \\ - A \cap (A \cup B) &= A \end{aligned}$$

- **Complement Laws**

$$\begin{aligned} - A \cup \overline{A} &= U \\ - A \cap \overline{A} &= \emptyset \end{aligned}$$

Set identities

- **Set identities can be proved using membership tables.**
- List each combination of sets that an element can belong to. Then show that for each such a combination the element either belongs or does not belong to both sets in the identity.
- Prove: $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$

A	B	$\overline{A}$	$\overline{B}$	$\overline{A \cap B}$	$\overline{A} \cup \overline{B}$
1	1	0	0	0	0
1	0	0	1	0	0
0	1	1	0	0	0
0	0	1	1	1	1

Generalized unions and intersections

Definition: The **union of a collection of sets** is the set that contains those elements that are members of at least one set in the collection.

$$\bigcup_{i=1}^n A_i = \{A_1 \cup A_2 \cup \dots \cup A_n\}$$

Example:

- Let $A_i = \{1, 2, \dots, i\}$ $i = 1, 2, \dots, n$

-

$$\bigcup_{i=1}^n A_i = \{1, 2, \dots, n\}$$

Generalized unions and intersections

Definition: The **intersection of a collection of sets** is the set that contains those elements that are members of all sets in the collection.

$$\bigcap_{i=1}^n A_i = \{A_1 \cap A_2 \cap \dots \cap A_n\}$$

Example:

- Let $A_i = \{1, 2, \dots, i\}$ $i = 1, 2, \dots, n$

$$\bigcap_{i=1}^n A_i = \{1\}$$

Computer representation of sets

- How to represent sets in the computer?
- One solution: Data structures like a list
- A better solution:
- Assign a bit in a bit string to each element in the universal set and set the bit to 1 if the element is present otherwise use 0

Example:

All possible elements: $U=\{1\ 2\ 3\ 4\ 5\}$

- Assume $A=\{2,5\}$
 - Computer representation: $A = 01001$
- Assume $B=\{1,5\}$
 - Computer representation: $B = 10001$

Computer representation of sets

Example:

- $A = 01001$
- $B = 10001$
- The **union** is modeled with a bitwise **or**
- $A \vee B = 11001$
- The **intersection** is modeled with a bitwise **and**
- $A \wedge B = 00001$
- The **complement** is modeled with a bitwise **negation**
- $\bar{A} = 10110$

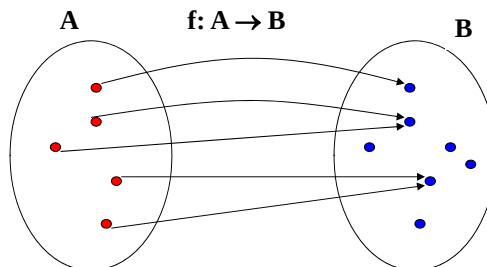
Functions

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Functions

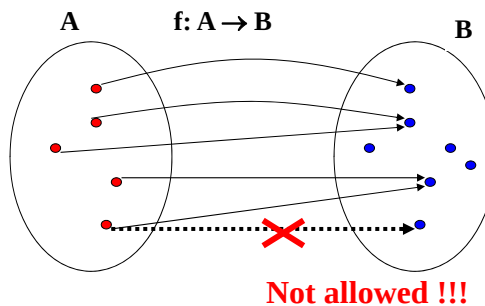
- **Definition:** Let A and B be two sets. A **function from A to B** , denoted $f: A \rightarrow B$, is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ to denote the assignment of b to an element a of A by the function f .



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Functions

- **Definition:** Let A and B be two sets. A **function from A to B**, denoted $f: A \rightarrow B$, is an assignment of exactly one element of B to each element of A. We write $f(a) = b$ to denote the assignment of b to an element a of A by the function f.



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Representing functions

Representations of functions:

1. Explicitly state the assignments in between elements of the two sets
2. Compactly by a formula. (using 'standard' functions)

Example1:

- Let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$
- Assume f is defined as:
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow c$
- Is f a function ?
- **Yes.** since $f(1)=c$, $f(2)=a$, $f(3)=c$. each element of A is assigned an element from B

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Representing functions

Representations of functions:

1. Explicitly state the assignments in between elements of the two sets
2. Compactly by a formula. (using 'standard' functions)

Example 2:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
- Assume g is defined as
 - $1 \rightarrow c$
 - $1 \rightarrow b$
 - $2 \rightarrow a$
 - $3 \rightarrow c$
- Is g a function?
- **No.** $g(1)$ is assigned both c and b .

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Representing functions

Representations of functions:

1. Explicitly state the assignments in between elements of the two sets
2. Compactly by a formula. (using 'standard' functions)

Example 3:

- $A = \{0,1,2,3,4,5,6,7,8,9\}$, $B = \{0,1,2\}$
- Define $h: A \rightarrow B$ as:
 - $h(x) = x \bmod 3$.
 - (the result is the remainder after the division by 3)
- Assignments:

• $0 \rightarrow 0$	$3 \rightarrow 0$
• $1 \rightarrow 1$	$4 \rightarrow 1$
• $2 \rightarrow 2$	$\dots$

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Important sets

Definitions: Let f be a function from A to B .

- We say that A is the **domain** of f and B is the **codomain** of f .
- If $f(a) = b$, **b is the image of a** and **a is a pre-image of b** .
- **The range of f** is the set of all images of elements of A . Also, if f is a function from A to B , we say f maps A to B .

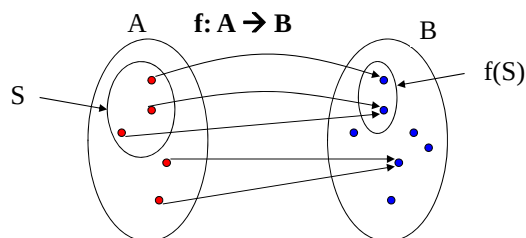
Example: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

- Assume f is defined as: $1 \rightarrow c, 2 \rightarrow a, 3 \rightarrow c$
- What is the image of 1?
- $1 \rightarrow c$ c is the image of 1
- What is the pre-image of a ?
- $2 \rightarrow a$ 2 is a pre-image of a .
- Domain of f ? $\{1,2,3\}$
- Codomain of f ? $\{a,b,c\}$
- Range of f ? $\{a,c\}$

M. Hauskrecht

Image of a subset

Definition: Let f be a function from set A to set B and let S be a subset of A . The image of S is a subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$, so that $f(S) = \{ f(s) \mid s \in S \}$.



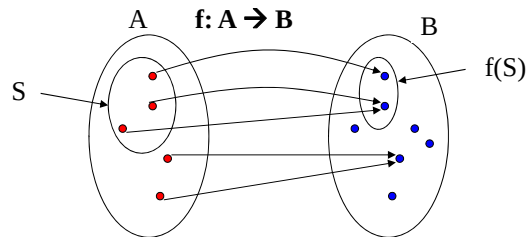
Example:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$ and $f: 1 \rightarrow c, 2 \rightarrow a, 3 \rightarrow c$
- Let $S = \{1,3\}$ then image $f(S) = ?$

M. Hauskrecht

Image of a subset

Definition: Let f be a function from set A to set B and let S be a subset of A . The image of S is a subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$, so that $f(S) = \{ f(s) \mid s \in S \}$.



Example:

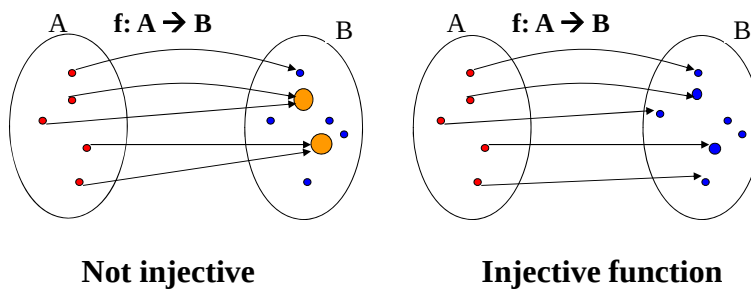
- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$ and $f: 1 \rightarrow c, 2 \rightarrow a, 3 \rightarrow c$
- Let $S = \{1,3\}$ then image $f(S) = \{c\}$.

M. Hauskrecht

Injective function

Definition: A function f is said to be **one-to-one, or injective**, if and only if $f(x) = f(y)$ implies $x = y$ for all x, y in the domain of f . A function is said to be an **injection if it is one-to-one**.

Alternate: A function is one-to-one if and only if $f(x) \neq f(y)$, whenever $x \neq y$. This is the contrapositive of the definition.



M. Hauskrecht

Injective functions

Example 1: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

- Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow c$
- Is f one to one? **No**, it is not one-to-one since $f(1) = f(3) = c$, and $1 \neq 3$.

Example 2: Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$, where $g(x) = 2x - 1$.

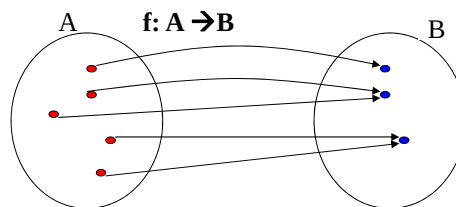
- Is g is one-to-one (why?)
- **Yes.**
- Suppose $g(a) = g(b)$, i.e., $2a - 1 = 2b - 1 \Rightarrow 2a = 2b$
 $\Rightarrow a = b$.

M. Hauskrecht

Surjective function

Definition: A function f from A to B is called **onto**, or **surjective**, if and only if for every $b \in B$ there is an element $a \in A$ such that $f(a) = b$.

Alternative: all co-domain elements are covered



M. Hauskrecht

Surjective functions

Example 1: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

– Define f as

- $1 \rightarrow c$
- $2 \rightarrow a$
- $3 \rightarrow c$

• Is f an onto?

• **No.** f is not onto, since $b \in B$ has no pre-image.

Example 2: $A = \{0,1,2,3,4,5,6,7,8,9\}$, $B = \{0,1,2\}$

– Define $h: A \rightarrow B$ as $h(x) = x \bmod 3$.

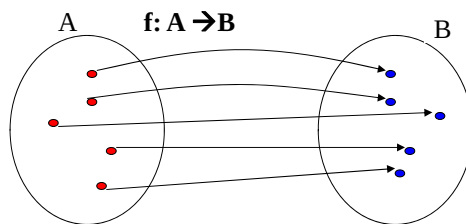
• Is h an onto function?

• **Yes.** h is onto since a pre-image of 0 is 6, a pre-image of 1 is 4, a pre-image of 2 is 8.

M. Hauskrecht

Bijjective functions

Definition: A function f is called a **bijection** if it is **both one-to-one and onto**.



M. Hauskrecht

Bijjective functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
 - Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow b$
- Is f is a bijection? **Yes.** It is both one-to-one and onto.
- **Note:** Let f be a function from a set A to itself, where A is finite. f is one-to-one if and only if f is onto.
- This is not true for A an infinite set. Define $f : \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(z) = 2 * z$. f is one-to-one but not onto (3 has no pre-image).

M. Hauskrecht

Bijjective functions

Example 2:

- Define $g : \mathbb{W} \rightarrow \mathbb{W}$ (whole numbers), where $g(n) = \lfloor n/2 \rfloor$ (floor function).
 - $0 \rightarrow \lfloor 0/2 \rfloor = \lfloor 0 \rfloor = 0$
 - $1 \rightarrow \lfloor 1/2 \rfloor = \lfloor 1/2 \rfloor = 0$
 - $2 \rightarrow \lfloor 2/2 \rfloor = \lfloor 1 \rfloor = 1$
 - $3 \rightarrow \lfloor 3/2 \rfloor = \lfloor 3/2 \rfloor = 1$
 - ...
- Is g a bijection?
 - **No.** g is onto but not 1-1 ($g(0) = g(1) = 0$ however $0 \neq 1$).

M. Hauskrecht

CS 441 Discrete Mathematics for CS
Lecture 9

Functions II

Milos Hauskrecht

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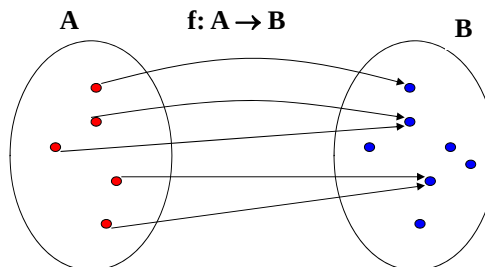
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Functions

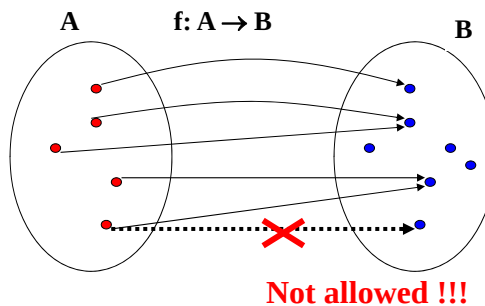
- **Definition:** Let A and B be two sets. A **function from A to B** , denoted $f: A \rightarrow B$, is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ to denote the assignment of b to an element a of A by the function f .



M. Hauskrecht

Functions

- **Definition:** Let A and B be two sets. A **function from A to B** , denoted $f: A \rightarrow B$, is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ to denote the assignment of b to an element a of A by the function f .



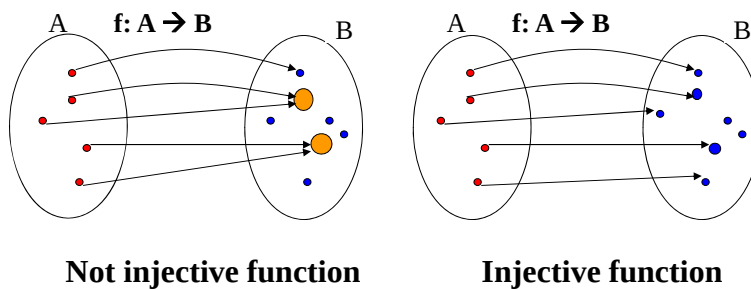
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Injective function

Definition: A function f is said to be **one-to-one, or injective**, if and only if $f(x) = f(y)$ implies $x = y$ for all x, y in the domain of f . A function is said to be an **injection if it is one-to-one**.

Alternative: A function is one-to-one if and only if $f(x) \neq f(y)$, whenever $x \neq y$. This is the contrapositive of the definition.



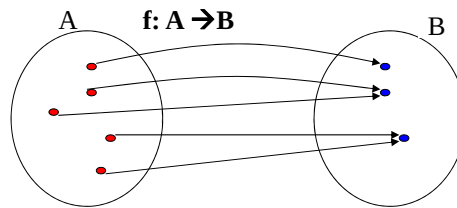
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Surjective function

Definition: A function f from A to B is called **onto, or surjective**, if and only if for every $b \in B$ there is an element $a \in A$ such that $f(a) = b$.

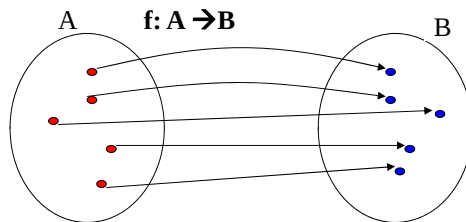
Alternative: all co-domain elements are covered



M. Hauskrecht

Bijjective functions

Definition: A function f is called **a bijection** if it is **both one-to-one (injection) and onto (surjection)**.



M. Hauskrecht

Bijjective functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
 - Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow b$
- Is f a bijection?
- ?

M. Hauskrecht

Bijjective functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
 - Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow b$
- Is f a bijection?
- **Yes.** It is both one-to-one and onto.

M. Hauskrecht

Bijjective functions

Example 2:

- Define $g : W \rightarrow W$ (whole numbers), where $g(n) = \lfloor n/2 \rfloor$ (floor function).
 - $0 \rightarrow \lfloor 0/2 \rfloor = \lfloor 0 \rfloor = 0$
 - $1 \rightarrow \lfloor 1/2 \rfloor = \lfloor 1/2 \rfloor = 0$
 - $2 \rightarrow \lfloor 2/2 \rfloor = \lfloor 1 \rfloor = 1$
 - $3 \rightarrow \lfloor 3/2 \rfloor = \lfloor 3/2 \rfloor = 1$
- ...
- Is g a bijection?

M. Hauskrecht

Bijjective functions

Example 2:

- Define $g : W \rightarrow W$ (whole numbers), where $g(n) = \lfloor n/2 \rfloor$ (floor function).
 - $0 \rightarrow \lfloor 0/2 \rfloor = \lfloor 0 \rfloor = 0$
 - $1 \rightarrow \lfloor 1/2 \rfloor = \lfloor 1/2 \rfloor = 0$
 - $2 \rightarrow \lfloor 2/2 \rfloor = \lfloor 1 \rfloor = 1$
 - $3 \rightarrow \lfloor 3/2 \rfloor = \lfloor 3/2 \rfloor = 1$
- ...
- Is g a bijection?
 - **No.** g is onto but not 1-1 ($g(0) = g(1) = 0$ however $0 \neq 1$).

M. Hauskrecht

Bijjective functions

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Assume

→ A is finite and f is one-to-one (injective)

- Is f an onto function (surjection)?

Bijjective functions

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Proof:

→ A is finite and f is one-to-one (injective)

- Is f an onto function (surjection)?
- **Yes.** Every element points to exactly one element. Injection assures they are different. So we have $|A|$ different elements A points to. Since $f: A \rightarrow A$ the co-domain is covered thus the function is also a surjection (and a bijection)

← A is finite and f is an onto function

- Is the function one-to-one?

Bijjective functions

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Proof:

→ **A is finite and f is one-to-one (injective)**

- Is f an onto function (surjection)?
- **Yes.** Every element points to exactly one element. Injection assures they are different. So we have $|A|$ different elements A points to. Since $f: A \rightarrow A$ the co-domain is covered thus the function is also a surjection (and a bijection)

← **A is finite and f is an onto function**

- Is the function one-to-one?
- **Yes.** Every element maps to exactly one element and all elements in A are covered. Thus the mapping must be one-to-

one

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Bijjective functions

Theorem. Let f be a function from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Please note the above is not true when A is an infinite set.

• **Example:**

- $f: \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(z) = 2 * z$.
- f is one-to-one but not onto.
 - $1 \rightarrow 2$
 - $2 \rightarrow 4$
 - $3 \rightarrow 6$
- 3 has no pre-image.

M. Hauskrecht

Functions on real numbers

Definition: Let f_1 and f_2 be functions from A to $\mathbf{R}$ (reals). Then $f_1 + f_2$ and $f_1 * f_2$ are also functions from A to $\mathbf{R}$ defined by

- $(f_1 + f_2)(x) = f_1(x) + f_2(x)$
- $(f_1 * f_2)(x) = f_1(x) * f_2(x).$

Examples:

- **Assume**

- $f_1(x) = x - 1$
- $f_2(x) = x^3 + 1$

then

- $(f_1 + f_2)(x) = x^3 + x$
- $(f_1 * f_2)(x) = x^4 - x^3 + x - 1.$

Increasing and decreasing functions

Definition: A function f whose domain and codomain are subsets of real numbers is **strictly increasing** if $f(x) > f(y)$ whenever $x > y$ and x and y are in the domain of f . Similarly, f is called **strictly decreasing** if $f(x) < f(y)$ whenever $x > y$ and x and y are in the domain of f .

Example:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$. Is it increasing ?

Increasing and decreasing functions

Definition: A function f whose domain and codomain are subsets of real numbers is **strictly increasing** if $f(x) > f(y)$ whenever $x > y$ and x and y are in the domain of f . Similarly, f is called **strictly decreasing** if $f(x) < f(y)$ whenever $x > y$ and x and y are in the domain of f .

Example:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$. Is it increasing ?

- **Proof .**

For $x > y$ holds $2x > 2y$ and subsequently $2x - 1 > 2y - 1$

Thus g is strictly increasing.

Increasing and decreasing functions

Definition: A function f whose domain and codomain are subsets of real numbers is **strictly increasing** if $f(x) > f(y)$ whenever $x > y$ and x and y are in the domain of f . Similarly, f is called **strictly decreasing** if $f(x) < f(y)$ whenever $x > y$ and x and y are in the domain of f .

Note: Strictly increasing and strictly decreasing functions are one-to-one.

Why?

Increasing and decreasing functions

Definition: A function f whose domain and codomain are subsets of real numbers is **strictly increasing** if $f(x) > f(y)$ whenever $x > y$ and x and y are in the domain of f . Similarly, f is called **strictly decreasing** if $f(x) < f(y)$ whenever $x > y$ and x and y are in the domain of f .

Note: Strictly increasing and strictly decreasing functions are one-to-one.

Why?

One-to-one function: A function is one-to-one if and only if $f(x) \neq f(y)$, whenever $x \neq y$.

Identity function

Definition: Let A be a set. The **identity function** on A is the function $i_A: A \rightarrow A$ where $i_A(x) = x$.

Example:

- Let $A = \{1, 2, 3\}$

Then:

- $i_A(1) = ?$

Identity function

Definition: Let A be a set. The **identity function** on A is the function $i_A: A \rightarrow A$ where $i_A(x) = x$.

Example:

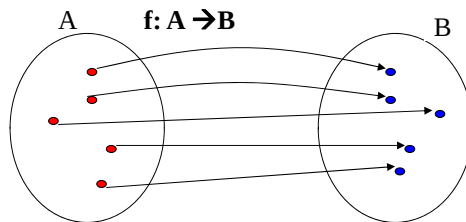
- Let $A = \{1, 2, 3\}$

Then:

- $i_A(1) = 1$
- $i_A(2) = 2$
- $i_A(3) = 3$.

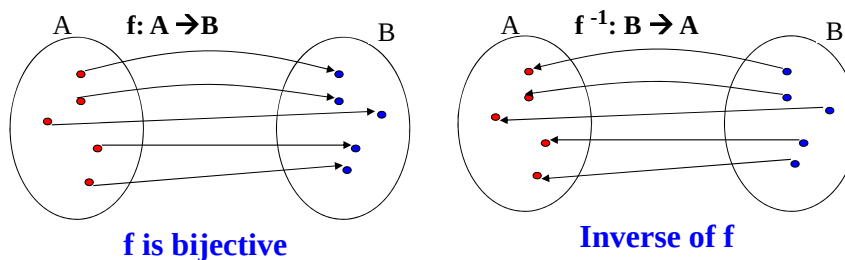
Bijjective functions

Definition: A function f is called **a bijection** if it is **both one-to-one and onto**.



Inverse functions

Definition: Let f be a **bijection** from set A to set B . The **inverse function of f** is the function that assigns to an element b from B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$, when $f(a) = b$. If the inverse function of f exists, f is called **invertible**.



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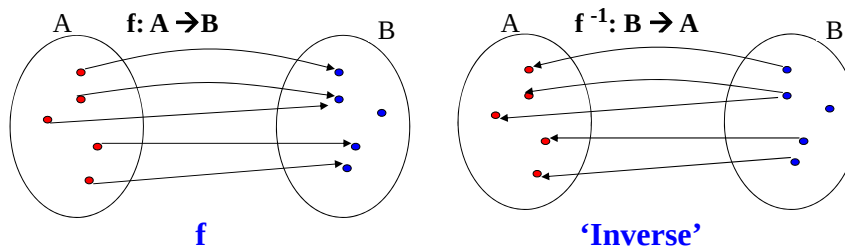
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Inverse functions

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is not one-to-one:

?



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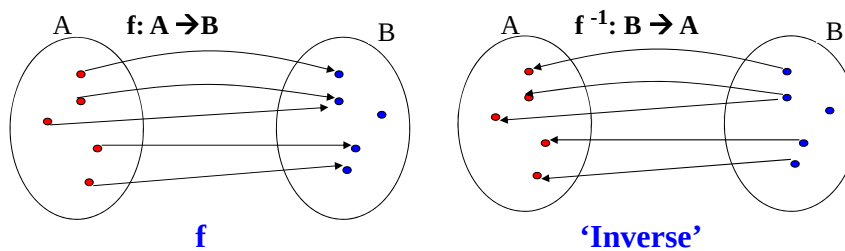
M. Hauskrecht

Inverse functions

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is not one-to-one:

Inverse is not a function. One element of B is mapped to two different elements.



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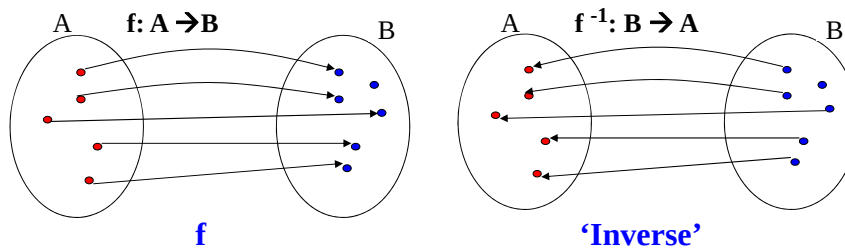
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Inverse functions

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is not onto:

?



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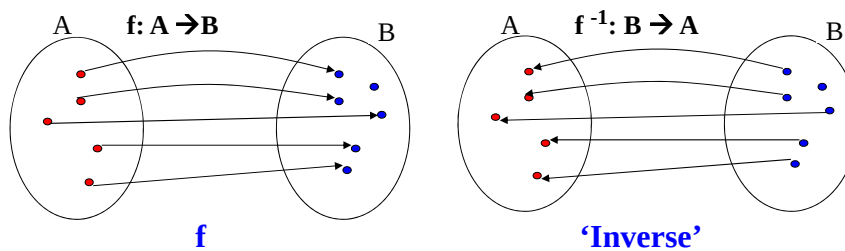
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Inverse functions

Note: if f is not a bijection then it is not possible to define the inverse function of f . Why?

Assume f is not onto:

Inverse is not a function. One element of B is not assigned any value in B .



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Inverse functions

Example 1:

- Let $A = \{1, 2, 3\}$ and i_A be the identity function

- | | | |
|---|--------------|-------------------|
| • | $i_A(1) = 1$ | $i_A^{-1}(1) = 1$ |
| • | $i_A(2) = 2$ | $i_A^{-1}(2) = 2$ |
| • | $i_A(3) = 3$ | $i_A^{-1}(3) = 3$ |

- Therefore, the inverse function of i_A is i_A .

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Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = ..$

Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = 2 \cdot 3 - 1 = 5$
- $g^{-1}(5) =$

Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = 2 \cdot 3 - 1 = 5$
- $g^{-1}(5) = (5+1)/2 = 3$
- $g(10) =$

Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = 2 \cdot 3 - 1 = 5$
- $g^{-1}(5) = (5+1)/2 = 3$
- $g(10) = 2 \cdot 10 - 1 = 19$
- $g^{-1}(19) =$

Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

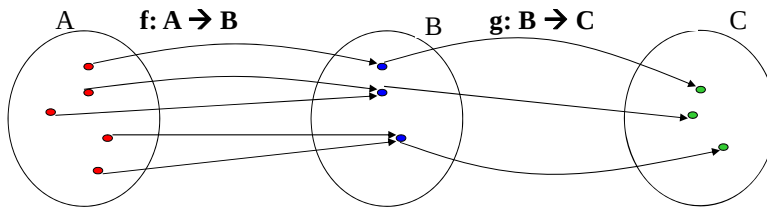
Test the correctness of inverse:

- $g(3) = 2 \cdot 3 - 1 = 5$
- $g^{-1}(5) = (5+1)/2 = 3$
- $g(10) = 2 \cdot 10 - 1 = 19$
- $g^{-1}(19) = (19+1)/2 = 10.$

Composition of functions

Definition: Let f be a function from set A to set B and let g be a function from set B to set C . The **composition of the functions g and f** , denoted by $g \circ f$ is defined by

- $(g \circ f)(a) = g(f(a))$.



Composition of functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$$g : A \rightarrow A, \quad f: A \rightarrow B$$

$$1 \rightarrow 3 \quad 1 \rightarrow b$$

$$2 \rightarrow 1 \quad 2 \rightarrow a$$

$$3 \rightarrow 2 \quad 3 \rightarrow d$$

$$f \circ g : A \rightarrow B:$$

- $1 \rightarrow$

Composition of functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$$\begin{array}{ll} g : A \rightarrow A, & f: A \rightarrow B \\ 1 \rightarrow 3 & 1 \rightarrow b \\ 2 \rightarrow 1 & 2 \rightarrow a \\ 3 \rightarrow 2 & 3 \rightarrow d \end{array}$$

$$f \circ g : A \rightarrow B:$$

- $1 \rightarrow d$
- $2 \rightarrow$

Composition of functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$$\begin{array}{ll} g : A \rightarrow A, & f: A \rightarrow B \\ 1 \rightarrow 3 & 1 \rightarrow b \\ 2 \rightarrow 1 & 2 \rightarrow a \\ 3 \rightarrow 2 & 3 \rightarrow d \end{array}$$

$$f \circ g : A \rightarrow B:$$

- $1 \rightarrow d$
- $2 \rightarrow b$
- $3 \rightarrow$

Composition of functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$$g : A \rightarrow A, \quad f: A \rightarrow B$$

$$1 \rightarrow 3 \quad 1 \rightarrow b$$

$$2 \rightarrow 1 \quad 2 \rightarrow a$$

$$3 \rightarrow 2 \quad 3 \rightarrow d$$

$$f \circ g : A \rightarrow B:$$

- $1 \rightarrow d$
- $2 \rightarrow b$
- $3 \rightarrow a$

Composition of functions

Example 2:

- Let f and g be two functions from Z to Z , where
- $f(x) = 2x$ and $g(x) = x^2$.
- $f \circ g : Z \rightarrow Z$
- $$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(x^2) \\ &= 2(x^2) \end{aligned}$$
- $g \circ f : Z \rightarrow Z$
- $(g \circ f)(x) = ?$

Composition of functions

Example 2:

- Let f and g be two functions from Z to Z , where
 - $f(x) = 2x$ and $g(x) = x^2$.
 - $f \circ g : Z \rightarrow Z$
 - $$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(x^2) \\ &= 2(x^2)\end{aligned}$$
 - $g \circ f : Z \rightarrow Z$
 - $$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(2x) \\ &= (2x)^2 \\ &= 4x^2\end{aligned}$$
- Note that the order of the function composition matters

Composition of functions

Example 3:

- $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$, for all x .
- Let $f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = 2x - 1$ and $f^{-1}(x) = (x+1)/2$.
- $$\begin{aligned}(f \circ f^{-1})(x) &= f(f^{-1}(x)) \\ &= f((x+1)/2) \\ &= 2((x+1)/2) - 1 \\ &= (x+1) - 1 \\ &= x\end{aligned}$$

Composition of functions

Example 3:

- $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$, for all x .
- Let $f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = 2x - 1$ and $f^{-1}(x) = (x+1)/2$.
- $(f \circ f^{-1})(x) = f(f^{-1}(x))$
= $f((x+1)/2)$
= $2((x+1)/2) - 1$
= $(x+1) - 1$
= x
- $(f^{-1} \circ f)(x) = f^{-1}(f(x))$
= $f^{-1}(2x - 1)$
= $(2x)/2$
= x

Some functions

Definitions:

- The **floor function** assigns a real number x the largest integer that is less than or equal to x . The floor function is denoted by $\lfloor x \rfloor$.
- The **ceiling function** assigns to the real number x the smallest integer that is greater than or equal to x . The ceiling function is denoted by $\lceil x \rceil$.

Other important functions:

- Factorials: $n! = n(n-1) \dots 1$ such that $1! = 1$

CS 441 Discrete Mathematics for CS

Lecture 10

Sequences and summations

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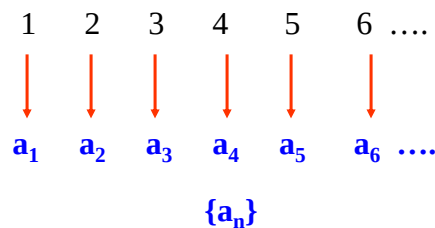
CS 441 Discrete mathematics for CS

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Sequences

Definition: A **sequence** is a **function** from a subset of the set of integers (typically the set $\{0,1,2,\dots\}$ or the set $\{1,2,3,\dots\}$) to a set S . We use the notation a_n to denote the image of the integer n . We call a_n a term of the sequence.

Notation: $\{a_n\}$ is used to represent the sequence (note $\{\}$ is the same notation used for sets, so be careful). $\{a_n\}$ represents the ordered list $a_1, a_2, a_3, \dots$.



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Sequences

Examples:

- (1) $a_n = n^2$, where $n = 1, 2, 3, \dots$
 - What are the elements of the sequence?
1, 4, 9, 16, 25, ...
- (2) $a_n = (-1)^n$, where $n = 0, 1, 2, 3, \dots$
 - Elements of the sequence?
1, -1, 1, -1, 1, ...
- 3) $a_n = 2^n$, where $n = 0, 1, 2, 3, \dots$
 - Elements of the sequence?
1, 2, 4, 8, 16, 32, ...

Arithmetic progression

Definition: An **arithmetic progression** is a sequence of the form
 $a, a+d, a+2d, \dots, a+nd$
where a is the *initial term* and d is *common difference*, such that
both belong to $\mathbb{R}$.

Example:

- $s_n = -1 + 4n$ for $n = 0, 1, 2, 3, \dots$
- members: -1, 3, 7, 11, ...

Geometric progression

Definition A **geometric progression** is a sequence of the form:

$$a, ar, ar^2, \dots, ar^k,$$

where a is the *initial term*, and r is the *common ratio*. Both a and r belong to $\mathbb{R}$.

Example:

- $a_n = \left(\frac{1}{2}\right)^n$ for $n = 0, 1, 2, 3, \dots$
members: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$

Sequences

- Given a sequence finding a rule for generating the sequence is not always straightforward

Example:

- Assume the sequence: $1, 3, 5, 7, 9, \dots$
- What is the formula for the sequence?
- Each term is obtained by adding 2 to the previous term.
 $1, 1+2=3, 3+2=5, 5+2=7$
- What type of progression this suggest?

Sequences

- Given a sequence finding a rule for generating the sequence is not always straightforward

Example:

- Assume the sequence: 1, 3, 5, 7, 9,
- What is the formula for the sequence?
- Each term is obtained by adding 2 to the previous term.
- 1, $1+2=3$, $3+2=5$, $5+2=7$
- It suggests an **arithmetic progression**: $a+nd$
with $a=1$ and $d=2$
 - $a_n=1+2n$

Sequences

- Given a sequence finding a rule for generating the sequence is not always straightforward

Example 2:

- Assume the sequence: 1, $1/3$, $1/9$, $1/27$, ...
- What is the sequence?
- The denominators are powers of 3.
 1 , $1/3=1/3$, $(1/3)/3=1/(3*3)=1/9$, $(1/9)/3=1/27$
- This suggests a **geometric progression**: ar^k
with $a=1$ and $r=1/3$
 - $(1/3)^n$

Recursively defined sequences

- The n -th element of the sequence $\{a_n\}$ is defined recursively in terms of the previous elements of the sequence and the initial elements of the sequence.

Example :

- $a_n = a_{n-1} + 2$ assuming $a_0 = 1$;
- $a_0 = 1$;
- $a_1 = 3$;
- $a_2 = 5$;
- $a_3 = 7$;
- Can you write a_n non-recursively using n ?
- $a_n = 1 + 2n$

Fibonacci sequence

- Recursively defined sequence, where
- $f_0 = 0$;
- $f_1 = 1$;
- $f_n = f_{n-1} + f_{n-2}$ for $n = 2, 3, \dots$

- $f_2 = 1$
- $f_3 = 2$
- $f_4 = 3$
- $f_5 = 5$

Summations

Summation of the terms of a sequence:

$$\sum_{j=m}^n a_j = a_m + a_{m+1} + \dots + a_n$$

The variable j is referred to as the index of summation.

- m is the *lower limit* and
- n is the *upper limit* of the summation.

Summations

Example:

- 1) Sum the first 7 terms of $\{n^2\}$ where $n=1,2,3, \dots$.

$$\sum_{j=1}^7 a_j = \sum_{j=1}^7 j^2 = 1 + 4 + 16 + 25 + 36 + 49 = 140$$

- 2) What is the value of

$$\sum_{k=4}^8 a_j = \sum_{k=4}^8 (-1)^j = 1 + (-1) + 1 + (-1) + 1 = 1$$

Arithmetic series

Definition: The sum of the terms of the arithmetic progression $a, a+d, a+2d, \dots, a+nd$ is called an **arithmetic series**.

Theorem: The sum of the terms of the arithmetic progression $a, a+d, a+2d, \dots, a+nd$ is

$$S = \sum_{j=1}^n (a + jd) = na + d \sum_{j=1}^n j = na + d \frac{n(n+1)}{2}$$

- Why?

Arithmetic series

Theorem: The sum of the terms of the arithmetic progression $a, a+d, a+2d, \dots, a+nd$ is

$$S = \sum_{j=1}^n (a + jd) = na + d \sum_{j=1}^n j = na + d \frac{n(n+1)}{2}$$

Proof:

$$S = \sum_{j=1}^n (a + jd) = \sum_{j=1}^n a + \sum_{j=1}^n jd = na + d \sum_{j=1}^n j$$

$$\sum_{j=1}^n j = 1 + 2 + 3 + 4 + \dots + (n-2) + (n-1) + n$$

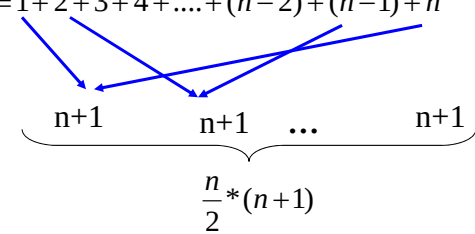
Arithmetic series

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$$S = \sum_{j=1}^n (a + jd) = na + d \sum_{j=1}^n j = na + d \frac{n(n+1)}{2}$$

Proof:

$$S = \sum_{j=1}^n (a + jd) = \sum_{j=1}^n a + \sum_{j=1}^n jd = na + d \sum_{j=1}^n j$$

$$\sum_{j=1}^n j = 1 + 2 + 3 + 4 + \dots + (n-2) + (n-1) + n$$


$$\underbrace{\qquad\qquad\qquad}_{n+1} \quad \underbrace{\qquad\qquad\qquad}_{n+1} \quad \dots \quad \underbrace{\qquad\qquad\qquad}_{n+1}$$

$$\frac{n}{2} * (n+1)$$

Arithmetic series

Example:

$$\begin{aligned} S &= \sum_{j=1}^5 (2 + j3) = \\ &= \sum_{j=1}^5 2 + \sum_{j=1}^5 j3 = \\ &= 2 \sum_{j=1}^5 1 + 3 \sum_{j=1}^5 j = \\ &= 2 * 5 + 3 \sum_{j=1}^5 j = \\ &= 10 + 3 \frac{(5+1)}{2} * 5 = \\ &= 10 + 45 = 55 \end{aligned}$$

Arithmetic series

Example 2: $S = \sum_{j=3}^5 (2 + j3) =$

$$= \left[\sum_{j=1}^5 (2 + j3) \right] - \left[\sum_{j=1}^2 (2 + j3) \right] \quad \leftarrow \quad \text{Trick}$$

$$= \left[2 \sum_{j=1}^5 1 + 3 \sum_{j=1}^5 j \right] - \left[2 \sum_{j=1}^2 1 + 3 \sum_{j=1}^2 j \right]$$

$$= 55 - 13 = 42$$

Double summations

Example: $S = \sum_{i=1}^4 \sum_{j=1}^2 (2i - j) =$

$$= \sum_{i=1}^4 \left[\sum_{j=1}^2 2i - \sum_{j=1}^2 j \right] =$$

$$= \sum_{i=1}^4 \left[2i \sum_{j=1}^2 1 - \sum_{j=1}^2 j \right] =$$

$$= \sum_{i=1}^4 \left[2i * 2 - \sum_{j=1}^2 j \right] =$$

$$= \sum_{i=1}^4 [2i * 2 - 3] =$$

$$= \sum_{i=1}^4 4i - \sum_{i=1}^4 3 =$$

$$= 4 \sum_{i=1}^4 i - 3 \sum_{i=1}^4 1 = 4 * 10 - 3 * 4 = 28$$

Geometric series

Definition: The sum of the terms of a geometric progression $a, ar, ar^2, \dots, ar^k$ is called a **geometric series**.

Theorem: The sum of the terms of a geometric progression $a, ar, ar^2, \dots, ar^n$ is

$$S = \sum_{j=0}^n (ar^j) = a \sum_{j=0}^n r^j = a \left[\frac{r^{n+1} - 1}{r - 1} \right]$$

Geometric series

Theorem: The sum of the terms of a geometric progression $a, ar, ar^2, \dots, ar^n$ is

$$S = \sum_{j=0}^n (ar^j) = a \sum_{j=0}^n r^j = a \left[\frac{r^{n+1} - 1}{r - 1} \right]$$

Proof:

$$S = \sum_{j=0}^n ar^j = a + ar + ar^2 + ar^3 + \dots + ar^n$$

- multiply S by r

$$rS = r \sum_{j=0}^n ar^j = ar + ar^2 + ar^3 + \dots + ar^{n+1}$$

- Subtract $rS - S = [ar + ar^2 + ar^3 + \dots + ar^{n+1}] - [a + ar + ar^2 + \dots + ar^n]$
 $= ar^{n+1} - a$



$$S = \frac{ar^{n+1} - a}{r - 1} = a \left[\frac{r^{n+1} - 1}{r - 1} \right]$$

Geometric series

Example:

$$S = \sum_{j=0}^3 2(5)^j =$$

General formula:

$$S = \sum_{j=0}^n (ar^j) = a \sum_{j=0}^n r^j = a \left[\frac{r^{n+1} - 1}{r - 1} \right]$$

$$S = \sum_{j=0}^3 2(5)^j = 2 * \frac{5^4 - 1}{5 - 1} =$$

$$= 2 * \frac{625 - 1}{4} = 2 * \frac{624}{4} = 2 * 156 = 312$$

Infinite geometric series

- Infinite geometric series can be computed in the closed form for $x < 1$
- How?

$$\sum_{n=0}^{\infty} x^n = \lim_{k \rightarrow \infty} \sum_{n=0}^k x^n = \lim_{k \rightarrow \infty} \frac{x^{k+1} - 1}{x - 1} = -\frac{1}{x - 1} = \frac{1}{1 - x}$$

- Thus:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}$$

CS 441 Discrete Mathematics for CS
Lecture 11

Countable and uncountable sets.
Matrices.

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Arithmetic series

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Theorem: The sum of the terms of the arithmetic progression
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$$S = \sum_{j=1}^n (a + jd) = na + d \sum_{j=1}^n j = na + d \frac{n(n+1)}{2}$$

Geometric series

Definition: The sum of the terms of a **geometric progression** $a, ar, ar^2, \dots, ar^k$ is called a **geometric series**.

Theorem: The sum of the terms of a geometric progression $a, ar, ar^2, \dots, ar^n$ is

$$S = \sum_{j=0}^n (ar^j) = a \sum_{j=0}^n r^j = a \left[\frac{r^{n+1} - 1}{r - 1} \right]$$

Infinite geometric series

- Infinite geometric series can be computed in the closed form for $x < 1$
- How?

$$\sum_{n=0}^{\infty} x^n = \lim_{k \rightarrow \infty} \sum_{n=0}^k x^n = \lim_{k \rightarrow \infty} \frac{x^{k+1} - 1}{x - 1} = -\frac{1}{x - 1} = \frac{1}{1 - x}$$

- Thus:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}$$

Cardinality

Recall: The cardinality of a finite set is defined by the number of elements in the set.

Definition: The sets A and B have **the same cardinality** if there is a one-to-one correspondence between elements in A and B . In other words if there is a bijection from A to B . Recall bijection is one-to-one and onto.

Example: Assume $A = \{a, b, c\}$ and $B = \{\alpha, \beta, \gamma\}$ and function f defined as:

- $a \rightarrow \alpha$
- $b \rightarrow \beta$
- $c \rightarrow \gamma$

f defines a bijection. Therefore A and B have the same cardinality, i.e. $|A| = |B| = 3$.

Cardinality

Definition: A set that is either finite or has the same cardinality as the set of positive integers $\mathbb{Z}^+$ is called **countable**. A set that is not countable is called **uncountable**.

Why these are called countable?

- The elements of the set can be enumerated and listed.

Countable sets

Example:

- Assume $A = \{0, 2, 4, 6, \dots\}$ set of even numbers. Is it countable?

Countable sets

Example:

- Assume $A = \{0, 2, 4, 6, \dots\}$ set of even numbers. Is it countable?
- Using the definition: Is there a bijective function $f: \mathbb{Z}^+ \rightarrow A$
 $\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$

Countable sets

Example:

- Assume $A = \{0, 2, 4, 6, \dots\}$ set of even numbers. Is it countable?
- Using the definition: Is there a bijective function $f: \mathbb{Z}^+ \rightarrow A$
 $\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$
- Define a function $f: x \rightarrow 2x - 2$ (an arithmetic progression)
 - $1 \rightarrow 2(1) - 2 = 0$
 - $2 \rightarrow 2(2) - 2 = 2$
 - $3 \rightarrow 2(3) - 2 = 4 \quad \dots$

Countable sets

Example:

- Assume $A = \{0, 2, 4, 6, \dots\}$ set of even numbers. Is it countable?
- Using the definition: Is there a bijective function $f: \mathbb{Z}^+ \rightarrow A$
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- Define a function $f: x \rightarrow 2x - 2$ (an arithmetic progression)
 - $1 \rightarrow 2(1) - 2 = 0$
 - $2 \rightarrow 2(2) - 2 = 2$
 - $3 \rightarrow 2(3) - 2 = 4 \quad \dots$
- one-to-one (why?)

Countable sets

Example:

- Assume $A = \{0, 2, 4, 6, \dots\}$ set of even numbers. Is it countable?
- Using the definition: Is there a bijective function $f: \mathbb{Z}^+ \rightarrow A$
 $\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$
- Define a function $f: x \rightarrow 2x - 2$ (an arithmetic progression)
 - $1 \rightarrow 2(1) - 2 = 0$
 - $2 \rightarrow 2(2) - 2 = 2$
 - $3 \rightarrow 2(3) - 2 = 4 \quad \dots$
- one-to-one (why?) $2x - 2 = 2y - 2 \Rightarrow 2x = 2y \Rightarrow x = y$.
- onto (why?) $\forall a \in A, (a+2)/2$ is the pre-image in $\mathbb{Z}^+$.
- Therefore $|A| = |\mathbb{Z}^+|$.

Countable sets

Theorem:

- **The set of integers $\mathbb{Z}$ is countable.**

Solution:

Can list a sequence:

$0, 1, -1, 2, -2, 3, -3, \dots$

Or can define a bijection from $\mathbb{Z}^+$ to $\mathbb{Z}$:

- When n is even: $f(n) = n/2$
- When n is odd: $f(n) = -(n-1)/2$

Countable sets

Definition:

- A rational number can be expressed as the ratio of two integers p and q such that $q \neq 0$.
 - $\frac{3}{4}$ is a rational number
 - $\sqrt{2}$ is not a rational number.

Theorem:

- The positive rational numbers are countable.

Solution:

The positive rational numbers are countable since they can be arranged in a sequence:

$$r_1, r_2, r_3, \dots$$

Countable sets

Theorem:

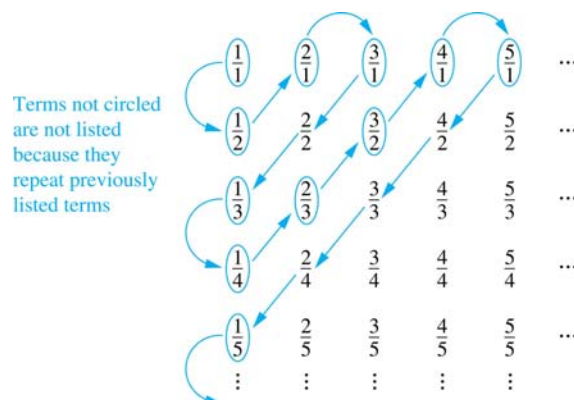
- The positive rational numbers are countable.

First row $q = 1$.
Second row $q = 2$.
etc.

Constructing the List

First list p/q with $p + q = 2$.
Next list p/q with $p + q = 3$

And so on.



Cardinality

Theorem: The set of real numbers ($\mathbb{R}$) is an uncountable set.

Proof by a contradiction.

- 1) Assume that the real numbers are countable.
- 2) Then every subset of the reals is countable, in particular, the interval from 0 to 1 is countable. This implies the elements of this set can be listed say $r_1, r_2, r_3, \dots$ where
 - $r_1 = 0.d_{11}d_{12}d_{13}d_{14} \dots$
 - $r_2 = 0.d_{21}d_{22}d_{23}d_{24} \dots$
 - $r_3 = 0.d_{31}d_{32}d_{33}d_{34} \dots$
 - where the $d_{ij} \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Real numbers are uncountable

Proof cont.

- 3) Want to show that not all reals in the interval between 0 and 1 are in this list.

- Form a new number called
 - $r = 0.d_1d_2d_3d_4 \dots$ where

$$d_i = \begin{cases} 2, & \text{if } d_{ii} \neq 2 \\ 3 & \text{if } d_{ii} = 2 \end{cases}$$

- Example: suppose

$r_1 = 0.\textcolor{red}{7}5243\dots$	$d_1 = 2$
$r_2 = 0.5\textcolor{red}{2}4310\dots$	$d_2 = 3$
$r_3 = 0.13\textcolor{red}{1}257\dots$	$d_3 = 2$
$r_4 = 0.936\textcolor{red}{3}633\dots$	$d_4 = 2$
$\dots$	$\dots$
$r_t = 0.23222\textcolor{red}{2}22\dots$	$d_t = 3$

Real numbers are uncountable

- $r = 0.d_1d_2d_3d_4 \dots$ where
$$d_i = \begin{cases} 2, & \text{if } d_{ii} \neq 2 \\ 3 & \text{if } d_{ii} = 2 \end{cases}$$
- **Claim:** r is different than each member in the list.
- Is each expansion unique? Yes, if we exclude an infinite string of 9s.
-
- Example: $\overline{.02850} = \overline{.02849}$
- Therefore r and r_i differ in the i -th decimal place for all i .

Matrices

Matrices

Definitions:

- A **matrix** is a rectangular array of numbers.
- A matrix with **m rows and n columns** is called an **$m \times n$** matrix.

Note: The plural of matrix is *matrices*.

Matrices

Definitions:

- A **matrix** is a rectangular array of numbers.
- A matrix with **m rows and n columns** is called an **$m \times n$** matrix.

Note: The plural of matrix is *matrices*.

Definitions:

- A matrix with the same number of rows as columns is called a *square matrix*.
- Two matrices are *equal* if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.

Matrices

- Let m and n be positive integers and let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- The i th row of $\mathbf{A}$ is the $1 \times n$ matrix $[a_{i1}, a_{i2}, \dots, a_{in}]$. The j th column of $\mathbf{A}$ is the $m \times 1$ matrix:

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \cdot \\ \cdot \\ a_{mj} \end{bmatrix}$$

- The (i,j) th *element* or *entry* of $\mathbf{A}$ is the element a_{ij} . We can use $\mathbf{A} = [a_{ij}]$ to denote the matrix with its (i,j) th element equal to a_{ij} .

Matrix addition

Defintion:

Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be $m \times n$ matrices. The sum of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} + \mathbf{B}$, is the $m \times n$ matrix that has $a_{ij} + b_{ij}$ as its (i,j) th element. In other words, $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]$.

Example:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & -3 \\ 3 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 & -1 \\ 1 & -3 & 0 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & -2 \\ 3 & -1 & -3 \\ 2 & 5 & 2 \end{bmatrix}$$

Note: matrices of different sizes can not be added.

Matrix multiplication

Definition:

- Let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ be a $k \times n$ matrix. The *product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{AB}$, is the $m \times n$ matrix that has its (i,j) th element equal to the sum of the products of the corresponding elements from the i th row of $\mathbf{A}$ and the j th column of $\mathbf{B}$. In other words, if

$$\mathbf{AB} = [c_{ij}] \text{ then } c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}.$$

Example:

$$\begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 14 & 4 \\ 8 & 9 \\ 7 & 13 \\ 8 & 2 \end{bmatrix}$$

- The product is not defined when the number of columns in the first matrix is not equal to the number of rows in the second matrix

Matrix multiplication

The Product of $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ik} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1j} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2j} & \dots & b_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{k1} & b_{k2} & \dots & b_{kj} & \dots & b_{kn} \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & c_{ij} & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$$

Matrix multiplication

- Example:

$$\begin{bmatrix} 2 & 4 & 3 \\ 1 & 1 & 2 \\ 2 & 4 & 6 \end{bmatrix} * \begin{bmatrix} 1 & 4 & 2 \\ 1 & 2 & 2 \\ 1 & 5 & 3 \end{bmatrix} = \begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}$$

Matrix multiplication

Properties of matrix multiplication:

- Does $\mathbf{AB} = \mathbf{BA}$?

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\mathbf{AB} \neq \mathbf{BA}$$

Matrix multiplication

Properties of matrix multiplication:

- Does $\mathbf{AB} = \mathbf{BA}$?

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

- $\mathbf{AB}$: ?

Matrix multiplication

Properties of matrix multiplication:

- Does $\mathbf{AB} = \mathbf{BA}$?

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

- $\mathbf{AB}$:

$$\mathbf{AB} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

Matrix multiplication

Properties of matrix multiplication:

- Does $\mathbf{AB} = \mathbf{BA}$?

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

- $\mathbf{AB}$:

$$\mathbf{AB} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

$\mathbf{BA}$: ?

Matrix multiplication

Properties of matrix multiplication:

- Does $\mathbf{AB} = \mathbf{BA}$?

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

- $\mathbf{AB}$:

$$\mathbf{AB} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

$\mathbf{BA}$:

$$\mathbf{BA} = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

- **Conclusion:** $\mathbf{AB} \neq \mathbf{BA}$

Matrices

Definition:

- **The identity matrix (of order n)** is the $n \times n$ matrix $\mathbf{I}_n = [\delta_{ij}]$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

Properties:

- Assume A is an $m \times n$ matrix. Then:
 $\mathbf{A}\mathbf{I}_n = \mathbf{A}$ and $\mathbf{I}_m\mathbf{A} = \mathbf{A}$
- Assume A is an $n \times n$ matrix. Then: $\mathbf{A}^0 = \mathbf{I}_n$

Matrices

Definition: Powers of square matrices

- When A is an $n \times n$ matrix, we have:

$$\mathbf{A}^0 = \mathbf{I}_n \quad \mathbf{A}^r = \underbrace{\mathbf{A}\mathbf{A}\mathbf{A}\cdots\mathbf{A}}_r$$

Matrix transpose

Definition:

- Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix. The **transpose** of $\mathbf{A}$, denoted by $\mathbf{A}^T$, is the $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$.

If $\mathbf{A}^T = [b_{ij}]$, then $b_{ij} = a_{ji}$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

The transpose of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is the matrix $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$.

Matrix inverse

Definition:

- Let $\mathbf{A} = [a_{ij}]$ be an $n \times n$ matrix. The **inverse** of $\mathbf{A}$, denoted by $\mathbf{A}^{-1}$, is the $n \times m$ matrix such that $\mathbf{A} \mathbf{A}^{-1} = \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}$
- Note: the inverse of the matrix $\mathbf{A}$ may not exist.

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Lecture 12

Integers and division

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Symmetric matrix

Definition:

- A square matrix $\mathbf{A}$ is called **symmetric** if $\mathbf{A} = \mathbf{A}^T$.
- Thus $\mathbf{A} = [a_{ij}]$ is symmetric if $a_{ij} = a_{ji}$ for i and j with $1 \leq i \leq n$ and $1 \leq j \leq n$.

• **Example:**

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

- Is it a symmetric matrix? yes

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Zero-one matrix

Definition:

- A matrix with entries that are either 0 or 1 is called **a zero-one matrix**.
- Algorithms operating on discrete structures represented by zero-one matrices are based on Boolean arithmetic defined by the Boolean operations **and** and **or** :

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{and}$$

$$b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{or}$$

Join and meet of matrices

Definition: Let A and B be two matrices:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

- **The join of A and B is:**

$$A \vee B = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

- **The meet of A and B is**

$$A \wedge B = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

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Integers and division

- **Number theory** is a branch of mathematics that explores integers and their properties.
- **Integers:**
 - $\mathbb{Z}$ integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$
 - $\mathbb{Z}^+$ positive integers $\{1, 2, \dots\}$
- Number theory has many applications within computer science, including:
 - Indexing - Storage and organization of data
 - Encryption
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Division

Definition: Assume 2 integers a and b , such that $a \neq 0$ (a is not equal 0). We say that **a divides b** if there is an integer c such that $b = ac$. If a divides b we say that **a is a factor of b** and that **b is multiple of a** .

- The fact that a divides b is denoted as **$a \mid b$** .

Examples:

- $4 \mid 24$ True or False ? **True**
 - 4 is a factor of 24
 - 24 is a multiple of 4
- $3 \mid 7$ True or False ? **False**

Divisibility

All integers divisible by $d > 0$ can be enumerated as:

– $\dots, -kd, \dots, -2d, -d, 0, d, 2d, \dots, kd, \dots$

- **Question:**

Let n and d be two positive integers. How many positive integers not exceeding n are divisible by d ?

- $0 < kd \leq n$

- **Answer:**

Count the number of integers kd that are less than n . What is the number of integers k such that $0 < kd \leq n$?

$0 < kd \leq n \rightarrow 0 < k \leq n/d$. Therefore, there are $\lfloor n/d \rfloor$ positive integers not exceeding n that are divisible by d .

Divisibility

Properties:

- Let a, b, c be integers. Then the following hold:
 1. if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$
 2. if $a \mid b$ then $a \mid bc$ for all integers c
 3. if $a \mid b$ and $b \mid c$ then $a \mid c$

Proof of 1: if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$

- from the definition of divisibility we get:
- $b = au$ and $c = av$ where u, v are two integers. Then
- $(b + c) = au + av = a(u + v)$
- **Thus a divides $b + c$.**

Divisibility

Properties:

- Let a, b, c be integers. Then the following hold:
 1. if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$
 2. if $a \mid b$ then $a \mid bc$ for all integers c
 3. if $a \mid b$ and $b \mid c$ then $a \mid c$

Proof of 2: if $a \mid b$ then $a \mid bc$ for all integers c

- If $a \mid b$, then there is some integer u such that $b = au$.
- Multiplying both sides by c gives us $bc = auc$, so by definition, $a \mid bc$.
- **Thus a divides bc .**

Primes

Definition: A positive integer p that is greater than 1 and that is divisible only by 1 and by itself (p) is called **a prime**.

Examples: 2, 3, 5, 7, ...

$1 \mid 2$ and $2 \mid 2$, $1 \mid 3$ and $3 \mid 3$, etc

Primes

Definition: A positive integer p that is greater than 1 and that is divisible only by 1 and by itself (p) is called **a prime**.

Examples: 2, 3, 5, 7, ...

$1 \mid 2$ and $2 \mid 2$, $1 \mid 3$ and $3 \mid 3$, etc

What is the next prime after 7?

- 11

Next?

- 13

Primes

Definition: A positive integer that is greater than 1 and is not a prime is called **a composite**.

Examples: 4, 6, 8, 9, ...

Why?

$2 \mid 4$

Why 6 is a composite?

Primes

Definition: A positive integer that is greater than 1 and is not a prime is called **a composite**.

Examples: 4, 6, 8, 9, ...

Why?

$2 \mid 4$

$3 \mid 6$ or $2 \mid 6$

$2 \mid 8$ or $4 \mid 8$

$3 \mid 9$

The Fundamental theorem of Arithmetic

Fundamental theorem of Arithmetic:

- Any positive integer greater than 1 can be expressed as a product of prime numbers.

Examples:

- $12 = ?$

The Fundamental theorem of Arithmetic

Fundamental theorem of Arithmetic:

- Any positive integer greater than 1 can be expressed as a product of prime numbers.

Examples:

- $12 = 2 * 2 * 3$
- $21 = 3 * 7$
- Process of finding out factors of the product: **factorization**.

Primes and composites

Factorization of composites to primes:

- $100 = 2*2*5*5 = 2^2*5^2$
- $99 = 3*3*11 = 3^2*11$

Important question:

- How to determine whether the number is a prime or a composite?

Primes and composites

- How to determine whether the number is a prime or a composite?

Simple approach (1):

- Let n be a number. To determine whether it is a prime we can test if any number $x < n$ divides it. If yes it is a composite. If we test all numbers $x < n$ and do not find the proper divisor then n is a prime.

Primes and composites

- How to determine whether the number is a prime or a composite?

Simple approach (1):

- Let n be a number. To determine whether it is a prime we can test if any number $x < n$ divides it. If yes it is a composite. If we test all numbers $x < n$ and do not find the proper divisor then n is a prime.
- **Example:**
 - Assume we want to check if 17 is a prime?
 - The approach would require us to check:
 - 2,3,4,5,6,7,8,9,10,11,12,13,14,15,16

Primes and composites

- **Example approach 1:**
 - Assume we want to check if 17 is a prime?
 - The approach would require us to check:
 - 2,3,4,5,6,7,8,9,10,11,12,13,14,15,16
- **Is this the best we can do?**
- **No.** The problem here is that we try to test all the numbers. But this is not necessary.
- **Idea:** Every composite factorizes to a product of primes. So it is sufficient to test only the primes $x < n$ to determine the primality of n .

Primes and composites

- How to determine whether the number is a prime or a composite?

Approach 2:

- Let n be a number. To determine whether it is a prime we can test if any prime number $x < n$ divides it. If yes it is a composite. If we test all primes $x < n$ and do not find a proper divisor then n is a prime.

Primes and composites

- How to determine whether the number is a prime or a composite?

Approach 2:

- Let n be a number. To determine whether it is a prime we can test if any prime number $x < n$ divides it. If yes it is a composite. If we test all primes $x < n$ and do not find a proper divisor then n is a prime.
- **Example:** Is 31 a prime?
- Check if 2,3,5,7,11,13,17,23,29 divide it
- It is a prime !!

Primes and composites

Example approach 2:

Is 91 a prime number?

- Easy primes 2,3,5,7,11,13,17,19 ..
- But how many primes are there that are smaller than 91?

Caveat:

- If n is relatively small the test is good because we can enumerate (memorize) all small primes
- But if n is large there can be larger not obvious primes

Primes and composites

Theorem: If n is a composite then n has a prime divisor less than or equal to $\sqrt{n}$.

Primes and composites

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Proof:

- If n is composite, then it has a positive integer factor a such that $1 < a < n$ by definition. This means that $n = ab$, where b is an integer greater than 1.
- Assume $a > \sqrt{n}$ and $b > \sqrt{n}$. Then $ab > \sqrt{n}\sqrt{n} = n$, which is a contradiction. So either $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.
- Thus, n has a divisor less than $\sqrt{n}$.
- By the fundamental theorem of arithmetic, this divisor is either prime, or is a product of primes. In either case, n has a prime divisor less than $\sqrt{n}$.

Primes and composites

Theorem: If n is a composite that n has a prime divisor less than or equal to $\sqrt{n}$.

Approach 3:

- Let n be a number. To determine whether it is a prime we can test if any prime number $x < \sqrt{n}$ divides it.

Example 1: Is 101 a prime?

- Primes smaller than $\sqrt{101} = 10.xxx$ are: 2,3,5,7
- 101 is not divisible by any of them
- **Thus 101 is a prime**

Example 2: Is 91 a prime?

- Primes smaller than $\sqrt{91}$ are: 2,3,5,7
- 91 is divisible by 7
- **Thus 91 is a composite**

Primes

Question: How many primes are there?

Theorem: There are infinitely many primes.

Primes

Question: How many primes are there?

Theorem: There are infinitely many primes.

Proof by Euclid.

- Proof by contradiction:
 - Assume there is a finite number of primes: $p_1, p_2, \dots, p_n$
- Let $Q = p_1 p_2 \dots p_n + 1$ be a number.
- None of the numbers $p_1, p_2, \dots, p_n$ divides the number Q .
- This is a contradiction since we assumed that we have listed all primes.

Division

Let a be an integer and d a positive integer. Then there are unique integers, q and r , with $0 \leq r < d$, such that

$$a = dq + r.$$

Definitions:

- a is called the **dividend**,
- d is called the **divisor**,
- q is called the **quotient** and
- r the **remainder** of the division.

Example: $a = 14, d = 3$

$$14 = 3 \cdot 4 + 2$$

$$14/3 = 3.666$$

$$14 \text{ div } 3 = 4$$

$$14 \text{ mod } 3 = 2$$

Relations:

- $q = a \text{ div } d, \quad r = a \text{ mod } d$

Greatest common divisor

Definition: Let a and b be integers, not both 0. Then the largest integer d such that $d \mid a$ and $d \mid b$ is called **the greatest common divisor** of a and b . The greatest common divisor is denoted as $\text{gcd}(a, b)$.

Examples:

- $\text{gcd}(24, 36) = ?$
- Check 2, 3, 4, 6, 12 $\text{gcd}(24, 36) = 12$
- $\text{gcd}(11, 23) = ?$

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Lecture 13

Integers and division

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Examples:

- $4 \mid 24$ True or False ? **True**
 - 4 is a factor of 24
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Definition: A positive integer p that greater than 1 and that is divisible only by 1 and by itself (p) is called **a prime**.

Examples: 2, 3, 5, 7, ...

$1 \mid 2$ and $2 \mid 2$, $1 \mid 3$ and $3 \mid 3$, etc

The Fundamental theorem of Arithmetic

Fundamental theorem of Arithmetic:

- Any positive integer greater than 1 can be expressed as a product of prime numbers.

Examples:

- $12 = 2 \cdot 2 \cdot 3$
- $21 = 3 \cdot 7$
- Process of finding out factors of the product: **factorization**.

Primes and composites

- How to determine whether the number is a prime or a composite?**

Let n be a number. Then in order to determine whether it is a **prime** we can test:

- Approach 1:** if any number $x < n$ divides it. If yes it is a composite. If we test all numbers $x < n$ and do not find the proper divisor then n is a prime.
- Approach 2:** if any prime number $x < n$ divides it. If yes it is a composite. If we test all primes $x < n$ and do not find a proper divisor then n is a prime.
- Approach 3:** if any prime number $x < \sqrt{n}$ divides it. If yes it is a composite. If we test all primes $x < \sqrt{n}$ and do not find a proper divisor then n is a prime.

Division

Let a be an integer and d a positive integer. Then there are unique integers, q and r , with $0 \leq r < d$, such that

$$a = dq + r.$$

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$$14 = 3 \cdot 4 + 2$$

$$14/3 = 3.666$$

$$14 \text{ div } 3 = 4$$

$$14 \bmod 3 = 2$$

Relations:

- $q = a \text{ div } d$, $r = a \bmod d$

Greatest common divisor

A systematic way to find the gcd using factorization:

- Let $a = p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_k^{a_k}$ and $b = p_1^{b_1} p_2^{b_2} p_3^{b_3} \dots p_k^{b_k}$
- $\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} p_3^{\min(a_3, b_3)} \dots p_k^{\min(a_k, b_k)}$

Examples:

- $\gcd(24, 36) = ?$
- $24 = 2 \cdot 2 \cdot 2 \cdot 3 = 2^3 \cdot 3$
- $36 = 2 \cdot 2 \cdot 3 \cdot 3 = 2^2 \cdot 3^2$
- $\gcd(24, 36) =$

Greatest common divisor

A systematic way to find the gcd using factorization:

- Let $a = p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_k^{a_k}$ and $b = p_1^{b_1} p_2^{b_2} p_3^{b_3} \dots p_k^{b_k}$
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Examples:

- $\gcd(24, 36) = ?$
- $24 = 2 \cdot 2 \cdot 2 \cdot 3 = 2^3 \cdot 3$
- $36 = 2 \cdot 2 \cdot 3 \cdot 3 = 2^2 \cdot 3^2$
- $\gcd(24, 36) = 2^2 \cdot 3 = 12$

Least common multiple

Definition: Let a and b are two positive integers. The least common multiple of a and b is the smallest positive integer that is divisible by both a and b . The **least common multiple** is denoted as **$\text{lcm}(a, b)$** .

Example:

- **What is $\text{lcm}(12, 9)$ =?**
- Give me a common multiple: ...

Least common multiple

Definition: Let a and b are two positive integers. The least common multiple of a and b is the smallest positive integer that is divisible by both a and b . The **least common multiple** is denoted as **$\text{lcm}(a,b)$** .

Example:

- **What is $\text{lcm}(12,9)$ =?**
- Give me a common multiple: ... $12 \cdot 9 = 108$
- **Can we find a smaller number?**

Least common multiple

Definition: Let a and b are two positive integers. The least common multiple of a and b is the smallest positive integer that is divisible by both a and b . The **least common multiple** is denoted as **$\text{lcm}(a,b)$** .

Example:

- **What is $\text{lcm}(12,9)$ =?**
- Give me a common multiple: ... $12 \cdot 9 = 108$
- **Can we find a smaller number?**
- **Yes.** Try 36. Both 12 and 9 cleanly divide 36.

Least common multiple

A systematic way to find the lcm using factorization:

- Let $a = p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_k^{a_k}$ and $b = p_1^{b_1} p_2^{b_2} p_3^{b_3} \dots p_k^{b_k}$
- $\text{lcm}(a,b) = p_1^{\max(a_1,b_1)} p_2^{\max(a_2,b_2)} p_3^{\max(a_3,b_3)} \dots p_k^{\max(a_k,b_k)}$

Example:

- What is $\text{lcm}(12,9)$ =?
- $12 = 2 \cdot 2 \cdot 3 = 2^2 \cdot 3$
- $9 = 3 \cdot 3 = 3^2$
- $\text{lcm}(12,9) = 2^2 \cdot 3^2 = 4 \cdot 9 = \mathbf{36}$

Euclid algorithm

Finding the greatest common divisor requires factorization

- $a = p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_k^{a_k}$, $b = p_1^{b_1} p_2^{b_2} p_3^{b_3} \dots p_k^{b_k}$
- $\text{gcd}(a,b) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} p_3^{\min(a_3,b_3)} \dots p_k^{\min(a_k,b_k)}$
- Factorization can be cumbersome and time consuming since we need to find all factors of the two integers that can be very large.
- Luckily a more efficient method for computing the gcd exists:
- It is called **Euclid's algorithm**
 - the method is known from ancient times and named after Greek mathematician Euclid.

Euclid algorithm

Assume two numbers 287 and 91. We want $\gcd(287,91)$.

- First divide the larger number (287) by the smaller one (91)
- We get $287 = 3 \cdot 91 + 14$

(1) Any divisor of 91 and 287 must also be a divisor of 14:

- $287 - 3 \cdot 91 = 14$
- Why? $[ak - cbk] = r \rightarrow (a - cb)k = r \rightarrow (a - cb) = r/k$ (must be an integer and thus k divides r)

(2) Any divisor of 91 and 14 must also be a divisor of 287

- Why? $287 = 3b + dk \rightarrow 287 = k(3b + d) \rightarrow 287/k = (3b + d) \leftarrow 287/k$ must be an integer
- But then $\gcd(287,91) = \gcd(91,14)$

Euclid algorithm

- We know that $\gcd(287,91) = \gcd(91,14)$
- But the same trick can be applied again:
 - $\gcd(91,14)$
 - $91 = 14 \cdot 6 + 7$
- and therefore
 - $\gcd(91,14) = \gcd(14,7)$
- And one more time:
 - $\gcd(14,7) = 7$
 - trivial
- The result: $\gcd(287,91) = \gcd(91,14) = \gcd(14,7) = 7$

Euclid algorithm

Example 1:

- Find the greatest common divisor of 666 & 558
 - $\gcd(666, 558)$
 $= \gcd(558, 108)$
 $= \gcd(108, 18)$
 $= 18$
- | |
|---------------------------|
| $666 = 1 \cdot 558 + 108$ |
| $558 = 5 \cdot 108 + 18$ |
| $108 = 6 \cdot 18 + 0$ |

Euclid algorithm

Example 2:

- Find the greatest common divisor of 286 & 503:
- $\gcd(503, 286)$
 $503 =$

Euclid algorithm

Example 2:

- Find the greatest common divisor of 286 & 503:
 - $\gcd(503, 286)$
 $= \gcd(286, 217)$
- $$503 = 1 \cdot 286 + 217$$
- $$286 =$$

Euclid algorithm

Example 2:

- Find the greatest common divisor of 286 & 503:
 - $\gcd(503, 286)$
 $= \gcd(286, 217)$
 $= \gcd(217, 69)$
 $= \gcd(69, 10)$
 $= \gcd(10, 9)$
 $= \gcd(9, 1) = \mathbf{1}$
- $$503 = 1 \cdot 286 + 217$$
- $$286 = 1 \cdot 217 + 69$$
- $$217 = 3 \cdot 69 + 10$$
- $$69 = 6 \cdot 10 + 9$$
- $$10 = 1 \cdot 9 + 1$$

Modular arithmetic

- In computer science we often care about the remainder of an integer when it is divided by some positive integer.

Problem: Assume that it is a midnight. What is the time on the 24 hour clock after 50 hours?

Answer: the result is 2am

How did we arrive to the result:

- Divide 50 with 24. The remainder is the time on the 24 hour clock.
 - $50 = 2 \cdot 24 + 2$
 - so the result is 2am.

Congruency

Definition: If a and b are integers and m is a positive integer, then **a is congruent to b modulo n** if m divides $a-b$. We use the notation **$a = b \pmod{m}$** to denote the congruency. If a and b are not congruent we write $a \not\equiv b \pmod{m}$.

Example:

- Determine if 17 is congruent to 5 modulo 6?

Congruency

Theorem. If a and b are integers and m a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

Example:

- Determine if 17 is congruent to 5 modulo 6?
- $17 \bmod 6 = 5$
- $5 \bmod 6 = 5$
- Thus 17 is congruent to 5 modulo 6.

Congruencies

Theorem 1. Let m be a positive integer. The integers a and b are congruent modulo m if and only if there exists an integer k such that $a = b + mk$.

Theorem2 . Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then:

$$a + c \equiv b + d \pmod{m} \text{ and } ac \equiv bd \pmod{m}.$$

Modular arithmetic in CS

Modular arithmetic and congruencies are used in CS:

- Pseudorandom number generators
- Hash functions
- Cryptology

Pseudorandom number generators

- Some problems we want to program need to simulate a random choice.
- Examples: flip of a coin, roll of a dice

We need a way to generate random outcomes

Basic problem:

- assume outcomes: $0, 1, \dots, N$
- generate the random sequences of outcomes
- Pseudorandom number generators let us generate sequences that look random
- **Next:** linear congruential method

Pseudorandom number generators

Linear congruential method

- We choose 4 numbers:
 - the modulus m ,
 - multiplier a ,
 - increment c , and
 - seed x_0 ,such that $2 \leq a < m$, $0 \leq c < m$, $0 \leq x_0 < m$.
- We generate a sequence of numbers $x_1, x_2, x_3 \dots x_n \dots$ such that $0 \leq x_n < m$ for all n by successively using the congruence:
 - $x_{n+1} = (a \cdot x_n + c) \bmod m$

Pseudorandom number generators

Linear congruential method:

- $x_{n+1} = (a \cdot x_n + c) \bmod m$

Example:

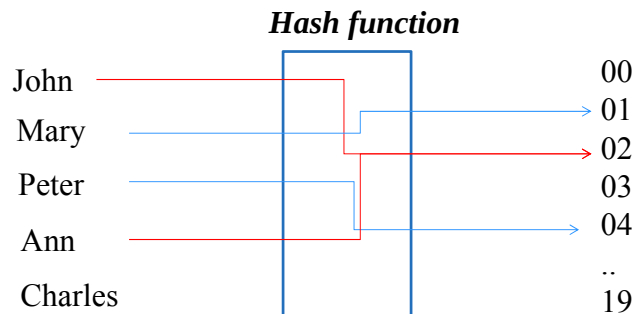
- Assume : $m=9, a=7, c=4, x_0 = 3$
- $x_1 = 7 \cdot 3 + 4 \bmod 9 = 25 \bmod 9 = 7$
- $x_2 = 53 \bmod 9 = 8$
- $x_3 = 60 \bmod 9 = 6$
- $x_4 = 46 \bmod 9 = 1$
- $x_5 = 11 \bmod 9 = 2$
- $x_6 = 18 \bmod 9 = 0$
-

Hash functions

A **hash function** is an algorithm that maps data of arbitrary length to data of a fixed length.

The values returned by a hash function are called **hash values** or **hash codes**.

Example:



Hash function

An example of a hash function that maps integers (including very large ones) to a subset of integers $0, 1, \dots, m-1$ is:

$$h(k) = k \bmod m$$

Example: Assume we have a database of employees, each with a unique ID – a social security number that consists of 8 digits. We want to store the records in a smaller table with m entries. Using $h(k)$ function we can map a social security number in the database of employees to indexes in the table.

Assume: $h(k) = k \bmod 111$

Then:

$$h(064212848) = 064212848 \bmod 111 = 14$$

$$h(037149212) = 037149212 \bmod 111 = 65$$

CS 441 Discrete Mathematics for CS
Lecture 14

Integers: applications, base conversions.

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Modular arithmetic in CS

Modular arithmetic and congruencies are used in CS:

- **Pseudorandom number generators**
- **Hash functions**
- **Cryptology**

Pseudorandom number generators

- Some problems we want to program need to simulate a random choice.
- Examples: flip of a coin, roll of a dice

We need a way to generate random outcomes

Basic problem:

- assume outcomes: $0, 1, \dots, N$
- generate the random sequences of outcomes
- Pseudorandom number generators let us generate sequences that look random
- **Next:** linear congruential method

Pseudorandom number generators

Linear congruential method

- We choose 4 numbers:
 - the modulus m ,
 - multiplier a ,
 - increment c , and
 - seed x_0 ,such that $2 \leq a < m$, $0 \leq c < m$, $0 \leq x_0 < m$.
- We generate a sequence of numbers $x_1, x_2, x_3, \dots, x_n, \dots$ such that $0 \leq x_n < m$ for all n by successively using the congruence:
 - $x_{n+1} = (a \cdot x_n + c) \bmod m$

Pseudorandom number generators

Linear congruential method:

- $x_{n+1} = (a \cdot x_n + c) \bmod m$

Example:

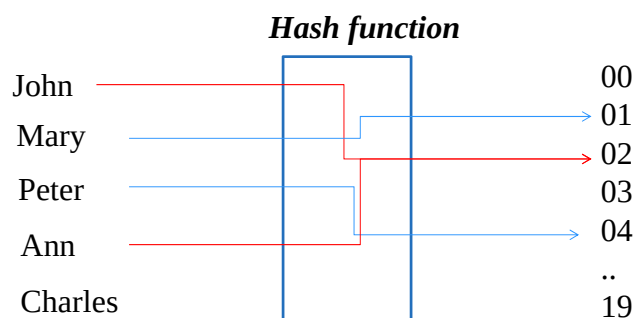
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- $x_1 = 7 \cdot 3 + 4 \bmod 9 = 25 \bmod 9 = 7$
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-

Hash functions

A **hash function** is an algorithm that maps data of arbitrary length to data of a fixed length.

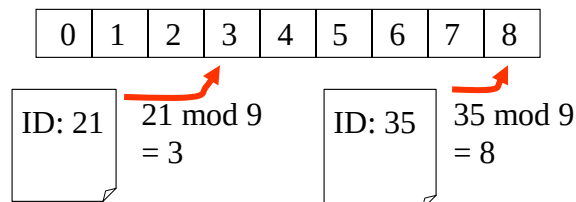
The values returned by a hash function are called **hash values** or **hash codes**.

Example:



Hash functions

- **Problem:** Given a large collection of records, how can we store and find a record quickly?
- **Solution:** Use a hash function calculate the location of the record based on the record's ID.
- **Example:** A common hash function is
 - $h(k) = k \bmod n$,where n is the number of available storage locations.



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Hash function

An example of a hash function that maps integers (including very large ones) to a subset of integers 0, 1, .. $m-1$ is:

$$h(k) = k \bmod m$$

Example: Assume we have a database of employees, each with a unique ID – a social security number that consists of 8 digits. We want to store the records in a smaller table with m entries. Using $h(k)$ function we can map a social security number in the database of employees to indexes in the table.

Assume: $h(k) = k \bmod 111$

Then:

$$h(064212848) = 064212848 \bmod 111 = 14$$

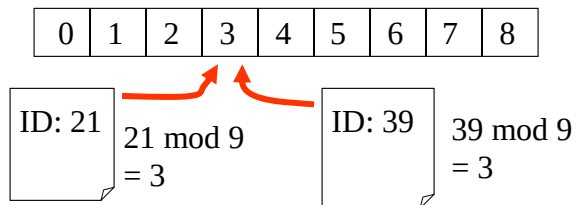
$$h(037149212) = 037149212 \bmod 111 = 65$$

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Hash functions

- **Problem:** two documents mapped to the same location

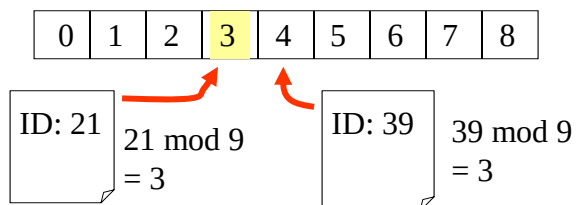


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Hash functions

- **Solution 1:** move the next available location
 - Method is represented by a sequence of hash functions to try

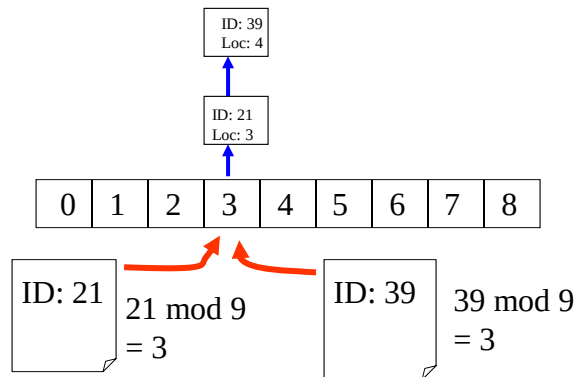
$$\begin{aligned}h_0(k) &= k \bmod n \\h_1(k) &= (k+1) \bmod n \\&\vdots \\h_m(k) &= (k+m) \bmod n\end{aligned}$$



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Hash functions

- **Solution 2:** remember the exact location in a secondary structure that is searched sequentially



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Cryptology

Encryption of messages.

- **Cesar cipher:**
- Shift letters in the message by 3, last three letters mapped to the first 3 letters, e.g. A is shifted to D, X is shifted to A

How to represent the idea of a shift by 3?

- There are 26 letters in the alphabet. Assign each of them a number from 0,1, 2, 3, .. 25 according to the alphabetical order.

A B C D E F G H I J K L M N O P Q R S T U V X W Z
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

- The encryption of the letter with an index p is represented as:
 - $f(p) = (p + 3) \bmod 26$

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Cryptology

Encryption of messages using a shift by 3.

- The encryption of the letter with an index p is represented as:
 - $f(p) = (p + 3) \bmod 26$

Coding of letters:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	X	W	Z	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- **Encrypt message:**
 - **I LIKE DISCRETE MATH**

—

Cryptology

Encryption of messages using a shift by 3.

- The encryption of the letter with an index p is represented as:
 - $f(p) = (p + 3) \bmod 26$

Coding of letters:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	X	W	Z	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- **Encrypt message:**
 - **I LIKE DISCRETE MATH**
 - **L 0LNH GLYFUHVH PDVK.**

Cryptology

How to decode the message ?

- The encryption of the letter with an index p is represented as:
 - $f(p) = (p + 3) \bmod 26$

Coding of letters:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	X	W	Z	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- What method would you use to decode the message:
 - $f^{-1}(p) = (p-3) \bmod 26$

Representations of Integers

- In the modern world, we use *decimal*, or *base 10, notation* to represent integers. For example when we write 965, we mean $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$.
- We can represent numbers using any base b , where b is a positive integer greater than 1.
- The bases $b = 2$ (*binary*), $b = 8$ (*octal*), and $b = 16$ (*hexadecimal*) are important for computing and communications
- The ancient Mayans used base 20 and the ancient Babylonians used base 60.

Base b Representations

- We can use positive integer b greater than 1 as a base

Theorem 1: Let b be a positive integer greater than 1. Then if n is a positive integer, it can be expressed uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$$

where k is a nonnegative integer, $a_0, a_1, \dots, a_k$ are nonnegative integers less than b , and $a_k \neq 0$. The $a_j, j = 0, \dots, k$ are called the base- b digits of the representation.

- The representation of n given in **Theorem 1** is called the **base b expansion of n** and is denoted by $(a_k a_{k-1} \dots a_1 a_0)_b$.
- We usually omit the subscript 10 for base 10 expansions.

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Binary Expansions

Most computers represent integers and do arithmetic with binary (base 2) expansions of integers. In these expansions, the only digits used are 0 and 1.

Example: What is the decimal expansion of the integer that has $(1\ 0101\ 1111)_2$ as its binary expansion?

Solution:

$$(1\ 0101\ 1111)_2 = 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 351.$$

Example: What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?

Solution: $(11011)_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 27$

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Octal Expansions

The octal expansion (base 8) uses the digits $\{0,1,2,3,4,5,6,7\}$.

Example: What is the decimal expansion of the number with octal expansion $(7016)_8$?

Solution: $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

Example: What is the decimal expansion of the number with octal expansion $(111)_8$?

Solution: $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$

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Hexadecimal Expansions

- The hexadecimal expansion uses 16 digits:
 $\{0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F\}$.
 - The letters A through F represent the decimal numbers 10 through 15.

Example: What is the decimal expansion of the number with hexadecimal expansion $(2AE0B)_{16}$?

Solution:

$$2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 175627$$

Example: What is the decimal expansion of the number with hexadecimal expansion $(E5)_{16}$?

Solution: $14 \cdot 16^1 + 5 \cdot 16^0 = 224 + 5 = 229$

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Base Conversion

To construct the base b expansion of an integer n :

- Divide n by b to obtain a quotient and remainder.

$$n = bq_0 + a_0 \quad 0 \leq a_0 \leq b$$

- The remainder, a_0 , is the rightmost digit in the base b expansion of n . Next, divide q_0 by b .

$$q_0 = bq_1 + a_1 \quad 0 \leq a_1 \leq b$$

- The remainder, a_1 , is the second digit from the right in the base b expansion of n .
- Continue by successively dividing the quotients by b , obtaining the additional base b digits as the remainder. The process terminates when the quotient is 0.

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Base Conversion

Example: Find the octal expansion of $(12345)_{10}$

Solution: Successively dividing by 8 gives:

- $12345 = 8 \cdot 1543 + 1$
- $1543 = 8 \cdot 192 + 7$
- $192 = 8 \cdot 24 + 0$
- $24 = 8 \cdot 3 + 0$
- $3 = 8 \cdot 0 + 3$

The remainders are the digits from right to left yielding $(30071)_8$.

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Lecture 14

**Mathematical induction
& Recursion**

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Proofs

Basic proof methods:

- Direct, Indirect, Contradiction, By Cases, Equivalences

Proof of quantified statements:

- **There exists x with some property $P(x)$.**
 - It is sufficient to find one element for which the property holds.
- **For all x some property $P(x)$ holds.**
 - Proofs of ‘For all x some property $P(x)$ holds’ must cover all x and can be harder.
- **Mathematical induction** is a technique that can be applied to prove the universal statements for sets of positive integers or their associated sequences.

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Mathematical induction

- Used to prove statements of the form $\forall x P(x)$ where $x \in \mathbb{Z}^+$

Mathematical induction proofs consists of two steps:

- 1) **Basis:** The proposition $P(1)$ is true.
- 2) **Inductive Step:** The implication $P(n) \rightarrow P(n+1)$, is true for all positive n .

- Therefore we conclude $\forall x P(x)$.
- **Based on the well-ordering property:** Every nonempty set of nonnegative integers has a **least element**.

Mathematical induction

Example: Prove the sum of first n odd integers is n^2 .

i.e. $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$ for all positive integers.

Proof:

- What is $P(n)$? $P(n)$: $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$

Basis Step Show $P(1)$ is true

- Trivial: $1 = 1^2$

Inductive Step Show if $P(n)$ is true then $P(n+1)$ is true for all n .

- Suppose $P(n)$ is true, that is $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$
- Show $P(n+1)$: $1 + 3 + 5 + 7 + \dots + (2n - 1) + (2n + 1) = (n+1)^2$ follows:
$$\underbrace{1 + 3 + 5 + 7 + \dots + (2n - 1)}_{n^2} + (2n + 1) = (n+1)^2$$

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Lecture 15

**Mathematical induction
& Recursion**

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Proofs

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- Direct, Indirect, Contradiction, By Cases, Equivalences

Proof of quantified statements:

- **There exists x with some property $P(x)$.**
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- Suppose $P(n)$ is true, that is $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$
- Show $P(n+1)$: $1 + 3 + 5 + 7 + \dots + (2n - 1) + (2n + 1) = (n+1)^2$ follows:
$$\underbrace{1 + 3 + 5 + 7 + \dots + (2n - 1)}_{n^2} + (2n + 1) = (n+1)^2$$

Correctness of the mathematical induction

Suppose **P(1) is true** and **P(n) $\rightarrow$ P(n+1) is true** for all positive integers n. Want to show **$\forall x P(x)$** .

Assume there is at least one n such that P(n) is false. Let S be the set of nonnegative integers where P(n) is false. Thus $S \neq \emptyset$.

Well-Ordering Property: Every nonempty set of nonnegative integers has a least element.

By the Well-Ordering Property, S has a least member, say k. $k > 1$, since P(1) is true. This implies $k - 1 > 0$ and P(k-1) is true (since k is the smallest integer where P(k) is false).

Now: $P(k-1) \rightarrow P(k)$ is true

thus, P(k) must be true (a contradiction).

- **Therefore $\forall x P(x)$.**

Mathematical induction

Example: Prove $n < 2^n$ for all positive integers n.

- P(n): $n < 2^n$

Basis Step: $1 < 2^1$ (obvious)

Inductive Step: If P(n) is true then P(n+1) is true for each n.

- Suppose P(n): $n < 2^n$ is true
- Show P(n+1): $n+1 < 2^{n+1}$ is true.

$$\begin{aligned} n + 1 &< 2^n + 1 \\ &< 2^n + 2^n \\ &= 2^n (1 + 1) \\ &= 2^n (2) \\ &= 2^{n+1} \end{aligned}$$

Mathematical induction

Example: Prove $n^3 - n$ is divisible by 3 for all positive integers.

- $P(n)$: $n^3 - n$ is divisible by 3

Basis Step: $P(1)$: $1^3 - 1 = 0$ is divisible by 3 (obvious)

Inductive Step: If $P(n)$ is true then $P(n+1)$ is true for each positive integer.

- Suppose $P(n)$: $n^3 - n$ is divisible by 3 is true.
- Show $P(n+1)$: $(n+1)^3 - (n+1)$ is divisible by 3.

$$\begin{aligned}(n+1)^3 - (n+1) &= n^3 + 3n^2 + 3n + 1 - n - 1 \\&= (n^3 - n) + 3n^2 + 3n \\&= \underbrace{(n^3 - n)}_{\text{divisible by 3}} + 3\underbrace{(n^2 + n)}_{\text{divisible by 3}}\end{aligned}$$

Strong induction

- **The regular induction:**
 - uses the basic step $P(1)$ and
 - inductive step $P(n-1) \rightarrow P(n)$
- **Strong induction uses:**
 - Uses the basis step $P(1)$ and
 - inductive step $P(1) \text{ and } P(2) \dots P(n-1) \rightarrow P(n)$

Example: Show that a positive integer greater than 1 can be written as a product of primes.

Strong induction

Example: Show that a positive integer greater than 1 can be written as a product of primes.

Assume $P(n)$: an integer n can be written as a product of primes.

Basis step: $P(2)$ is true

Inductive step: Assume true for $P(2), P(3), \dots, P(n)$

Show that $P(n+1)$ is true as well.

2 Cases:

- If $n+1$ is a prime then $P(n+1)$ is trivially true
- If $n+1$ is a composite then it can be written as a product of two integers $(n+1) = a \cdot b$ such that $1 < a, b < n+1$
- From the assumption $P(a)$ and $P(b)$ holds.
- Thus, $n+1$ can be written as a product of primes
- **End of proof**

Recursive Definitions

- Sometimes it is possible to define an object (function, sequence, algorithm, structure) in terms of itself. This process is called **recursion**.

Examples:

- Recursive definition of an **arithmetic sequence**:

- $a_n = a + nd$

- $a_n = a_{n-1} + d$, $a_0 = a$

- Recursive definition of a **geometric sequence**:

- $x_n = ar^n$

- $x_n = rx_{n-1}$, $x_0 = a$

Recursive Definitions

- In some instances recursive definitions **of objects may be much easier to write**

Examples:

- **Algorithm for computing the gcd:**
 - $\text{gcd}(79, 35) = \text{gcd}(35, 9)$
 - More general:
 $\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$
- **Factorial function:**
 - $n! = n (n-1)! \text{ and } 0!=1$

Recursively Defined Functions

To define a function on the set of nonnegative integers

1. Specify the value of the function at 0
2. Give a rule for finding the function's value at $n+1$ in terms of the function's value at integers $i \leq n$.

Example: factorial function definition

- $0! = 1$
- $n! = n (n-1)!$
- **recursive or inductive definition of a function on nonnegative integers**

Recursively defined functions

Example: Assume a recursive function on positive integers:

- $f(0) = 3$
- $f(n+1) = 2f(n) + 3$
- **What is the value of $f(0)$?** 3
- $f(1) = 2f(0) + 3 = 2(3) + 3 = 6 + 3 = 9$
- $f(2) = f(1 + 1) = 2f(1) + 3 = 2(9) + 3 = 18 + 3 = 21$
- $f(3) = f(2 + 1) = 2f(2) + 3 = 2(21) = 42 + 3 = 45$
- $f(4) = f(3 + 1) = 2f(3) + 3 = 2(45) + 3 = 90 + 3 = 93$

Recursive definitions

- **Example:**

Define the function:

$$f(n) = 2n + 1 \quad n = 0, 1, 2, \dots$$

recursively.

- $f(0) = 1$
- $f(n+1) = f(n) + 2$

Recursive definitions

- **Example:**

Define the sequence:

$$a_n = n^2 \quad \text{for } n = 1, 2, 3, \dots$$

recursively.

- $a_1 = 1$
- $a_{n+1} = a_n^2 + (2n + 1), \quad n \geq 1$

Recursive definitions

- **Example:**

Define a recursive definition of the sum of the first n positive integers:

$$F(n) = \sum_{i=1}^n i$$

- $F(1) = 1$
- $F(n+1) = F(n) + (n+1), \quad n \geq 1$

Recursive definitions

Some important functions or sequences in mathematics are defined recursively

Factorials

- $n! = 1$ if $n=1$
- $n! = n \cdot (n-1)!$ if $n \geq 1$

Fibonacci numbers:

- $F(0)=0$, $F(1)=1$ and
- $F(n) = F(n-1) + F(n-2)$ for $n=2,3, \dots$

Recursive definitions

Methods (algorithms)

- **Greatest common divisor**

$$\begin{aligned}\gcd(a,b) &= b \quad \text{if } b \mid a \\ &= \gcd(b, a \bmod b)\end{aligned}$$

- **Pseudorandom number generators:**

- $x_{n+1} = (ax_n + c) \bmod m$

Recursive definitions

- **Assume the alphabet Σ**
 - Example: $\Sigma = \{a,b,c,d\}$
- **A set of all strings containing symbols in Σ : Σ^***
 - Example: $\Sigma^* = \{\epsilon, a, aa, aaa, aaa\dots, ab, \dots b, bb, bbb, \dots\}$

Recursive definition of Σ^*

- **Basis step:**
 - empty string $\lambda \in \Sigma^*$
- **Recursive step:**
 - If $w \in \Sigma^*$ and $x \in \Sigma$ then $wx \in \Sigma^*$

Length of a String

Example:

Give a recursive definition of $l(w)$, the length of the string w .

Solution:

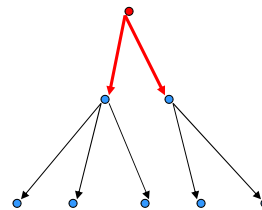
The length of a string can be recursively defined by:

$l(\epsilon) = 0$; the length of the empty string

$l(wx) = l(w) + 1$ if $w \in \Sigma^*$ and $x \in \Sigma$.

Recursive definitions

- **Data structures**
 - **Example: Rooted tree**
- **A basis step:**
 - a single node (vertex)
is a rooted tree
- **Recursive step:**
 - Assume $T_1, T_2, \dots, T_k$ are rooted trees, then the graph with a root r connected to $T_1, T_2, \dots, T_k$ is a rooted tree



CS 441 Discrete Mathematics for CS
Lecture 16

Counting

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Course administration

Midterm exam:

- Thursday, March 6, 2014

Homework assignment 8:

- due after Spring break
- An opportunity to practice mathematical induction, and basic counting for the midterm

Course web page:

<http://www.cs.pitt.edu/~milos/courses/cs441/>

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Counting

- Assume we have a set of **objects with certain properties**
- **Counting** is used to determine **the number of these objects**

Examples:

- Number of available phone numbers with 7 digits in the local calling area
- Number of possible match starters (football, basketball) given the number of team members and their positions

Basic counting rules

- Counting problems may be very hard, not obvious
- **Solution:**
 - **simplify the solution by decomposing the problem**
- **Two basic decomposition rules:**
 - **Product rule**
 - A count decomposes into a sequence of dependent counts (“each element in the first count is associated with all elements of the second count”)
 - **Sum rule**
 - A count decomposes into a set of independent counts (“elements of counts are alternatives”)

Product rule

A count can be broken down into a sequence of dependent counts

- “each element in the first count is associated with all elements of the second count”

Example:

- Assume an auditorium with a seat labeled by a letter and numbers in between 1 to 50 (e.g. A23). We want the total number of seats in the auditorium.
- 26 letters and 50 numbers
- How to count?
- **One solution: write down all seats (objects) and count them**

A-1 A-2 A-3 ...A-50 B-1... Z-49 Z-50

1 2 3 50 51 ... (n-1) n ← eventually we get it

Product rule

A count can be broken down into a sequence of dependent counts

- “each element in the first count is associated with all elements of the second count”

Example:

- assume an auditorium with a seat labeled by a letter and numbers in between 1 to 50 (e.g. A23). We want the total number of seats in the auditorium.
- 26 letters and 50 numbers
- **A better solution?**
- For each letter there are 50 numbers
- So the number of seats is $26 \cdot 50 = 1300$
- **Product rule:** number of letters * number of integers in $[1, 50]$

Product rule

A count can be broken down into a sequence of dependent counts

- “each element in the first count is associated with all elements of the second count”
- **Product rule:** If a count of elements can be broken down into a sequence of dependent counts where the first count yields n_1 elements, the second n_2 elements, and k th count n_k elements, by the product rule the total number of elements is:
 - $n = n_1 * n_2 * \dots * n_k$

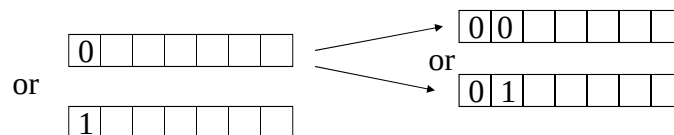
Product rule

Example:

- How many different bit strings of length 7 are there?
 - E.g. 1011010
- Is it possible to decompose the count problem and if yes how?

Yes.

- Count the number of possible assignments to bit 1
- For the first bit assignment (say 0) count assignments to bit 2



Total assignments to first 2 bits: $2 * 2 = 4$

Product rule

Example:

- How many different bit strings of length 7 are there?
 - E.g. 1011010
- Is it possible to decompose the count problem and if yes how?
- **Yes.**
 - Count the number of possible assignments to bit 1
 - For the specific first bit count possible assignments to bit 2
 - For the specific first two bits count assignments to bit 3
 - Number of assignments to the first 3 bits: $2*2*2=8$

Product rule

Example:

- How many different bit strings of length 7 are there?
 - E.g. 1011010
- Is it possible to decompose the count problem and if yes how?
- **Yes.**
 - Count the number of possible assignments to bit 1
 - For the specific first bit count possible assignments to bit 2
 - For the specific first two bits count assignments to bit 3
 - Gives a sequence of n dependent counts and by the product rule we have:
$$n = 2 * 2 * 2 * 2 * 2 * 2 * 2 = 2^7$$

Product rule

Example:

The number of subsets of a set S with k elements.

- How to count them?
- **Hint:** think in terms of bitstring representation of a set?
- Assume each element in S is assigned a bit position.
- If A is a subset it can be encoded as a bitstring: if an element is in A then use 1 else put 0
- How many different bitstrings are there?

$$n = \underbrace{2^* 2^* \dots 2^*}_{k \text{ bits}} = 2^k$$

k bits

Sum rule

A count decomposes into a set of independent counts

- “elements of counts are alternatives”, they do not depend on each other

Example:

- You need to travel in between city A and B. You can either fly, take a train, or a bus. There are 12 different flights in between A and B, 5 different trains and 10 buses. How many options do you have to get from A to B?
- We can take only one type of transportation and for each only one option. The number of options:
 - $n = 12+5+10$

Sum rule:

- $n = \text{number of flights} + \text{number of trains} + \text{number of buses}$

Sum rule

A count decomposes into a set of independent counts

- “elements of counts are alternatives”
- **Sum rule:** If a count of elements can be broken down into a set of independent counts where the first count yields n_1 elements, the second n_2 elements, and k th count n_k elements, by the sum rule the total number of elements is:
 - $n = n_1 + n_2 + \dots + n_k$

Beyond basic counting rules

- **More complex counting problems** typically require a combination of the sum and product rules.

Example: A login password:

- The minimum password length is 6 and the maximum is 8. The password can consist of either an uppercase letter or a digit. There must be at least one digit in the password.
- How many different passwords are there?

Beyond basic counting rules

Example: A password for the login name.

- The minimum password length is 6 and the maximum is 8. The password can consist of either an uppercase letter or a digit. There must be at least one digit in the password.
- How to compute the number of possible passwords?

Step 1:

- The password we select has either 6,7 or 8 characters.
- So the total number of valid passwords is by the sum rule:

- $P = P_6 + P_7 + P_8$

The number of passwords of length 6,7 and 8 respectively

Beyond basic counting rules

Step 1:

- The password we select has either 6,7 or 8 characters.
- So the total number of valid passwords is by the sum rule:

- $P = P_6 + P_7 + P_8$

The number of passwords of length 6,7 and 8 respectively

Step 2

- Assume passwords with 6 characters (upper-case letters):
- How many are there?
- If we let each character to be at any position we have:
 - $P_6\text{-nodigits} = 26^6$ different passwords of length 6

Beyond basic counting rules

Step 1:

- The password we select has either 6,7 or 8 characters.
- So the total number of valid passwords is by the sum rule:
 - $P = P_6 + P_7 + P_8$

The number of passwords of length 6,7 and 8 respectively

Step 2

- Assume passwords with 6 characters
(either digits + upper case letters):
- How many are there?
- If we let each character to be at any position we have:
 - $P_6\text{-all} = (26+10)^6 = (36)^6$ different passwords of length 6

Beyond basic counting rules

Step 2

But we must have a password with at least one digit. How to account for it?

A trick. Split the count of all passwords of length 6 into two mutually exclusive groups:

- **$P_6\text{-all} = P_6\text{-digits} + P_6\text{-nodigits}$**
 1. $P_6\text{-digits}$ – count when the password has one or more digits
 2. $P_6\text{-nodigits}$ – count when the password has no digits
- We know how to easily compute $P_6\text{-all}$ and $P_6\text{-nodigits}$
 - **$P_6\text{-all} = 36^6$ and $P_6\text{-nodigits} = 26^6$**
 - Then **$P_6\text{-digits} = P_6\text{-all} - P_6\text{-nodigits}$**

Beyond basic counting rules

Step 1:

the total number of valid passwords is by the sum rule:

- $P = P6 + P7 + P8$
- The number of passwords of length 6, 7 and 8 respectively

Step 2

The number of valid passwords of length 6:

$$\begin{aligned} P6 &= P6\text{-digits} = P6\text{-all} - P6\text{-nodigits} \\ &= 36^6 - 26^6 \end{aligned}$$

Analogically:

$$\begin{aligned} P7 &= P7\text{-digits} = P7\text{-all} - P7\text{-nodigits} \\ &= 36^7 - 26^7 \end{aligned}$$

$$\begin{aligned} P8 &= P8\text{-digits} = P8\text{-all} - P8\text{-nodigits} \\ &= 36^8 - 26^8 \end{aligned}$$

Inclusion-Exclusion principle

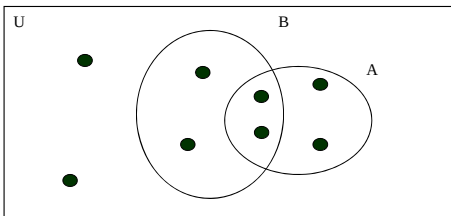
Used in counts where the decomposition yields two dependent count tasks with overlapping elements

- If we used the sum rule some elements would be counted twice

Inclusion-exclusion principle: uses a sum rule and then corrects for the overlapping elements.

We used the principle for the cardinality of the set union.

- $|A \cup B| = |A| + |B| - |A \cap B|$



Inclusion-exclusion principle

Example: How many bitstrings of length 8 start either with a bit 1 or end with 00?

- It is easy to count **strings that start with 1**:
- How many are there? 2^7
- It is easy to count the **strings that end with 00**.
- How many are there? 2^6
- Is it OK to add the two numbers to get the answer? $2^7 + 2^6$
- **No. Overcount.** There are some strings that can both start with 1 and end with 00. These strings are counted in twice.
- How to deal with it? How to correct for overlap?
- How many of strings were counted twice? 2^5 (1 xxxxx 00)
- Thus we can correct for the overlap simply by using:
- $2^7 + 2^6 - 2^5 = 128 + 64 - 32 = 160$

Tree diagrams

Tree: is a structure that consists of a root, branches and leaves.

- Can be useful to represent a counting problem and record the choices we made for alternatives. The count appears on the leaf nodes.

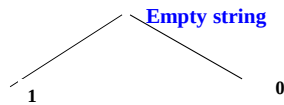
Example:

What is the number of bit strings of length 4 that do not have two consecutive ones.

Tree diagrams

Example:

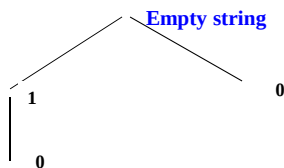
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Tree diagrams

Example:

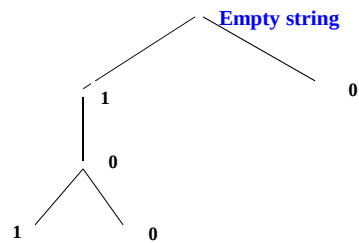
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Tree diagrams

Example:

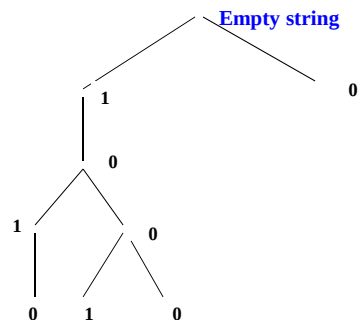
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Tree diagrams

Example:

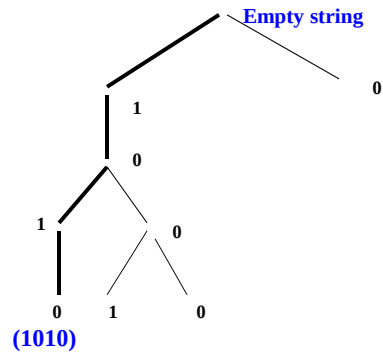
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Tree diagrams

Example:

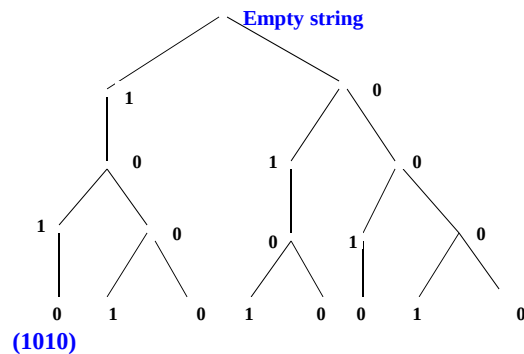
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Tree diagrams

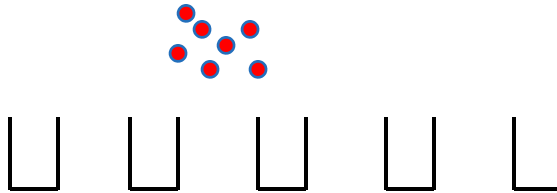
Example:

What is the number of bit strings of length 4 that do not have two consecutive ones?



Pigeonhole principle

- Assume you have a set of objects and a set of bins used to store objects.
- **The pigeonhole principle** states that if there are more objects than bins then there is at least one bin with more than one object.
- **Example:** 7 balls and 5 bins to store them

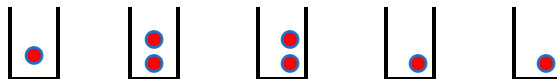


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M. Hauskrecht

Pigeonhole principle

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- At least one bin with more than 1 ball exists.



CS 441 Discrete mathematics for CS

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Pigeonhole principle

- Assume you have a set of objects and a set of bins used to store objects. The pigeonhole principle states that if there are more objects than bins then there is at least one bin with more than one object.
- **Theorem.** If there are $k+1$ objects and k bins. Then there is at least one bin with two or more objects.



k bins

Pigeonhole principle

- Assume you have a set of objects and a set of bins used to store objects. The pigeonhole principle states that if there are more objects than bins then there is at least one bin with more than one object.
- **Theorem.** If there are $k+1$ objects and k bins. Then there is at least one bin with two or more objects.
- **Proof. (by contradiction)**
 - Assume that we have $k + 1$ objects and every bin has at most one element. Then the total number of elements is k which is a contradiction.
 - End of proof

Pigeonhole principle

Example:

- Assume 367 people. Are there any two people who has the same birthday?
- How many days are in the year? 365.
- Then there must be at least two people with the same birthday.

Generalized pigeonhole principle

- We can often say more about the number of objects.
- Say we have 5 bins and 12 objects. What is it we can say about the bins and number of elements they hold?
- There must be **a bin with at least 3 elements**.
- Why?
- Assume there is no bin with more than 3 elements. Then the max number of elements we can have in 5 bins is 10. We need to place 13 so at least one bin should have at least 3 elements.

Generalized pigeonhole principle

Theorem. If N objects are placed into k bins then there is at least one bin containing at least $\lceil N/k \rceil$ objects.

Example. Assume 100 people. Can you tell something about the number of people born in the same month.

- Yes. There exists a month in which at least $\lceil 100/12 \rceil = \lceil 8.3 \rceil = 9$ people were born.

CS 441 Discrete Mathematics for CS
Lecture 17

Counting

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Counting

- Assume we have a set of **objects with certain properties**
- **Counting** is used to determine **the number of these objects**

Examples:

- Number of available phone numbers with 7 digits in the local calling area
- Number of possible match starters (football, basketball) given the number of team members and their positions

Basic counting rules

- Counting problems may be hard, and easy solutions are not obvious
- **Approach:**
 - simplify the solution by decomposing the problem
- **Two basic decomposition rules:**
 - **Product rule**
 - A count decomposes into a sequence of dependent counts (“each element in the first count is associated with all elements of the second count”)
 - **Sum rule**
 - A count decomposes into a set of independent counts (“elements of counts are alternatives”)

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Inclusion-Exclusion principle

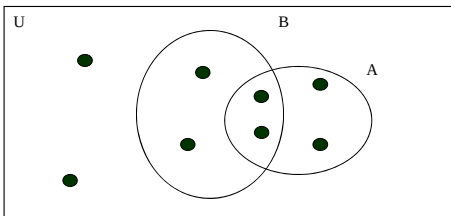
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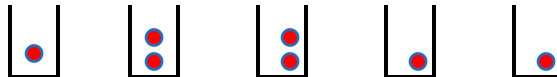
Inclusion-exclusion principle

Example: How many bitstrings of length 8 start either with a bit 1 or end with 00?

- **Count strings that start with 1:**
- How many are there? 2^7
- **Count the strings that end with 00.**
- How many are there? 2^6
- **The two counts overlap !!!**
- How many of strings were counted twice? 2^5 (1 xxxxx 00)
- Thus we can correct for the overlap simply by using:
- $2^7 + 2^6 - 2^5 = 128 + 64 - 32 = 160$

Pigeonhole principle

- Assume you have a set of objects and a set of bins used to store objects.
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- Yes. There exists a month in which at least $\lceil 100/12 \rceil = \lceil 8.3 \rceil = 9$ people were born.

Generalized pigeonhole principle

Example.

- Show that among any set of 5 integers, there are 2 with the same remainder when divided by 4.

Answer:

- Let there be 4 boxes, one for each remainder when divided by 4.
- After 5 integers are sorted into the boxes, there are $\lceil 5/4 \rceil = 2$ in one box.

Generalized pigeonhole principle

Example:

- How many students, each of whom comes from one of the 50 states, must be enrolled in a university to guaranteed that there are at least 100 who come from the same state?

Answer:

- Let there be 50 boxes, one per state.
- We want to find the minimal N so that $\lceil N/50 \rceil = 100$.
- Letting $N=5000$ is too much, since the remainder is 0.
- We want a remainder of 1 so that let $N=50 \cdot 99 + 1 = 4951$.

Permutations

A permutation of a set of distinct objects is an ordered arrangement of the objects. Since the objects are distinct, they cannot be selected more than once. Furthermore, the order of the arrangement matters.

Example:

- Assume we have a set S with n elements. $S = \{a, b, c\}$.
- **Permutations of S :**
- **a b c a c b b a c b c a c a b c b a**

Number of permutations

- Assume we have a set S with n elements. $S = \{a_1 a_2 \dots a_n\}$.
 - **Question:** How many different permutations are there?
 - In how many different ways we can choose the first element of the permutation? **n** (**either** a_1 or $a_2 \dots$ or a_n)
 - Assume we picked a_2 .
 - In how many different ways we can choose the remaining elements? **$n-1$** (**either** a_1 or $a_3 \dots$ or a_n **but not** a_2)
 - **Assume** we picked a_j .
 - In how many different ways we can choose the remaining elements? **$n-2$** (**either** a_1 or $a_3 \dots$ or a_n **but not** a_2 **and not** a_j)
- $P(n,n) = n.(n-1)(n-2)\dots 1 = n!$**

Permutations

Example 1.

- How many permutations of letters $\{a,b,c\}$ are there?
- Number of permutations is:
$$P(n,n) = P(3,3) = 3! = 6$$

- Verify:

$abc \quad acb \quad bac \quad bca \quad cab \quad cba$

Permutations

Example 2

- How many permutations of letters A B C D E F G H contain a substring ABC.

Idea: consider ABC as one element and D,E,F,G,H as other 5 elements for the total of 6 elements.

Then we need to count the number of permutation of these elements.

$$6! = 720$$

k-permutations

- **k-permutation** is an ordered arrangement of k elements of a set.
- The number of k -permutations of a set with n distinct elements is:

$$P(n,k) = n(n-1)(n-2)\dots(n-k+1) = n!/(n-k)!$$

k-permutations

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- The number of k -permutations of a set with n distinct elements is:

$$P(n,k) = n(n-1)(n-2)\dots(n-k+1) = n!/(n-k)!$$

Explanation:

- Assume we have a set S with n elements. $S = \{a_1, a_2, \dots, a_n\}$.
- The 1st element of the k -permutation may be any of the n elements in the set.
- The 2nd element of the k -permutation may be any of the $n-1$ remaining elements of the set.
- And so on. For last element of the k -permutation, there are $n-k+1$ elements remaining to choose from.

k-permutations

Example:

The 2-permutations of set $\{a,b,c\}$ are:

ab, ac, ba, bc, ca, cb .

The number of 2-permutations of this 3-element set is

$$P(n,k) = P(3,2) = 3(3-2+1) = 6.$$

k-permutations

Example:

Suppose that there are eight runners in a race. The winner receives a gold medal, the second-place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur?

Answer:

note that the runners are distinct and that the medals are ordered.

The solution is $P(8,3) = 8 * 7 * 6 = 8! / (8-3)! = 336$.

Combinations

A k -combination of elements of a set is an unordered selection of k elements from the set. Thus, a k -combination is simply a subset of the set with k elements.

Example:

- 2-combinations of the set $\{a,b,c\}$

a b a c b c



a b covers 2-permutations: **a b** and **b a**

Combinations

Theorem: The number of k -combinations of a set with n distinct elements, where n is a positive integer and k is an integer with $0 \leq k \leq n$ is

$$C(n, k) = \frac{n!}{(n-k)! k!}$$

Combinations

Theorem: The number of k -combinations of a set with n distinct elements, where n is a positive integer and k is an integer with $0 \leq k \leq n$ is

$$C(n, k) = \frac{n!}{(n-k)! k!}$$

Proof: The k -permutations of the set can be obtained by first forming the $C(n, k)$ k -combinations of the set, and then ordering the elements in each k -combination, which can be done in $P(k, k)$ ways. Consequently,

$$P(n, k) = C(n, k) * P(k, k).$$

This implies that

$$C(n, k) = P(n, k) / P(k, k) = P(n, k) / k! = n! / (k! (n-k)!)$$

Combinations

Intuition (example): Assume elements A1, A2, A3, A4 and A5 in the set. All 3-combinations of elements are:

- A1 A2 A3
- A1 A2 A4
- A1 A2 A5
- A1 A3 A4
- A1 A3 A5
- A1 A4 A5
- A2 A3 A4
- A2 A3 A5
- A2 A4 A5
- A3 A4 A5
- **Total of 10.**



Each combination cover many 3-permutations

A1 A2 A3
A1 A3 A2
A2 A1 A3
A2 A3 A1
A3 A1 A2
A3 A2 A1

Combinations

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- A1 A3 A5
- A1 A4 A5
- A2 A3 A4
- A2 A3 A5
- A2 A4 A5
- A3 A4 A5
- **Total of 10.**



Each 3-combination covers many 3-permutations

A1 A2 A3
A1 A3 A2
A2 A1 A3
A2 A3 A1
A3 A1 A2
A3 A2 A1

So: $P(5,3) = C(5,3) P(3,3)$

Combinations

Intuition (example): Assume elements A1, A2, A3, A4 and A5 in the set. All 3-combinations of elements are:

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- A1 A3 A5
- A1 A4 A5
- A2 A3 A4
- A2 A3 A5
- A2 A4 A5
- A3 A4 A5
- **Total of 10.**



Each 3-combination covers many 3-permutations

A1 A2 A3
A1 A3 A2
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A2 A3 A1
A3 A1 A2
A3 A2 A1

So: $P(5,3) = C(5,3) P(3,3)$

and: $C(5,3) = P(5,3)/P(3,3)$

Combinations

Example:

- We need to create a team of 5 player for the competition out of 10 team members. How many different teams is it possible to create?

Answer:

- When creating a team we do not care about the order in which players were picked. It is important that the player is in. Because of that we need to consider unordered sets of combinations.
- $C(10,5) = 10!/(10-5)!5! = (10 \cdot 9 \cdot 8 \cdot 7 \cdot 6) / (5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)$
 $= 2 \cdot 3 \cdot 2 \cdot 7 \cdot 3 = 6 \cdot 14 \cdot 3 = 6 \cdot 42 = 252$

Combinations

Corrolary:

- $C(n,k) = C(n,n-k)$

Proof.

- $$\begin{aligned} C(n,k) &= n! / (n-k)! k! \\ &= n! / (n-k)! (n - (n-k))! \\ &= C(n,n-k) \end{aligned}$$

Binomial coefficients

- The number of k-combinations out of n elements $C(n,k)$ is often denoted as:

$$\binom{n}{k}$$

and reads **n choose k**. The number is also called **a binomial coefficient**.

- Binomial coefficients occur as coefficients in the expansion of powers of binomial expressions such as

$$(a + b)^n$$

- Definition: a binomial expression is the sum of two terms $(a+b)$.

Binomial coefficients

Example:

- Expansion of the binomial expression $(a+b)^3$.

$$(a+b)^3 =$$

$$(a+b)(a+b)(a+b) =$$

$$(a^2 + 2ab + b^2)(a+b) =$$

$$a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3 =$$

$$1a^3 + 3a^2b + 3ab^2 + 1b^3$$

$$\begin{matrix} 1 & 3 & 3 & 1 \\ \binom{3}{0} & \binom{3}{1} & \binom{3}{2} & \binom{3}{3} \end{matrix} \quad \leftarrow \text{Binomial coefficients}$$

Binomial coefficients

Binomial theorem: Let a and b be variables and n be a nonnegative integer. Then:

$$\begin{aligned} (a+b)^n &= \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i \\ &= \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n-1} a b^{n-1} + \binom{n}{n} b^n \end{aligned}$$

Binomial coefficients

Binomial theorem: Let a and b be variables and n be a nonnegative integer. Then:

$$(a+b)^n = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i$$

- **Proof.** The products after the expansion include terms $a^{(n-i)} b^i$ for all $i=0,1, \dots, n$. To obtain the number of such coefficients note that we have to choose exactly $(n-i)$ a 's out of the product of n binomial expressions.

$(n-i)$ picks

$$(a+b)^n = \underbrace{(a+b)(a+b)(a+b)\dots(a+b)}_n$$

- The number of ways we pull a 's out of the product is given as:

Binomial coefficients

Binomial theorem: Let a and b be variables and n be a nonnegative integer. Then:

$$(a+b)^n = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i$$

- **Proof.** The products after the expansion include terms $a^{(n-i)} b^i$ for all $i=0,1, \dots, n$. To obtain the number of such coefficients note that we have to choose exactly $(n-i)$ a 's out of the product of n binomial expressions.

$(n-i)$ picks

$$(a+b)^n = \underbrace{(a+b)(a+b)(a+b)\dots(a+b)}_n$$

The number of ways we pull a 's out is:

$$\binom{n}{n-i} = \binom{n}{i}$$

CS 441 Discrete Mathematics for CS

Lecture 18

Counting

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Permutations

A permutation of a set of distinct objects is an ordered arrangement of the objects. Since the objects are distinct, they cannot be selected more than once. Furthermore, the order of the arrangement matters.

Example:

- Assume we have a set S with n elements. $S = \{a, b, c\}$.
- **Permutations of S :**
- **a b c a c b b a c b c a c a b c b a**

The number of permutations of n elements is:

$$P(n, n) = n!$$

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k-permutations

- **k-permutation** is an ordered arrangement of k elements of a set.

Example:

- Assume we have a set S with n elements. $S=\{a,b,c\}$.
 - **2 permutations of S :**
 - **ab ba ac ca bc cb**
-
- The number of k -permutations of a set with n distinct elements is:

$$P(n,k) = n(n-1)(n-2)\dots(n-k+1) = n!/(n-k)!$$

Combinations

A k -combination of elements of a set is an **unordered** selection of k elements from the set. Thus, a k -combination is simply a subset of the set with k elements.

Example:

- 2-combinations of the set $\{a,b,c\}$

a b a c b c



a b covers two of the 2-permutations: **a b** and **b a**

Combinations

Theorem: The number of k -combinations of a set with n distinct elements, where n is a positive integer and k is an integer with $0 \leq k \leq n$ is

$$C(n, k) = \frac{n!}{(n-k)! k!}$$

Combinations

Example:

- We need to create a team of 5 players for the competition out of 10 team members. How many different teams is it possible to create?

Answer:

- When creating a team we do not care about the order in which players were picked. It is important that the player is in. Because of that we need to consider unordered sets of people.
- $C(10,5) = 10!/(10-5)!5! = (10 \cdot 9 \cdot 8 \cdot 7 \cdot 6) / (5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)$
 $= 2 \cdot 3 \cdot 2 \cdot 7 \cdot 3 = 6 \cdot 14 \cdot 3 = 6 \cdot 42 = 252$

Binomial coefficients

- The number of k-combinations out of n elements $C(n,k)$ is often denoted as:

$$\binom{n}{k}$$

and reads **n choose k**. The number is also called **a binomial coefficient**.

- Binomial coefficients occur as coefficients in the expansion of powers of binomial expressions such as

$$(a+b)^n$$

- Definition:** a binomial expression is the sum of two terms $(a+b)$.

Binomial coefficients

Example:

- Expansion of the binomial expression $(a+b)^3$.

$$(a+b)^3 =$$

$$(a+b)(a+b)(a+b) =$$

$$(a^2 + 2ab + b^2)(a+b) =$$

$$a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3 =$$

$$1a^3 + 3a^2b + 3ab^2 + 1b^3$$

$$\begin{matrix} 1 & 3 & 3 & 1 \end{matrix} \quad \leftarrow \text{Binomial coefficients}$$

$$\begin{pmatrix} 3 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

Binomial coefficients

Binomial theorem: Let a and b be variables and n be a nonnegative integer. Then:

$$\begin{aligned}(a+b)^n &= \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i \\ &= \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n-1} a b^{n-1} + \binom{n}{n} b^n\end{aligned}$$

Binomial coefficients

Corrolary: Let n be a nonnegative integer. Then:

$$\sum_{i=0}^n \binom{n}{i} = 2^n$$

Binomial coefficients

Corrolary:

- Let n be a nonnegative integer. Then:

$$\sum_{i=0}^n (-1)^i \binom{n}{i} = 0$$

Proof:

$$\sum_{i=0}^n \binom{n}{i} (-1)^i = \sum_{i=0}^n \binom{n}{i} (-1)^i 1^{n-i} = ((-1) + 1)^n = 0^n = 0$$

Binomial coefficients

Example:

- Show that $\sum_{i=0}^n \binom{n}{i} 2^i = 3^n$

• Answer:

$$\sum_{i=0}^n \binom{n}{i} 2^i = \sum_{i=0}^n \binom{n}{i} (2)^i 1^{n-i} = (2+1)^n = 3^n$$

Binomial coefficients

We have binomial coefficients for expressions with the power n .

Question: Are binomial coefficients for powers of $n-1$ or $n+1$ in any way related to coefficients for n ?

- The answer is yes.

Theorem:

- Let n and k be two positive integers with $k \leq n$. Then it holds:

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

Pascal triangle

Drawing the binomial coefficients for different powers in increasing order gives a **Pascal triangle**:

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

powers

```
      1
     1 1
    1 2 1
   1 3 3 1
```

2
3
4

Pascal traingle

Drawing the binomial coefficients for different powers in increasing order gives a **Pascal triangle**:

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

powers

				1			
			1		1		
		1		2		1	
	1		3		3		1
1		4		6		4	1

2
3
4
5

Pascal traingle

Drawing the binomial coefficients for different powers in increasing order gives a **Pascal triangle**:

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

powers

					1		
				1		1	
		1		2		1	
	1		3		3		1
	1		4		6		4
1		5		10		10	5
1		5		10		10	5

2
3
4
5
6

Pascal traingle

Drawing the binomial coefficients for different powers in increasing order gives a **Pascal triangle**:

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

powers

				1				
			1		1			
		1		2		1		2
	1		3		3		1	3
	1	4		6		4	1	4
	1	5	10		10	5	1	5
1	6	15	20	15	6	1		6

Pascal traingle

Drawing the binomial coefficients for different powers in increasing order gives a **Pascal triangle**:

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

powers

				1				
			1		1			
		1		2		1		2
	1		3		3		1	3
	1	4		6		4	1	4
	1	5	10		10	5	1	5
1	6	15	20	15	6	1		6
			...					7

Permutations with repetitions

Assume we want count different ordered collections of objects such that we can pick the object from a set multiple times: that is, an object can be on the position 1, 2, 3, etc

Example:

- 26 letters of alphabet. How many different strings of length k are there?

Answer:

- 26^k

Permutations with repetitions

Assume we want count different ordered collections of objects such that we can pick the object from a set multiple times: that is, an object can be on the position 1, 2, 3, etc

Example:

- 26 letters of alphabet. How many different strings of length k are there?

Answer:

- 26^k

Theorem: The number of k -permutations of a set of n objects with repetitions is n^k .

Combinations with repetitions

Example:

- Pick four pieces of a fruit from three bowls with apples, pears and oranges. How many possible combinations are there? List /count all of them?

Answer:

- **Star and bar approach**

- Apples Pears Oranges
- 3 bowls separated by | |
- Choice 2 apples and 2 pears represented as: ** | ** |
- Choice of 1 apple and 3 oranges: * | | ***

Combinations with repetitions

Example:

- Pick four pieces of a fruit from three bowls with apples, pears and oranges. How many possible combinations are there? List /count all of them?

Answer:

- **Star and bar approach**

- Apples Pears Oranges
- 3 bowls separated by | |
- Choice 2 apples and 2 pears represented as: ** | ** |
- Choice of 1 apple and 3 oranges: * | | ***
- **Count:** How many different ways of arranging $(3-1)=2$ bars and 4 stars are there?
- Total number of positions: $2+4=6$
- **Count:** the number of ways to select the positions of 4 stars.

Combinations with repetitions

- **Theorem:** The number of ways to pick n elements from k different groups is:

$$\binom{n-1+k}{n}$$

Combinations with repetitions

- **Theorem:** The number of ways to pick n elements from k different groups is:

$$\binom{n-1+k}{n}$$

- $(n+k-1)$ positions
- n - stars
- **Count:** the number of ways to select the positions of 4 stars.

Probabilities

Probability

Discrete probability theory

- Dates back to 17th century.
- It was used to compute the **odds** of seeing some outcomes: e.g. in games, races etc.
- Odds are related to counting when the outcomes are equally likely.

Example: Coin flip

- Assume 2 outcomes (head and tail) and each of them is equally likely
- Odds: 50%, 50%
- the probability of seeing:
 - a head is 0.5
 - a tail is 0.5

Probability

Probability is related to relative **counts** of target outcomes with respect to all outcomes

$$P = \text{Number of target outcomes} / \text{Total number of outcomes}$$

Example: Coin flip

- the probability of seeing:
 - a head is 0.5
 - a tail is 0.5
- Probability of all (disjoint) events is 1
- **$P(\text{any outcome}) = P(\text{head}) + P(\text{tail}) = 1$**

Probability

Example: roll of a dice

- 6 different outcomes. Each of them is equally likely
- Probability of each outcome is:
 - 1/6
- How did we get the number?

Probability

Example: roll of a dice

- 6 different outcomes. Each of them is equally likely
- Probability of each outcome is:
 - $1/6$
 - How did we get the number?
- Assuming each outcome is equally likely and we have 6 outcomes, then the probability of an outcome is:
 $1/6$

Probability of aggregate outcomes

Example: roll of a dice

- Roll of the dice is odd or even. All outcomes are equally likely.
- Probability = number of outcomes when odd/ total number of outcomes.

Solution 1: all outcomes are equally likely and $= 1/6$

Odd numbers: 1,3,5 Even numbers: 2,4,6

- $P(\text{odd}) = 1/6 + 1/6 + 1/6 = 3/6 = 1/2$
- $P(\text{even}) = 3/6 = 1/2$

Solution 2:

Odd numbers are equally likely as even numbers (2 outcomes)

$P(\text{odd}) = 1/2$ and $P(\text{even}) = 1/2$

Probabilities

- **Experiment:** a procedure that yields one of the possible outcomes
- **Sample space:** a set of possible outcomes
- **Event:** a subset of possible outcomes (E is a subset of S)
- **Assuming the outcomes are equally likely, the probability of an event E, defined by a subset of outcomes from the sample space S is**
 - $P(\text{Event}) = |E| / |S|$
- The cardinality of the subset divided by the cardinality of the sample space.

Probabilities

Example 1:

- A box with 4 red balls and 6 blue balls. What is the probability that we pull the blue ball out.
- $P(E) = 6/10 = 0.6$

Probabilities

Example 2:

- roll of two dices.
- What is the probability that the outcome is 7.
- Possible outcomes:
 - (1,6) (2,6) ... (6,1), ... (6,6) total: 36
- Outcomes leading to
- (1,6) (2,5) ... (6,1) total: 6
- $P(\text{sum}=7) = 6/36 = 1/6$

Probabilities

More complex:

- Odd of winning a lottery: 6 numbers out of 40.
- Total number of outcomes:
 - $C(40,6) = \dots$
- Probability of winning: ?
 - $P(E) = 1/C(40,6) = 39! / 6! / 40! = 1/3,838,380$

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Lecture 19

Probabilities

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Probabilities

- **Experiment:**
 - a procedure that yields one of the possible outcomes
- **Sample space:** a set of possible outcomes
- **Event:** a subset of possible outcomes (E is a subset of S)
- **Assuming the outcomes are equally likely, the probability of an event E, defined by a subset of outcomes from the sample space S is**
 - $P(E) = |E| / |S|$
- The cardinality of the subset divided by the cardinality of the sample space.

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Probabilities

- Event E, Sample space S, all outcomes equally likely, then

$$P(E) = |E| / |S|$$

Example:

- roll of two dices
- What is the probability that the outcome is 7.
- All possible outcomes (sample space S):
- (1,6) (2,6) ... (6,1), ... (6,6) total: 36
- Outcomes leading to 7 (event E)
- (1,6) (2,5) ... (6,1) total: 6
- $P(\text{sum}=7) = 6/36 = 1/6$

Probabilities

- Event E, Sample space S, all outcomes equally likely, then

$$P(E) = |E| / |S|$$

Example:

- Odd of winning a lottery: 6 numbers out of 40.
- Total number of outcomes (sample space S):
 - $C(40,6) = 3,838,380$
- Winning combination (event E): 1
- Probability of winning:
 - $P(E) = 1/C(40,6) = 34! \cdot 6! / 40! = 1/3,838,380$

Probabilities

- Event E, Sample space S, all outcomes equally likely, then

$$P(E) = |E| / |S|$$

Example (cont):

- Odd of winning a second prize in lottery: hit 5 of 6 numbers selected from 40.
- Total number of outcomes (sample space S):
 - $C(40,6) = 3,838,380$
- Second prize (event E): $C(6,5) \cdot (40-6) = 6 \cdot 34$
- Probability of winning:
 - $P(E) = 6 \cdot 34 / C(40,6) = (6 \cdot 34) / 3,838,380$

Probabilities

- Event E, Sample space S, all outcomes equally likely, then

$$P(E) = |E| / |S|$$

Example (cont):

- Odd of winning a **third prize** in lottery: hit 4 of 6 numbers selected from 40.
- Total number of outcomes (sample space S):
 - $C(40,6) = 3,838,380$
- Third prize (event E): $C(6,4) \cdot C(40-6,2) = C(6,4) \cdot C(34,2)$
- Probability of winning:
 - $P(E) = C(6,4) \cdot C(34,2) / C(40,6)$

Probabilities

- Event E, Sample space S, all outcomes equally likely, then

$$P(E) = |E| / |S|$$

Another lottery:

- 6 numbers (ordered) selected out of 40 numbers (with repetitions)
- Total number of outcomes:
 - Permutations with repetitions: $= 40^6$
- Number of winning configuration: 1
 - $P(\text{win}) = 1/40^6$

And its modification:

- If the winning combination is order independent:
 - E.g. (1,5,17,25,5,13) is equivalent to (5,17,5,1,25,13)
 - Number of winning permutations = number of permutations of 6 = $6!$
 - $P(\text{win}) = 6! / 40^6$

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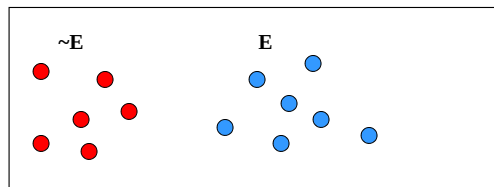
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Probabilities

Theorem: Let E be an event and $\sim E$ its complement with regard to S. Then:

- $P(\sim E) = 1 - P(E)$

Sample space



Proof.

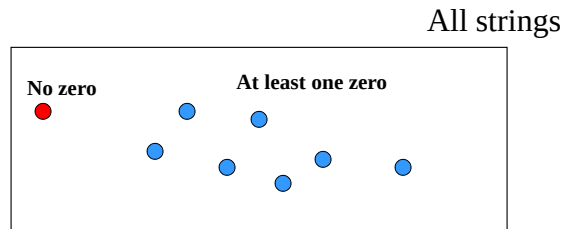
$$P(\sim E) = (|S| - |E|) / |S| = 1 - |E| / |S|$$

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Probabilities

Example:

- 10 randomly generated bits. What is the probability that there is at least one zero in the string.



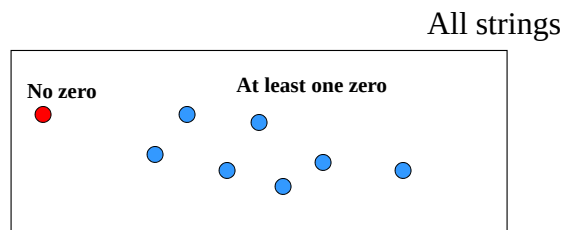
- Event: seeing no-zero string $P(E) = ?$
- $\sim$ Event: seeing at least one zero in the string

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Probabilities

Example:

- 10 randomly generated bits. What is the probability that there is at least one zero in the string.



- Event: seeing no-zero string $P(E) = 1/2^{10}$
- $\sim$ Event: seeing at least one zero in the string
 $P(\sim E) = 1 - P(E) = 1 - 1/2^{10}$

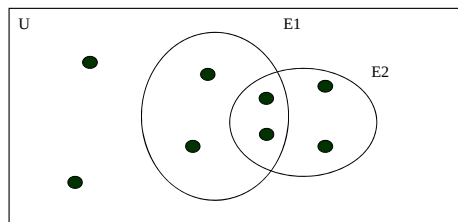
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Probabilities

Theorem. Let $E1$ and $E2$ be two events in the sample space S .

Then:

- $P(E1 \cup E2) = P(E1) + P(E2) - P(E1 \cap E2)$
- This is an example of the inclusion-exclusion principle



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Probabilities

Theorem. Let $E1$ and $E2$ be two events in the sample space S .

Then:

- $P(E1 \cup E2) = P(E1) + P(E2) - P(E1 \cap E2)$

Example: Probability that a positive integer ≤ 100 is divisible either by 2 or 5.

- $P(E1) = 50/100$
- $P(E2) = 20/100$
- $P(E1 \cap E2) = 10/100$
- $P(E1 \cup E2) = (5+2-1)/10 = 6/10$

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Probabilities

- Assumption applied so far:
 - **the probabilities of each outcome are equally likely.**
- However in many cases outcomes may not be equally likely.

Example: a biased coin or a biased dice.

- **Biased Coin:**
 - Probability of head 0.6,
 - probability of a tail 0.4.
- **Biased Dice:**
 - Probability of **6**: 0.4,
 - Probability of **1, 2, 3, 4, 5**: 0.12 each

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Probabilities

Three axioms of the probability theory:

- (1) Probability of a discrete outcome is:
 - $0 \leq P(s) \leq 1$
- (2) Sum of probabilities of all (disjoint) outcomes is = 1
- (3) For any two events E1 and E2 holds:
$$P(E1 \cup E2) = P(E1) + P(E2) - P(E1 \cap E2)$$

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Probability distribution

Definition: A function $p: S \rightarrow [0,1]$ satisfying the three conditions is called a **probability distribution**

Example: a biased coin

- Probability of head 0.6, probability of a tail 0.4
- **Probability distribution:**
 - Head $\rightarrow 0.6$ The sum of the probabilities sums to 1
 - Tail $\rightarrow 0.4$

Note: a **uniform distribution** is a special distribution that assigns equal probabilities to each outcome.

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Probability of an Event

Definition: The probability of the event E is the sum of the probabilities of the outcomes in E .

$$P(E) = \sum_{s \in E} P(s)$$

- Note that now no assumption is being made about the distribution.

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Example

Probability of an event

$$P(E) = \sum_{s \in E} P(s)$$

Example: Suppose that a die is biased so that 3 appears twice as often as each other number, but that the other five outcomes are equally likely. What is the probability that an odd number appears when we roll this die?

Solution: We want the probability of the event $E = \{1, 3, 5\}$. Probabilities of outcomes:

- $p(3) = 2/7$ and
 $p(1) = p(2) = p(4) = p(5) = p(6) = 1/7$.

$$\begin{aligned} \text{Then, } p(E) &= p(1) + p(3) + p(5) = \\ &1/7 + 2/7 + 1/7 = 4/7. \end{aligned}$$

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Probabilities of Complements and Unions

- Complements still hold. Since each outcome is in either E or $\overline{E}$ but not both,

$$p(\overline{E}) = 1 - p(E)$$

- Unions: $\sum_{s \in S} p(s) = 1 = p(E) + p(\overline{E})$.

also still holds under the new definition.

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

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Conditional probability

Definition: Let E and F be two events such that $P(F) > 0$. The **conditional probability** of E given F

- $P(E|F) = P(E \cap F) / P(F)$

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Conditional probability

Definition: Let E and F be two events such that $P(F) > 0$. The **conditional probability** of E given F

- $P(E|F) = P(E \cap F) / P(F)$

Example:

- What is the probability that a family has two boys given that they have at least one boy. Assume the probability of having a girl or a boy is equal.
- **Possibilities.** BB BG GB GG
- **Probability of having two boys** $P(BB) = \frac{1}{4}$
- **Probability of having one boy** $P(\text{one boy}) = \frac{3}{4}$
- $P(BB|\text{given a boy}) = \frac{1}{4} / \frac{3}{4} = \frac{1}{3}$

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Conditional probability

Definition: Let E and F be two events such that $P(F) > 0$. The **conditional probability** of E given F

- $P(E|F) = P(E \cap F) / P(F)$

Example:

- What is the probability that a family has two boys given that they have at least one boy. Assume the probability of having a girl or a boy is equal.
- Possibilities.** BB BG GB GG
- Probability of having two boys** $P(BB) = 1/4$
- Probability of having a boy** $P(\text{one boy}) = 3/4$
- $P(BB|\text{given a boy}) = 1/4 / 3/4 = 1/3$

BB BG GB GG

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Conditional probability

Corrolary: Let E and F be two events such that $P(F) > 0$. Then:

- $P(E \cap F) = P(E|F) * P(F)$

Proof:

- From the definition of the conditional probability:

$$P(E|F) = P(E \cap F) / P(F)$$

$\rightarrow$

$$P(E \cap F) = P(E|F) P(F)$$

- This result is also referred to as **the product rule**.

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Conditional probability

Corrolary: Let E and F be two events such that $P(F) > 0$. Then:

- $P(E \cap F) = P(E|F) P(F)$

Example:

- Assume the probability of getting a flu is 0.2
- Assume the probability of having a high fever given the flu: 0.9

What is the probability of getting a flu with fever?

$$P(\text{flu} \cap \text{fever}) = P(\text{fever} | \text{flu})P(\text{flu}) = 0.9 \cdot 0.2 = 0.18$$

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Conditional probability

Corrolary: Let E and F be two events such that $P(F) > 0$. Then:

- $P(E \cap F) = P(E|F) P(F)$

Example:

- Assume the probability of getting a flu is 0.2
- Assume the probability of having a high fever given the flu: 0.9

What is the probability of getting a flu with fever?

$$P(\text{flu} \cap \text{fever}) = P(\text{fever} | \text{flu})P(\text{flu}) = 0.9 \cdot 0.2 = 0.18$$

- When is this useful?

Sometimes conditional probabilities are easier to estimate.

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Lecture 20

Probabilities

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Probabilities

Three axioms of the probability theory:

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 - $0 \leq P(s) \leq 1$
- (2) Sum of probabilities of all (disjoint) outcomes is = 1
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Example: a biased coin

- Probability of head 0.6, probability of a tail 0.4
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Note: a **uniform distribution** is a special distribution that assigns equal probabilities to each outcome.

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Probability of an Event

Definition: The probability of the event E is the sum of the probabilities of the outcomes in E .

$$p(E) = \sum_{s \in E} p(s)$$

- Note that now no assumption is being made about the distribution.

Complement:

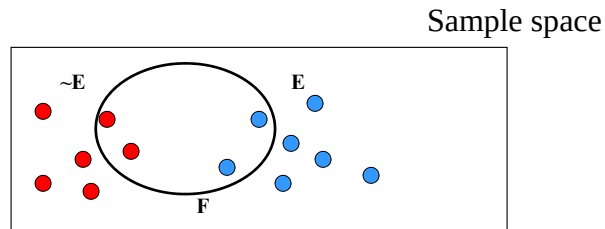
- $P(\sim E) = 1 - P(E)$

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Complements

Let E and F are two events. Then:

- $P(F) = P(F \cap E) + P(F \cap \sim E)$



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Conditional probability

Definition: Let E and F be two events such that $P(F) > 0$. The **conditional probability** of E given F

- $P(E|F) = P(E \cap F) / P(F)$

Corrolary: Let E and F be two events such that $P(F) > 0$. Then:

- $P(E \cap F) = P(E|F) * P(F)$

This result is also referred to as a **product rule**.

M. Hauskrecht

Conditional probability

Product rule:

- $P(E \cap F) = P(E|F) P(F)$

Example:

- Assume the probability of getting a flu is 0.2
- Assume the probability of having a high fever given the flu: 0.9

What is the probability of getting a flu with fever?

$$P(\text{flu} \wedge \text{fever}) = P(\text{fever} | \text{flu})P(\text{flu}) = 0.9 \cdot 0.2 = 0.18$$

- When is this useful?

Sometimes conditional probabilities are easier to estimate.

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Bayes theorem

Definition: Let E and F be two events such that $P(F) > 0$. Then:

- $$P(E|F) = P(F|E)P(E) / P(F)$$
$$= P(F|E)P(E) / (P(F|E)P(E) + P(F|\sim E)P(\sim E))$$

Proof:

$$P(E|F) = P(E \cap F) / P(F)$$

$$= P(F|E) P(E) / P(F)$$

$$P(F) = P(F \cap E) + P(F \cap \sim E)$$

$$= P(F|E) P(E) + P(F|\sim E) P(\sim E)$$

Hence:

$$P(E|F) = P(F|E)P(E) / (P(F|E)P(E) + P(F|\sim E)P(\sim E))$$

Idea: Simply switch the conditioning events.

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Bayes theorem

Definition: Let E and F be two events such that $P(F) > 0$. Then:

- $P(E|F) = P(F|E)P(E) / P(F)$
 $= P(F|E)P(E) / (P(F|E)P(E) + P(F|\sim E)P(\sim E))$

Example:

- Assume the probability of getting a flu is 0.2
- Assume the probability of getting a fever is 0.3
- Assume the probability of having a high fever given the flu: 0.9
- What is the probability of having a flu given the fever?

M. Hauskrecht

Bayes theorem

Definition: Let E and F be two events such that $P(F) > 0$. Then:

- $P(E|F) = P(F|E)P(E) / P(F)$
 $= P(F|E)P(E) / (P(F|E)P(E) + P(F|\sim E)P(\sim E))$

Example:

- Assume the probability of getting a flu is 0.2
- Assume the probability of getting a fever is 0.3
- Assume the probability of having a fever given the flu: 0.9
- What is the probability of having a flu given the fever?
- $P(\text{flu} | \text{fever}) = P(\text{fever} | \text{flu}) P(\text{flu}) / P(\text{fever}) =$
 $= 0.9 \times 0.2 / 0.3 = 0.18 / 0.3 = 0.6$

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Bayes theorem

Definition: Let E and F be two events such that $P(F) > 0$. Then:

$$\begin{aligned} \bullet \quad P(E|F) &= P(F|E)P(E) / P(F) \\ &= P(F|E)P(E) / (P(F|E)P(E) + P(F|\sim E)P(\sim E)) \end{aligned}$$

Example (same as above but different probabilities are given):

- Assume the probability of getting a flu is 0.2
- Assume the probability of having a fever given the flu: 0.9
- Assume the probability of having a fever given no flu: 0.15
- What is the probability of having a flu given the fever?

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Bayes theorem

Definition: Let E and F be two events such that $P(F) > 0$. Then:

$$\begin{aligned} \bullet \quad P(E|F) &= P(F|E)P(E) / P(F) \\ &= P(F|E)P(E) / (P(F|E)P(E) + P(F|\sim E)P(\sim E)) \end{aligned}$$

Example:

- Assume the probability of getting a flu is 0.2
- Assume the probability of having a fever given the flu: 0.9
- Assume the probability of having a fever given no flu: 0.15
- What is the probability of having a flu given the fever?
- $P(\text{flu} | \text{fever}) = P(\text{fever} | \text{flu}) P(\text{flu}) / P(\text{fever})$
- $P(\text{fever}) = P(\text{fever} | \text{flu}) P(\text{flu}) + P(\text{fever} | \sim \text{flu}) P(\sim \text{flu})$
 $= 0.9 \cdot 0.2 + 0.15 \cdot 0.8 = 0.3$

$$P(\text{flu} | \text{fever}) = 0.9 \times 0.2 / 0.3 = 0.18 / 0.3 = 0.6$$

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Independence

Definition: The two events E and F are said to be **independent** if:

- $P(E \cap F) = P(E)P(F)$

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Independence

Definition: The events E and F are said to be **independent** if:

- $P(E \cap F) = P(E)P(F)$

Example. Assume that E denotes the family has three children of both sexes and F the fact that the family has at most one boy. Are E and F independent?

- All combos = {BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG} the number of elements = 8
- Both sexes = {BBG BGB GBB BGG GBG GGB} # = 6
- At most one boy = {GGG GGB GBG BGG} # = 4
- $E \cap F = \{GGB GBG BGG\}$ # = 3
- $P(E \cap F) = 3/8$ and $P(E) \cdot P(F) = 6/8 \cdot 4/8 = 3/8$
- **The two probabilities are equal $\rightarrow$ E and F are independent**

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Independence

Example:

- Assume the probability of getting a flu is 0.2
- Assume the probability of getting a fever is 0.3
- Assume the probability of having a fever given the flu: 0.9
- Are flu and fever independent ?
- $P(\text{flu} \cap \text{fever}) = P(\text{fever} | \text{flu}) * P(\text{flu}) = 0.2 * 0.9 = 0.18$
- $P(\text{flu}) * P(\text{fever}) = 0.2 * 0.3 = 0.06$
- **Independent or not? Not independent**

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Bernoulli trial

Assume:

- $p = 0.6$ is a probability of seeing head
- 0.4 is the probability of seeing tail

Assume we see a sequence of independent coin flips:

- **HHHTTHTHHT**
- **The probability of seeing this sequence:**
 $0.6^6 * 0.4^4$
- **What is the probability of seeing a sequence of with 6 Heads and 4 tails?**
- The probability of each such sequence is $0.6^6 * 0.4^4$
- **How many such sequences are there:** $C(10,4)$
- $P(6H \text{ and } 4T) = C(10,4) * 0.6^6 * 0.4^4$

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Random variables

- **Definition: A random variable** is a function from the **sample space of an experiment** to the set of real numbers $f: S \rightarrow \mathbb{R}$.
A random variable assigns a number to each possible outcome.
- **The distribution of a random variable X on the sample space S** is a set of pairs $(r, p(X=r))$ for all r in S where r is the number and $p(X=r)$ is the probability that X takes a value r .

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Random variables

Example:

Let S be the outcomes of a two-dice roll

Let random variable X denotes the sum of outcomes

$(1,1) \rightarrow 2$

$(1,2)$ and $(2,1) \rightarrow 3$

$(1,3)$, $(3,1)$ and $(2,2) \rightarrow 4$

...

Distribution of X :

- $2 \rightarrow 1/36$,
- $3 \rightarrow 2/36$,
- $4 \rightarrow 3/36$...
- $12 \rightarrow 1/36$

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Probabilities

- **Assume a repeated coin flip**
- $P(\text{head}) = 0.6$ and the probability of a tail is 0.4. Each coin flip is independent of the previous.
- What is the probability of seeing:
 - HHHHH - 5 heads in a row
- $P(\text{HHHHH}) = 0.6^5 =$
 - Assume the outcome is HHTTT
- $P(\text{HHTTT}) = 0.6 * 0.6 * 0.4^3 = 0.6^2 * 0.4^3$
 - Assume the outcome is TTHHT
- $P(\text{TTHHT}) = 0.4^2 * 0.6^2 * 0.4 = 0.6^2 * 0.4^3$
- What is the probability of seeing three tails and two heads?
- The number of two-head-three tail combinations = $C(5,2)$
- $P(\text{two-heads-three tails}) = C(5,2) * 0.6^2 * 0.4^3$

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Probabilities

- **Assume a variant of a repeated coin flip problem**
- The space of possible outcomes is the count of occurrences of heads in 5 coin flips. For example:
- TTTTT yields outcome 0
- HTTTT or TTHTT yields 1
- HTHHT yields 3 ...
- What is the probability of an outcome 0?
- $P(\text{outcome}=0) = 0.6^0 * 0.4^5$
- $P(\text{outcome}=1) = C(5,1) 0.6^1 * 0.4^4$
- $P(\text{outcome}=2) = C(5,2) 0.6^2 * 0.4^3$
- $P(\text{outcome}=3) = C(5,3) 0.6^3 * 0.4^2$
- ...

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Expected value and variance

Definition: The **expected value** of the random variable $X(s)$ on the sample space is equal to:

$$E(X) = \sum_{s \in S} p(s)X(s)$$

Example: roll of a dice

- Outcomes: 1 2 3 4 5 6
- Expected value:
 $E(X) = ?$

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Expected value and variance

Definition: The **expected value** of the random variable $X(s)$ on the sample space is equal to:

$$E(X) = \sum_{s \in S} p(s)X(s)$$

Example: roll of a dice

- Outcomes: 1 2 3 4 5 6
- Expected value:
 $E(X) = 1 \cdot 1/6 + 2 \cdot 1/6 + 3 \cdot 1/6 + 4 \cdot 1/6 + 5 \cdot 1/6 + 6 \cdot 1/6 = 7/2$

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Expected value

Example:

Flip a fair coin 3 times. The outcome of the trial X is the number of heads. What is the expected value of the trial?

Answer:

Possible outcomes:

= {HHH HHT HTH THH HTT THT TTH TTT}
3 2 2 2 1 1 1 0

$E(X) = ?$

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Expected value

Example:

Flip a fair coin 3 times. The outcome of the trial X is the number of heads. What is the expected value of the trial?

Answer:

Possible outcomes:

= {HHH HHT HTH THH HTT THT TTH TTT}
3 2 2 2 1 1 1 0

$E(X) = 1/8 (3 + 3*2 + 3*1 + 0) = 12/8 = 3/2$

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Expected value

- **Theorem:** If X_i $i=1,2,3, n$ with n being a positive integer, are random variables on S , and a and b are real numbers then:
 - $E(X_1+X_2+ \dots X_n) = E(X_1)+E(X_2) + \dots E(X_n)$
 - $E(aX+b) = aE(X) +b$

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Expected value

Example:

- Roll a pair of dices. What is the expected value of the sum of outcomes?

- **Approach 1:**

- Outcomes: (1,1) (1,2) (1,3) (6,1)... (6,6)
2 3 4 7 12

Expected value: $1/36 (2*1 + \dots) = 7$

- **Approach 2 (theorem):**

- $E(X_1+X_2) = E(X_1) +E(X_2)$
- $E(X_1) = 7/2$ $E(X_2) = 7/2$
- $E(X_1+X_2) = 7$

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Lecture 21

**Probabilities:
Expected value**

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Probability basics

Sample space S: space of all possible outcomes

Event E: a subset of outcomes

Probability: a number in $[0,1]$ we can associate with an outcome or an event

$$p(E) = \sum_{s \in E} p(s)$$

Probability distribution

A function $p: S \rightarrow [0,1]$ that assigns a probability to every possible outcome in S

$$\sum_{s \in S} p(s) = 1$$

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Conditional probability

Definition: Let E and F be two events such that $P(F) > 0$. The **conditional probability** of E given F

- $P(E|F) = P(E \cap F) / P(F)$

Corrolary: Let E and F be two events such that $P(F) > 0$. Then:

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$$= P(F|E)P(E) / (P(F|E)P(E) + P(F|\sim E)P(\sim E))$$

- Expresses **$P(E|F)$** in terms of **$P(F|E)$**
- conditioning event are switched

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Independence

Definition: The two events E and F are said to be **independent** if:

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Random variables

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Expected value

Definition: The **expected value** of the random variable $X(s)$ on the sample space is equal to:

$$E(X) = \sum_{s \in S} p(s) X(s)$$

Example: roll of a dice

- Outcomes: 1 2 3 4 5 6
- Expected value:

$$E(X) = 1 \cdot 1/6 + 2 \cdot 1/6 + 3 \cdot 1/6 + 4 \cdot 1/6 + 5 \cdot 1/6 + 6 \cdot 1/6 = 7/2$$

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Expected value

Investment problem:

- You have 100 dollars and can invest into a stock. The returns are volatile and you may get either \$120 with probability of 0.4, or \$90 with probability 0.6.
- **What is the expected value of your investment?**
-

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Expected value

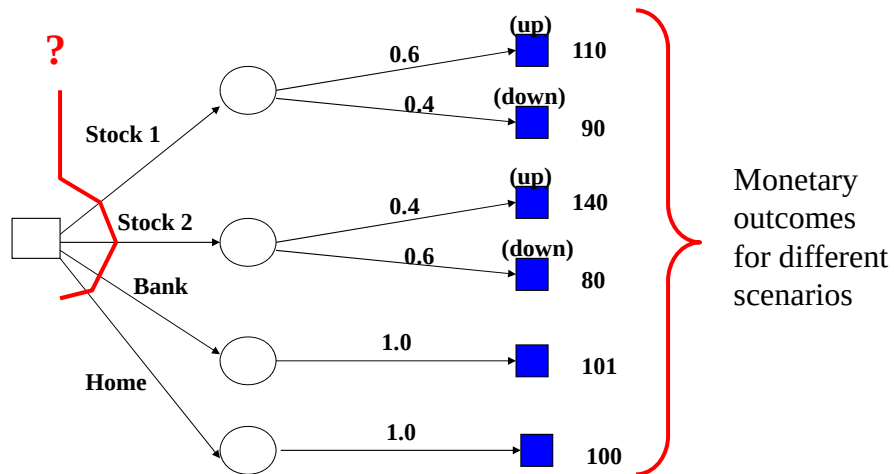
Investment problem:

- You have 100 dollars and can invest into a stock. The returns are volatile and you may get either \$120 with probability of 0.4, or \$90 with probability 0.6.
- **What is the expected value of your investment?**
- $E(X) = 0.4 \cdot 120 + 0.6 \cdot 90 = 48 + 54 = 102$
- **Is it OK to invest?**

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Decision making example

We need to make a choice whether to invest in Stock 1 or 2, put money into bank or keep them at home.

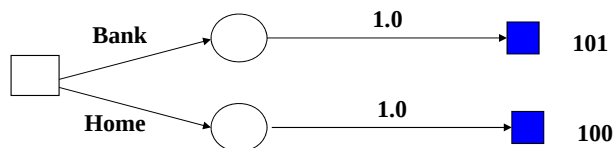


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Decision making example.

Assume the simplified problem with the Bank and Home choices only.

The result is guaranteed – the outcome is deterministic



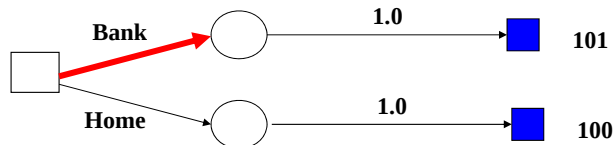
What is the rational choice assuming our goal is to make money?

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Decision making. Deterministic outcome.

Assume the simplified problem with the Bank and Home choices only.

These choices are deterministic.



Our goal is to make money. What is the rational choice?

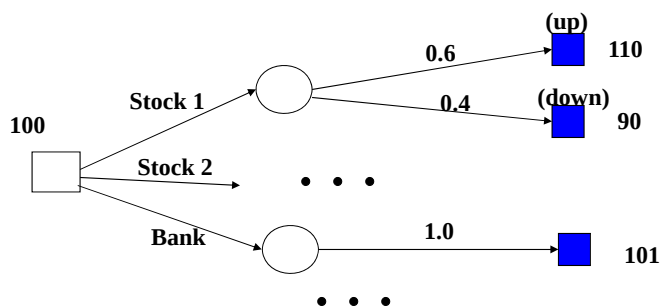
Answer: Put money into the bank. The choice is always strictly better in terms of the outcome

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Decision making

- How to quantify the goodness of the stochastic outcome?

We want to compare it to deterministic and other stochastic outcomes.

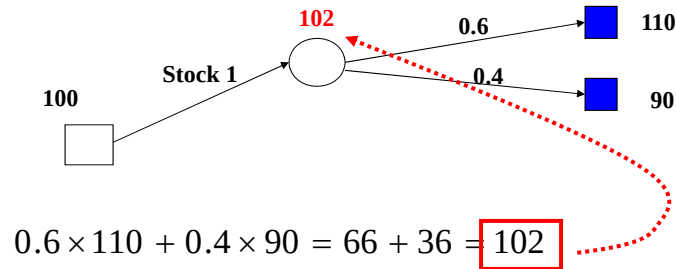


Idea: Use the expected value of the outcome

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Expected value

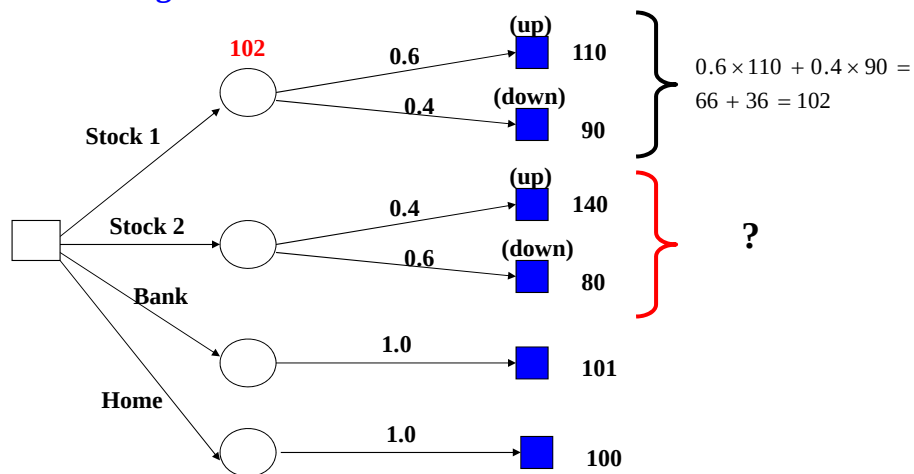
- **Expected value** summarizes all stochastic outcomes into a single quantity
 - Expected value for the outcome of the Stock 1 option is:



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Expected values

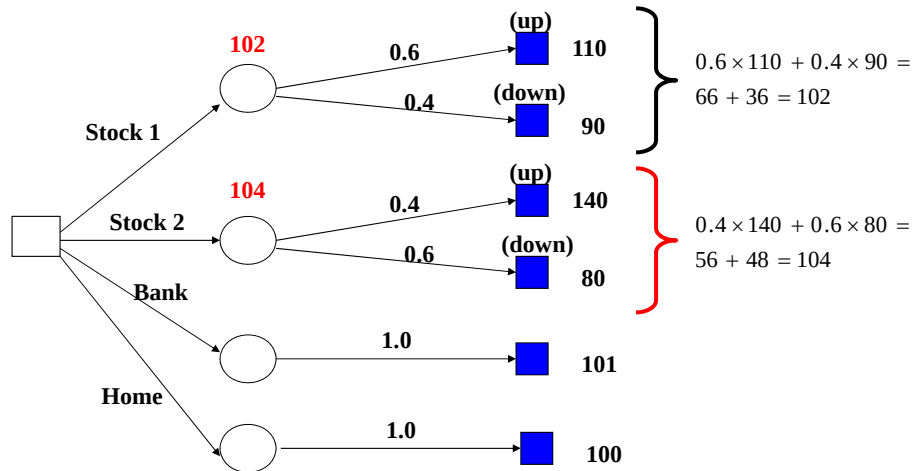
Investing \$100 for 6 months



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Expected values

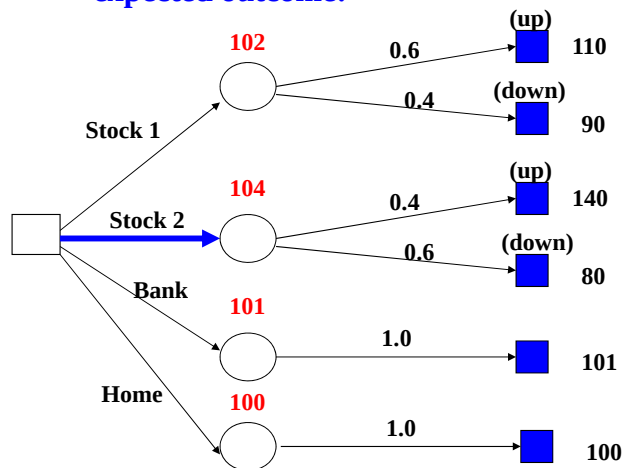
Investing \$100 for 6 months



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Selection based on expected values

The optimal action is the option that maximizes the expected outcome:



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Lecture 21b

Relations

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Cartesian product (review)

Let $A = \{a_1, a_2, \dots, a_k\}$ and $B = \{b_1, b_2, \dots, b_m\}$.

The Cartesian product $A \times B$ is defined by a set of pairs
 $\{(a_1, b_1), (a_1, b_2), \dots, (a_1, b_m), \dots, (a_k, b_m)\}$.

Cartesian product defines a product set, or a set of all ordered arrangements of elements in sets in the Cartesian product.

Binary relation

Definition: Let A and B be two sets. A **binary relation from A to B** is a subset of a Cartesian product $A \times B$.

- Let $R \subseteq A \times B$ means R is a set of ordered pairs of the form (a,b) where $a \in A$ and $b \in B$.
- We use the notation **$a R b$** to denote $(a,b) \in R$ and **$a \not R b$** to denote $(a,b) \notin R$. If **$a R b$** , we say a is related to b by R.

Example: Let $A = \{a,b,c\}$ and $B = \{1,2,3\}$.

- Is $R = \{(a,1), (b,2), (c,2)\}$ a relation from A to B? **Yes.**
- Is $Q = \{(1,a), (2,b)\}$ a relation from A to B? **No.**
- Is $P = \{(a,a), (b,c), (b,a)\}$ a relation from A to A? **Yes**

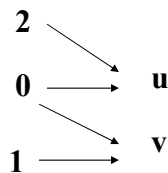
Representing binary relations

- We can graphically represent a binary relation R as follows:
 - if **$a R b$** then draw an arrow from a to b.

$$a \rightarrow b$$

Example:

- Let $A = \{0, 1, 2\}$, $B = \{u,v\}$ and $R = \{(0,u), (0,v), (1,v), (2,u)\}$
- Note: $R \subseteq A \times B$.
- **Graph:**



Representing binary relations

- We can represent a binary relation R by a **table** showing (marking) the ordered pairs of R .

Example:

- Let $A = \{0, 1, 2\}$, $B = \{u, v\}$ and $R = \{(0, u), (0, v), (1, v), (2, u)\}$
- Table:**

R	u	v
0	x	x
1		x
2	x	

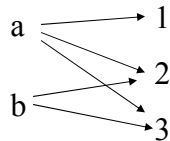
or

R	u	v
0	1	1
1	0	1
2	1	0

Relations and functions

- Relations represent **one to many relationships** between elements in A and B .

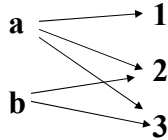
- Example:**



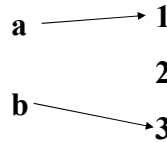
- What is the difference between a **relation** and a **function from A to B**?

Relations and functions

- Relations represent **one to many relationships** between elements in A and B.
- Example:**



- What is the difference between a **relation** and a **function from A to B**? A function defined on sets A,B $A \rightarrow B$ assigns to each element in the domain set A exactly one element from B. So it is a **special relation**.



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Relation on the set

Definition: A relation on the set A is a relation from A to itself.

Example 1:

- Let $A = \{1,2,3,4\}$ and $R_{\text{div}} = \{(a,b) \mid a \text{ divides } b\}$
- What does R_{div} consist of?
- $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

R	1	2	3	4
1	x	x	x	x
2		x		x
3			x	
4				x

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Relation on the set

Example:

- Let $A = \{1, 2, 3, 4\}$.
- Define $a R_{\neq} b$ if and only if $a \neq b$.

$$R_{\neq} = \{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$$

R	1	2	3	4
1		x	x	x
2	x		x	x
3	x	x		x
4	x	x	x	

Binary relations

- Theorem:** The number of binary relations on a set A , where $|A| = n$ is:

$$2^{n^2}$$

- Proof:**

- If $|A| = n$ then the cardinality of the Cartesian product $|A \times A| = n^2$.
- R is a binary relation on A if $R \subseteq A \times A$ (that is, R is a subset of $A \times A$).
- The number of subsets of a set with k elements : 2^k
- The number of subsets of $A \times A$ is : $2^{|A \times A|} = 2^{n^2}$

Binary relations

- **Example:** Let $A = \{1,2\}$
- What is $A \times A = \{(1,1),(1,2),(2,1),(2,2)\}$
- **List of possible relations (subsets of $A \times A$):**

• $\emptyset$		1	}	16
• $\{(1,1)\} \quad \{(1,2)\} \quad \{(2,1)\} \quad \{(2,2)\}$		4		
• $\{(1,1), (1,2)\} \quad \{(1,1), (2,1)\} \quad \{(1,1), (2,2)\}$		6		
• $\{(1,2), (2,1)\} \quad \{(1,2), (2,2)\} \quad \{(2,1), (2,2)\}$				
• $\{(1,1), (1,2), (2,1)\} \quad \{(1,1), (1,2), (2,2)\}$		4		
• $\{(1,1), (2,1), (2,2)\} \quad \{(1,2), (2,1), (2,2)\}$				
• $\{(1,1), (1,2), (2,1), (2,2)\}$		1		
- Use formula: $2^4 = 16$

Properties of relations

Definition (reflexive relation) : A relation R on a set A is called **reflexive** if $(a,a) \in R$ for every element $a \in A$.

Example 1:

- Assume relation $R_{\text{div}} = \{(a,b) \mid a \mid b\}$ on $A = \{1,2,3,4\}$
- **Is R_{div} reflexive?**
- $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$
- **Answer:** Yes. $(1,1), (2,2), (3,3),$ and $(4,4) \in A$.

Reflexive relation

Reflexive relation

- $R_{\text{div}} = \{(a, b), \text{ if } a \mid b\}$ on $A = \{1, 2, 3, 4\}$
- $R_{\text{div}} = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$

$$MR_{\text{div}} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

- **A relation R is reflexive** if and only if MR has 1 in every position on its main diagonal.

Properties of relations

Definition (reflexive relation) : A relation R on a set A is called **reflexive** if $(a, a) \in R$ for every element $a \in A$.

Example 2:

- Relation R_{fun} on $A = \{1, 2, 3, 4\}$ defined as:
 - $R_{\text{fun}} = \{(1, 2), (2, 2), (3, 3)\}$.
- **Is R_{fun} reflexive?**
- **No.** It is not reflexive since $(1, 1) \notin R_{\text{fun}}$.

Properties of relations

Definition (irreflexive relation): A relation R on a set A is called **irreflexive** if $(a,a) \notin R$ for every $a \in A$.

Example 1:

- Assume relation $R_{\neq}$ on $A = \{1,2,3,4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- Is $R_{\neq}$ irreflexive?
- $R_{\neq} = \dots$

Properties of relations

Definition (irreflexive relation): A relation R on a set A is called **irreflexive** if $(a,a) \notin R$ for every $a \in A$.

Example 1:

- Assume relation $R_{\neq}$ on $A = \{1,2,3,4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- Is $R_{\neq}$ irreflexive?
- $R_{\neq} = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}$
- **Answer:** Yes. Because $(1,1), (2,2), (3,3)$ and $(4,4) \notin R_{\neq}$

Irreflexive relation

Irreflexive relation

- $R_{\neq}$ on $A = \{1, 2, 3, 4\}$, such that $\mathbf{a} R_{\neq} \mathbf{b}$ if and only if $a \neq b$.
- $R_{\neq} = \{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$

		0	1	1	1
		1	0	1	1
MR	=	1	1	0	1
		1	1	1	0

- **A relation R is irreflexive** if and only if MR has 0 in every position on its main diagonal.

Properties of relations

Definition (irreflexive relation): A relation R on a set A is called **irreflexive** if $(a, a) \notin R$ for every $a \in A$.

Example 2:

- R_{fun} on $A = \{1, 2, 3, 4\}$ defined as:
 - $R_{\text{fun}} = \{(1, 2), (2, 2), (3, 3)\}$.
- **Is R_{fun} irreflexive?**
- **Answer: No.** Because $(2, 2)$ and $(3, 3) \in R_{\text{fun}}$

Properties of relations

Definition (symmetric relation): A relation R on a set A is called **symmetric** if

$$\forall a, b \in A \quad (a, b) \in R \rightarrow (b, a) \in R.$$

Example 1:

- $R_{\text{div}} = \{(a, b) \mid a \mid b\}$ on $A = \{1, 2, 3, 4\}$
- **Is R_{div} symmetric?**
- $R_{\text{div}} = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$
- **Answer: No.** It is not symmetric since $(1, 2) \in R$ but $(2, 1) \notin R$.

Properties of relations

Definition (symmetric relation): A relation R on a set A is called **symmetric** if

$$\forall a, b \in A \quad (a, b) \in R \rightarrow (b, a) \in R.$$

Example 2:

- $R_{\neq}$ on $A = \{1, 2, 3, 4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- **Is $R_{\neq}$ symmetric?**
- $R_{\neq} = \{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$
- **Answer: Yes.** If $(a, b) \in R_{\neq} \rightarrow (b, a) \in R_{\neq}$

Symmetric relation

Symmetric relation:

- $R_{\neq}$ on $A = \{1, 2, 3, 4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- $R_{\neq} = \{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$

$$MR = \begin{matrix} & \begin{matrix} 0 & 1 & 1 & 1 \end{matrix} \\ \begin{matrix} 1 \\ 1 \\ 1 \\ 1 \end{matrix} & \begin{matrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{matrix} \end{matrix}$$

- A relation R is symmetric if and only if $m_{ij} = m_{ji}$ for all i, j .

Properties of relations

Definition (symmetric relation): A relation R on a set A is called symmetric if

$$\forall a, b \in A \quad (a, b) \in R \rightarrow (b, a) \in R.$$

Example 3:

- Relation R_{fun} on $A = \{1, 2, 3, 4\}$ defined as:
 - $R_{\text{fun}} = \{(1, 2), (2, 2), (3, 3)\}$.
- Is R_{fun} symmetric?
- Answer: No. For $(1, 2) \in R_{\text{fun}}$ there is no $(2, 1) \in R_{\text{fun}}$

CS 441 Discrete Mathematics for CS
Lecture 22

Relations II

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Cartesian product (review)

- Let $A = \{a_1, a_2, \dots, a_k\}$ and $B = \{b_1, b_2, \dots, b_m\}$.
- **The Cartesian product** $A \times B$ is defined by a set of pairs $\{(a_1, b_1), (a_1, b_2), \dots, (a_1, b_m), \dots, (a_k, b_m)\}$.

Example:

Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$. What is $A \times B$?

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

Binary relation

Definition: Let A and B be sets. A **binary relation from A to B** is a subset of a Cartesian product $A \times B$.

Example: Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$.

- $R = \{(a, 1), (b, 2), (c, 2)\}$ is an example of a relation from A to B.

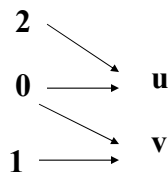
Representing binary relations

- We can graphically represent a binary relation R as follows:
 - if $a R b$ then draw an arrow from a to b.

$$a \rightarrow b$$

Example:

- Let $A = \{0, 1, 2\}$, $B = \{u, v\}$ and $R = \{(0, u), (0, v), (1, v), (2, u)\}$
- Note: $R \subseteq A \times B$.
- **Graph:**



Representing binary relations

- We can represent a binary relation R by a **table** showing (marking) the ordered pairs of R .

Example:

- Let $A = \{0, 1, 2\}$, $B = \{u, v\}$ and $R = \{(0, u), (0, v), (1, v), (2, u)\}$
- **Table:**

R	u	v
0	x	x
1		x
2	x	

or

R	u	v
0	1	1
1	0	1
2	1	0

Properties of relations

Properties of relations on A:

- Reflexive ✓
- Irreflexive ✓
- Symmetric ✓
- **Anti-symmetric**
- **Transitive**

Reflexive relation

Reflexive relation

- $R_{\text{div}} = \{(a, b), \text{ if } a \mid b\}$ on $A = \{1, 2, 3, 4\}$
- $R_{\text{div}} = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$

$$MR_{\text{div}} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

- **A relation R is reflexive** if and only if MR has 1 in every position on its main diagonal.

Irreflexive relation

Irreflexive relation

- $R_{\neq}$ on $A = \{1, 2, 3, 4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- $R_{\neq} = \{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$

$$MR = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

- **A relation R is irreflexive** if and only if MR has 0 in every position on its main diagonal.

Symmetric relation

Symmetric relation:

- $R_{\neq}$ on $A = \{1, 2, 3, 4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- $R_{\neq} = \{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$

$$MR = \begin{matrix} & \begin{matrix} 0 & 1 & 1 & 1 \end{matrix} \\ \begin{matrix} 1 \\ 1 \\ 1 \\ 1 \end{matrix} & \begin{matrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{matrix} \end{matrix}$$

- A relation R is symmetric if and only if $m_{ij} = m_{ji}$ for all i, j .

Anti-symmetric relation

Definition (anti-symmetric relation): A relation on a set A is called **anti-symmetric** if

- $[(a, b) \in R \text{ and } (b, a) \in R] \rightarrow a = b$ where $a, b \in A$.

Example 3:

- Relation R_{fun} on $A = \{1, 2, 3, 4\}$ defined as:
 - $R_{\text{fun}} = \{(1, 2), (2, 2), (3, 3)\}$.
- Is R_{fun} anti-symmetric?
- Answer: Yes. It is anti-symmetric

Anti-symmetric relation

Antisymmetric relation

- relation $R_{\text{fun}} = \{(1,2), (2,2), (3,3)\}$

$$MR_{\text{fun}} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

- A relation is **antisymmetric** if and only if $m_{ij} = 1 \rightarrow m_{ji} = 0$ for $i \neq j$.

Transitive relation

Definition (transitive relation): A relation R on a set A is called **transitive** if

- $[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R$ for all $a, b, c \in A$.

- Example 1:**

- $R_{\text{div}} = \{(a,b) \mid a \mid b\}$ on $A = \{1,2,3,4\}$
- $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$
- Is R_{div} transitive?**
- Answer:**

Transitive relation

Definition (transitive relation): A relation R on a set A is called **transitive** if

- $[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R$ for all $a, b, c \in A$.

- **Example 1:**

- $R_{\text{div}} = \{(a,b) \mid a \mid b\}$ on $A = \{1,2,3,4\}$
- $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$
- **Is R_{div} transitive?**
- **Answer: Yes.**

Transitive relation

Definition (transitive relation): A relation R on a set A is called **transitive** if

- $[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R$ for all $a, b, c \in A$.

- **Example 2:**

- $R_{\neq}$ on $A = \{1,2,3,4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- $R_{\neq} = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}$
- **Is $R_{\neq}$ transitive ?**
- **Answer:**

Transitive relation

Definition (transitive relation): A relation R on a set A is called **transitive** if

- $[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R$ for all $a, b, c \in A$.
- **Example 2:**
- $R_{\neq}$ on $A = \{1,2,3,4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- $R_{\neq} = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}$
- **Is $R_{\neq}$ transitive?**
- **Answer: No.** It is not transitive since $(1,2) \in R$ and $(2,1) \in R$ but $(1,1)$ is not an element of R .

Transitive relations

Definition (transitive relation): A relation R on a set A is called **transitive** if

- $[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R$ for all $a, b, c \in A$.
- **Example 3:**
- Relation R_{fun} on $A = \{1,2,3,4\}$ defined as:
 - $R_{\text{fun}} = \{(1,2), (2,2), (3,3)\}$.
- **Is R_{fun} transitive?**
- **Answer:**

Transitive relations

Definition (transitive relation): A relation R on a set A is called **transitive** if

- $[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R$ for all $a, b, c \in A$.
- **Example 3:**
- Relation R_{fun} on $A = \{1,2,3,4\}$ defined as:
 - $R_{\text{fun}} = \{(1,2), (2,2), (3,3)\}$.
- **Is R_{fun} transitive?**
- **Answer: Yes.** It is transitive.

Combining relations

Definition: Let A and B be sets. A **binary relation from A to B** is a subset of a Cartesian product $A \times B$.

- Let $R \subseteq A \times B$ means R is a set of ordered pairs of the form (a,b) where $a \in A$ and $b \in B$.

Combining Relations

- **Relations are sets $\rightarrow$ combinations via set operations**
- Set operations of: **union, intersection, difference and symmetric difference.**

Combining relations

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

What is:

- $R1 \cup R2 = ?$

Combining relations

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

What is:

- $R1 \cup R2 = \{(1,u), (1,v), (2,u), (2,v), (3,u), (3,v)\}$
- $R1 \cap R2 = ?$

Combining relations

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

What is:

- $R1 \cup R2 = \{(1,u), (1,v), (2,u), (2,v), (3,u), (3,v)\}$
- $R1 \cap R2 = \{(3,u)\}$
- $R1 - R2 = ?$

Combining relations

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

What is:

- $R1 \cup R2 = \{(1,u), (1,v), (2,u), (2,v), (3,u), (3,v)\}$
- $R1 \cap R2 = \{(3,u)\}$
- $R1 - R2 = \{(1,u), (2,u), (2,v)\}$
- $R2 - R1 = ?$

Combining relations

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

What is:

- $R1 \cup R2 = \{(1,u), (1,v), (2,u), (2,v), (3,u), (3,v)\}$
- $R1 \cap R2 = \{(3,u)\}$
- $R1 - R2 = \{(1,u), (2,u), (2,v)\}$
- $R2 - R1 = \{(1,v), (3,v)\}$

Combination of relations

Representation of operations on relations:

- **Question:** Can the relation be formed by taking the union or intersection or composition of two relations $R1$ and $R2$ be represented in terms of matrix operations?
- **Answer: Yes**

Combination of relations: implementation

Definition. The **join**, denoted by $\vee$, of two m-by-n matrices (a_{ij}) and (b_{ij}) of 0s and 1s is an m-by-n matrix (m_{ij}) where

- $m_{ij} = a_{ij} \vee b_{ij}$ for all i,j
= **pairwise or (disjunction)**

- **Example:**

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

- $$\begin{array}{cc} \text{MR1} = & \begin{matrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{matrix} & \text{MR2} = & \begin{matrix} 0 & 1 \\ 0 & 0 \\ 1 & 1 \end{matrix} & M(R1 \vee R2) = & \begin{matrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{matrix} \end{array}$$

Combination of relations: implementation

Definition. The **meet**, denoted by $\wedge$, of two m-by-n matrices (a_{ij}) and (b_{ij}) of 0s and 1s is an m-by-n matrix (m_{ij}) where

- $m_{ij} = a_{ij} \wedge b_{ij}$ for all i,j
= **pairwise and (conjunction)**

- **Example:**

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

- $$\begin{array}{cc} \text{MR1} = & \begin{matrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{matrix} & \text{MR2} = & \begin{matrix} 0 & 1 \\ 0 & 0 \\ 1 & 1 \end{matrix} & \text{MR1} \wedge \text{MR2} = & \begin{matrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{matrix} \end{array}$$

Composite of relations

Definition: Let R be a relation from a set A to a set B and S a relation from B to a set C . The **composite of R and S** is the relation consisting of the ordered pairs (a,c) where $a \in A$ and $c \in C$, and for which there is a $b \in B$ such that $(a,b) \in R$ and $(b,c) \in S$. We denote the composite of R and S by $S \circ R$.

Examples:

- Let $A = \{1,2,3\}$, $B = \{0,1,2\}$ and $C = \{a,b\}$.
- $R = \{(1,0), (1,2), (3,1), (3,2)\}$
- $S = \{(0,b), (1,a), (2,b)\}$
- $S \circ R = ?$

Composite of relations

Definition: Let R be a relation from a set A to a set B and S a relation from B to a set C . The **composite of R and S** is the relation consisting of the ordered pairs (a,c) where $a \in A$ and $c \in C$, and for which there is a $b \in B$ such that $(a,b) \in R$ and $(b,c) \in S$. We denote the composite of R and S by $S \circ R$.

Examples:

- Let $A = \{1,2,3\}$, $B = \{0,1,2\}$ and $C = \{a,b\}$.
- $R = \{(1,0), (1,2), (3,1), (3,2)\}$
- $S = \{(0,b), (1,a), (2,b)\}$
- $S \circ R = \{(1,b), (3,a), (3,b)\}$

Implementation of composite

Definition. The **Boolean product**, denoted by $\odot$, of an m -by- n matrix (a_{ij}) and n -by- p matrix (b_{jk}) of 0s and 1s is an m -by- p matrix (m_{ik}) where

- $m_{ik} = \begin{matrix} 1, & \text{if } a_{ij} = 1 \text{ and } b_{jk} = 1 \text{ for some } k=1,2,\dots,n \\ 0, & \text{otherwise} \end{matrix}$

Examples:

- Let $A = \{1,2,3\}$, $B = \{0,1,2\}$ and $C = \{a,b\}$.
- $R = \{(1,0), (1,2), (3,1), (3,2)\}$
- $S = \{(0,b), (1,a), (2,b)\}$
- $S \circ R = \{(1,b), (3,a), (3,b)\}$

Implementation of composite

Examples:

- Let $A = \{1,2\}$, $B = \{1,2,3\}$ $C = \{a,b\}$
- $R = \{(1,2), (1,3), (2,1)\}$ is a relation from A to B
- $S = \{(1,a), (3,b), (3,a)\}$ is a relation from B to C .
- $S \circ R = \{(1,b), (1,a), (2,a)\}$

$$\begin{array}{ccccc}
 & 0 & 1 & 1 & & 1 & 0 \\
 M_R = & 1 & 0 & 0 & M_S = & 0 & 0 \\
 & & & & & 1 & 1 \\
 M_R \odot M_S & = & ? & & & &
 \end{array}$$

Implementation of composite

Examples:

- Let $A = \{1,2\}$, $\{1,2,3\}$ $C = \{a,b\}$
- $R = \{(1,2),(1,3),(2,1)\}$ is a relation from A to B
- $S = \{(1,a),(3,b),(3,a)\}$ is a relation from B to C.
- $S \circ R = \{(1,b),(1,a),(2,a)\}$

$$\begin{array}{ccc}
 & 0 & 1 & 1 \\
 M_R = & 1 & 0 & 0
 \end{array}
 \quad
 \begin{array}{ccc}
 & 1 & 0 \\
 M_S = & 0 & 0 \\
 & 1 & 1
 \end{array}$$

$$M_R \odot M_S = \begin{array}{cc} x & x \\ x & x \end{array}$$

Implementation of composite

Examples:

- Let $A = \{1,2\}$, $\{1,2,3\}$ $C = \{a,b\}$
- $R = \{(1,2),(1,3),(2,1)\}$ is a relation from A to B
- $S = \{(1,a),(3,b),(3,a)\}$ is a relation from B to C.
- $S \circ R = \{(1,b),(1,a),(2,a)\}$

$$\begin{array}{ccc}
 \boxed{0} & \boxed{1} & \boxed{1} \\
 M_R = & 1 & 0 & 0
 \end{array}
 \quad
 \begin{array}{ccc}
 \boxed{1} & & 0 \\
 M_S = & \boxed{0} & 0 \\
 & \boxed{1} & 1
 \end{array}$$

$$M_R \odot M_S = \begin{array}{cc} \textcolor{red}{1} & x \\ x & x \end{array}$$

Implementation of composite

Examples:

- Let $A = \{1,2\}$, $\{1,2,3\}$ $C = \{a,b\}$
- $R = \{(1,2),(1,3),(2,1)\}$ is a relation from A to B
- $S = \{(1,a),(3,b),(3,a)\}$ is a relation from B to C.
- $S \circ R = \{(1,b),(1,a),(2,a)\}$

$$M_R = \begin{matrix} & \begin{matrix} 0 & 1 & 1 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{matrix} 1 & 0 & 0 \end{matrix} \end{matrix} \quad M_S = \begin{matrix} \begin{matrix} 1 \\ 0 \\ 1 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 1 \end{matrix} \end{matrix}$$

$$M_R \odot M_S = \begin{matrix} & \begin{matrix} 1 & 1 \end{matrix} \\ \begin{matrix} x & x \end{matrix} & \end{matrix}$$

Implementation of composite

Examples:

- Let $A = \{1,2\}$, $\{1,2,3\}$ $C = \{a,b\}$
- $R = \{(1,2),(1,3),(2,1)\}$ is a relation from A to B
- $S = \{(1,a),(3,b),(3,a)\}$ is a relation from B to C.
- $S \circ R = \{(1,b),(1,a),(2,a)\}$

$$M_R = \begin{matrix} & \begin{matrix} 0 & 1 & 1 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{matrix} 1 & 0 & 0 \end{matrix} \end{matrix} \quad M_S = \begin{matrix} \begin{matrix} 1 \\ 0 \\ 1 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 1 \end{matrix} \end{matrix}$$

$$M_R \odot M_S = \begin{matrix} & \begin{matrix} 1 & 1 \end{matrix} \\ \begin{matrix} 1 & x \end{matrix} & \end{matrix}$$

Implementation of composite

Examples:

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- $R = \{(1,2),(1,3),(2,1)\}$ is a relation from A to B
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$$M_{S \circ R} = \quad ?$$

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Implementation of composite

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- Let $A = \{1,2\}$, $\{1,2,3\}$ $C = \{a,b\}$
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Composite of relations

Definition: Let R be a relation on a set A . The **powers R^n** , $n = 1, 2, 3, \dots$ is defined inductively by

$$\bullet R^1 = R \quad \text{and} \quad R^{n+1} = R^n \circ R.$$

Examples

- $R = \{(1,2), (2,3), (2,4), (3,3)\}$ is a relation on $A = \{1, 2, 3, 4\}$.
- $R^1 = ?$

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- $R^1 = R = \{(1,2), (2,3), (2,4), (3,3)\}$
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- $R^4 = \{(1,3), (2,3), (3,3)\}$
- $R^k = ?$, $k > 3$.

Composite of relations

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- $R^4 = \{(1,3), (2,3), (3,3)\}$
- $R^k = R^3$, $k > 3$.

Transitive relation

Definition (transitive relation): A relation R on a set A is called **transitive** if

- $[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R$ for all $a, b, c \in A$.

- **Example 1:**

- $R_{\text{div}} = \{(a,b), \text{ if } a \mid b\}$ on $A = \{1,2,3,4\}$
- $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$
- **Is R_{div} transitive?**
- **Answer: ?**

Transitive relation

Definition (transitive relation): A relation R on a set A is called **transitive** if

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- **Is R_{div} transitive?**
- **Answer: Yes.**

Connection to R^n

Theorem: The relation R on a set A is transitive if and only if
 $R^n \subseteq R$ for $n = 1, 2, 3, \dots$.

Proof: biconditional (if and only if)

($\Leftarrow$) Suppose $R^n \subseteq R$, for $n = 1, 2, 3, \dots$.

- Let $(a, b) \in R$ and $(b, c) \in R$
- by the definition of $R \circ R$, $(a, c) \in R \circ R = R^2 \subseteq R$
- Therefore R is transitive.

Connection to R^n

Theorem: The relation R on a set A is transitive if and only if $R^n \subseteq R$
for $n = 1, 2, 3, \dots$.

Proof: biconditional (if and only if)

($\Rightarrow$) Suppose R is transitive. Show $R^n \subseteq R$, for $n = 1, 2, 3, \dots$.

- Let $P(n) : R^n \subseteq R$. Mathematical induction.
- **Basis Step:**

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- Let $P(n) : R^n \subseteq R$. Mathematical induction.
- **Basis Step:** $P(1)$ says $R^1 = R$ so, $R^1 \subseteq R$ is true.

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- **Inductive Step:** show $P(n) \rightarrow P(n+1)$
- Want to show if $R^n \subseteq R$ then $R^{n+1} \subseteq R$.

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- **Basis Step:** $P(1)$ says $R^1 = R$ so, $R^1 \subseteq R$ is true.
- **Inductive Step:** show $P(n) \rightarrow P(n+1)$
- Want to show if $R^n \subseteq R$ then $R^{n+1} \subseteq R$.
- Let $(a, b) \in R^{n+1}$ then by the definition of $R^{n+1} = R^n \circ R$ there is an element $x \in A$ so that $(a, x) \in R$ and $(x, b) \in R^n \subseteq R$ (inductive hypothesis). In addition to $(a, x) \in R$ and $(x, b) \in R$, R is transitive; so $(a, b) \in R$.
- Therefore, $R^{n+1} \subseteq R$.

Number of reflexive relations

Theorem: The number of reflexive relations on a set A , where $|A| = n$ is: $2^{n(n-1)}$.

Proof:

- A reflexive relation R on A **must contain** all pairs (a, a) where $a \in A$.
- All other pairs in R are of the form (a, b) , $a \neq b$, such that $a, b \in A$.
- How many of these pairs are there? Answer: $n(n-1)$.
- How many subsets on $n(n-1)$ elements are there?
- **Answer:** $2^{n(n-1)}$.

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Lecture 23

Relations III.

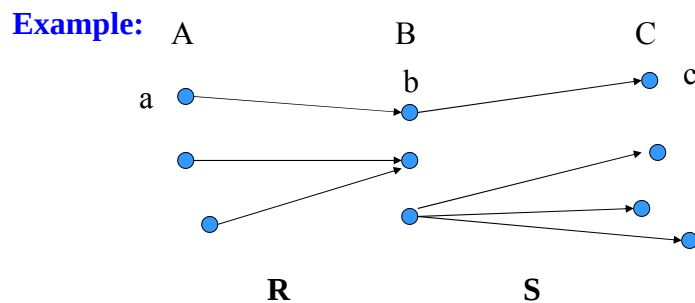
Milos Hauskrecht
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5329 Sennott Square

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Composite of relations

Definition: Let R be a relation from a set A to a set B and S a relation from B to a set C . The **composite of R and S** is the relation consisting of the ordered pairs (a,c) where $a \in A$ and $c \in C$, and for which there is a $b \in B$ such that $(a,b) \in R$ and $(b,c) \in S$. We denote the composite of R and S by $S \circ R$.

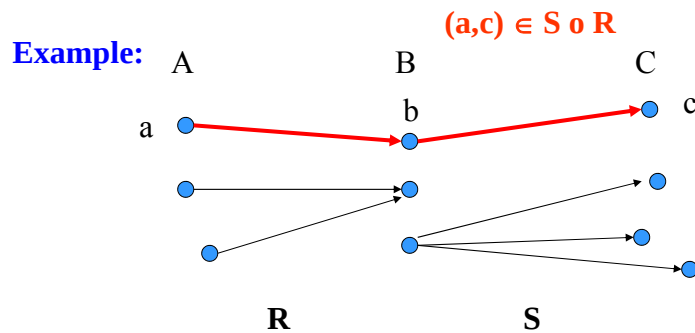


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Examples:

- Let $A = \{1,2,3\}$, $B = \{0,1,2\}$ and $C = \{a,b\}$.
- $R = \{(1,0), (1,2), (3,1), (3,2)\}$
- $S = \{(0,b), (1,a), (2,b)\}$
- $S \circ R = ?$

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Example:

- Let $A = \{1,2,3\}$, $B = \{0,1,2\}$ and $C = \{a,b\}$.
- $R = \{(1,0), (1,2), (3,1), (3,2)\}$
- $S = \{(0,b), (1,a), (2,b)\}$
- $S \circ R = \{(1,b), (3,a), (3,b)\}$

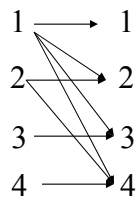
Representing binary relations with graphs

- We can graphically represent a binary relation R from A to B as follows:
 - if **$a R b$** then draw an arrow from a to b .

$$a \rightarrow b$$

Example:

- Relation R_{div} (from previous lectures) on $A = \{1,2,3,4\}$
- $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

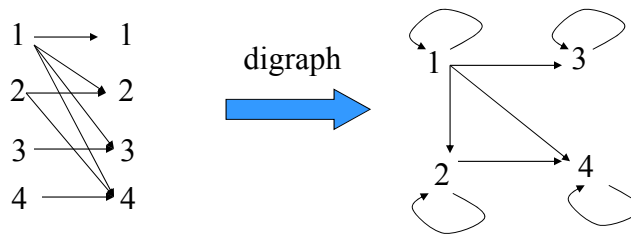


Representing relations on a set with digraphs

Definition: A **directed graph or digraph** consists of a set of vertices (or nodes) together with a set E of ordered pairs of elements of V called edges (or arcs). The vertex a is called the initial vertex of the edge (a,b) and vertex b is the terminal vertex of this edge. An edge of the form (a,a) is called a **loop**.

Example

- Relation $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$



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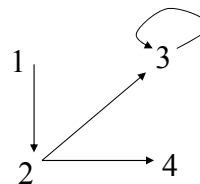
Powers of R

Definition: Let R be a relation on a set A . The **powers R^n** , $n = 1, 2, 3, \dots$ is defined inductively by

- $R^1 = R$ and $R^{n+1} = R^n \circ R$.

Examples

- $R = \{(1,2), (2,3), (2,4), (3,3)\}$ is a relation on $A = \{1, 2, 3, 4\}$.



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Powers of R

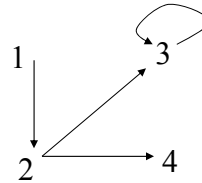
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Examples

- $R = \{(1,2), (2,3), (2,4), (3,3)\}$ is a relation on $A = \{1, 2, 3, 4\}$.

- $R^1 = R = \{(1,2), (2,3), (2,4), (3,3)\}$
- $R^2 = \{(1,3), (1,4), (2,3), (3,3)\}$
- What does R^2 represent?



Powers of R

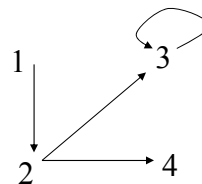
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- $R^2 = \{(1,3), (1,4), (2,3), (3,3)\}$
- What does R^2 represent?
- Paths of length 2



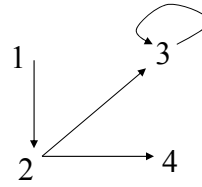
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Examples

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- $R^1 = R = \{(1,2), (2,3), (2,4), (3,3)\}$
- $R^2 = \{(1,3), (1,4), (2,3), (3,3)\}$
- What does R^2 represent?
- Paths of length 2
- $R^3 = \{(1,3), (2,3), (3,3)\}$

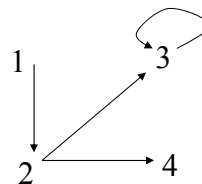
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Examples

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- $R^1 = R = \{(1,2), (2,3), (2,4), (3,3)\}$
- $R^2 = \{(1,3), (1,4), (2,3), (3,3)\}$
- What does R^2 represent?
- Paths of length 2
- $R^3 = \{(1,3), (2,3), (3,3)\}$ path of length 3

Powers of R

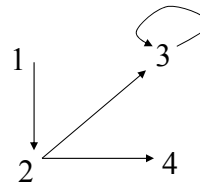
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Powers of R

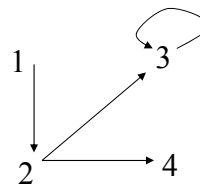
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- $R^4 = \{(1,3), (2,3), (3,3)\}$
- $R^k = \{(1,3), (2,3), (3,3)\} \quad k > 3$



Transitive relation and R^n

Theorem: The relation R on a set A is transitive if and only if
 $R^n \subseteq R$ for $n = 1, 2, 3, \dots$.

Proof: bi-conditional (if and only if)

Proved last lecture

Closures of relations

- Let $R = \{(1,1), (1,2), (2,1), (3,2)\}$ on $A = \{1, 2, 3\}$.
- Is this relation reflexive?
- Answer: ?

Closures of relations

- Let $R = \{(1,1), (1,2), (2,1), (3,2)\}$ on $A = \{1, 2, 3\}$.
- Is this relation reflexive?
- Answer: **No**. Why?

Closures of relations

- Let $R = \{(1,1), (1,2), (2,1), (3,2)\}$ on $A = \{1, 2, 3\}$.
- Is this relation reflexive?
- Answer: **No**. Why?
- **$(2,2)$ and $(3,3)$ is not in R .**
- The question is what is **the minimal relation $S \supseteq R$** that is reflexive?
- How to make R reflexive with minimum number of additions?
- Answer: ?

Closures of relations

- Let $R = \{(1,1), (1,2), (2,1), (3,2)\}$ on $A = \{1, 2, 3\}$.
- Is this relation reflexive?
- Answer: **No**. Why?
- **(2,2) and (3,3) is not in R.**
- The question is what is **the minimal relation $S \supseteq R$** that is reflexive?
- How to make R reflexive with minimum number of additions?
- **Answer:** Add (2,2) and (3,3)
 - Then $S = \{(1,1), (1,2), (2,1), (3,2), (2,2), (3,3)\}$
 - $R \subseteq S$
 - The minimal set $S \supseteq R$ is called **the reflexive closure of R**

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Reflexive closure

The set S is called **the reflexive closure of R** if it:

- contains R
- has reflexive property
- is contained in every reflexive relation Q that contains R ($R \subseteq Q$), that is $S \subseteq Q$

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Closures on relations

- Relations can have different properties:
 - reflexive,
 - symmetric
 - transitive
- Because of that we define:
 - symmetric,
 - reflexive and
 - transitiveclosures.

Closures

Definition: Let R be a relation on a set A . A relation S on A with property P is called **the closure of R with respect to P** if S is a subset of every relation Q ($S \subseteq Q$) with property P that contains R ($R \subseteq Q$).

Closures

Definition: Let R be a relation on a set A . A relation S on A with property P is called **the closure of R with respect to P** if S is a subset of every relation Q ($S \subseteq Q$) with property P that contains R ($R \subseteq Q$).

Example (symmetric closure):

- Assume $R = \{(1,2), (1,3), (2,2)\}$ on $A = \{1,2,3\}$.
- What is the symmetric closure S of R ?
- $S = ?$

Closures

Definition: Let R be a relation on a set A . A relation S on A with property P is called **the closure of R with respect to P** if S is a subset of every relation Q ($S \subseteq Q$) with property P that contains R ($R \subseteq Q$).

Example (a symmetric closure):

- Assume $R = \{(1,2), (1,3), (2,2)\}$ on $A = \{1,2,3\}$.
- What is the symmetric closure S of R ?
- $S = \{(1,2), (1,3), (2,2)\} \cup \{(2,1), (3,1)\}$
 $= \{(1,2), (1,3), (2,2), (2,1), (3,1)\}$

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- Assume $R = \{(1,2), (2,2), (2,3)\}$ on $A = \{1,2,3\}$.
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- **How to make it transitive?**
- **$S = ?$**

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- **How to make it transitive?**
- $S = \{(1,2), (2,2), (2,3)\} \cup \{(1,3)\}$
 $= \{(1,2), (2,2), (2,3), (1,3)\}$
- S is the transitive closure of R

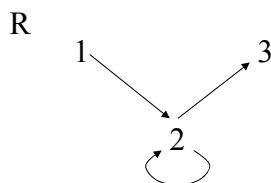
Transitive closure

We can represent the relation on the graph. Finding a transitive closure corresponds to finding all pairs of elements that are connected with a directed path (or digraph).

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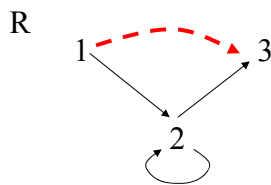
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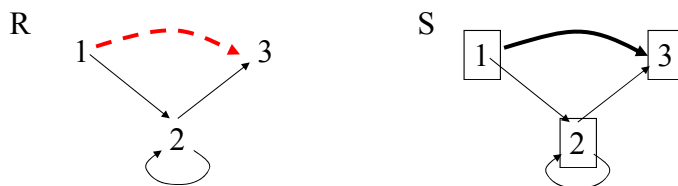
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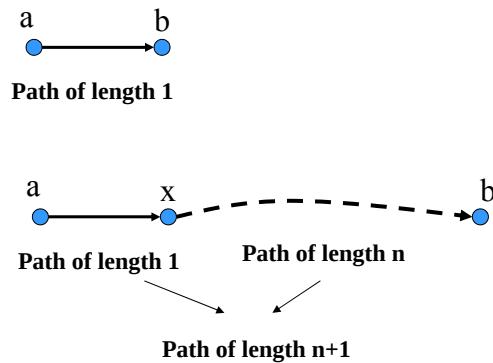
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Path length

Theorem: Let R be a relation on a set A . There is a path of length n from a to b if and only if $(a,b) \in R^n$.

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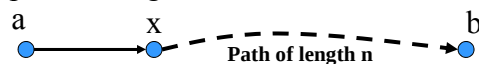
- **P(1):** There is a path of length 1 from a to b if and only if $(a,b) \in R^1$, by the definition of R .
- **Show $P(n) \rightarrow P(n+1)$:** Assume there is a path of length n from a to b if and only if $(a,b) \in R^n \rightarrow$ there is a path of length $n+1$ from a to b if and only if $(a,b) \in R^{n+1}$.

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- There is a path of length $n+1$ from a to b if and only if there exists an $x \in A$, such that $(a,x) \in R$ (a path of length 1) and $(x,b) \in R^n$ is a path of length n from x to b .



- $(x,b) \in R^n$ holds due to $P(n)$. Therefore, there is a path of length $n+1$ from a to b . This also implies that $(a,b) \in R^{n+1}$.

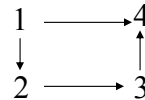
Connectivity relation

Definition: Let R be a relation on a set A . The **connectivity relation** R^* consists of all pairs (a,b) such that there is a path (of any length, ie. 1 or 2 or 3 or ...) between a and b in R .

$$R^* = \bigcup_{k=1}^{\infty} R^k$$

Example:

- $A = \{1,2,3,4\}$
- $R = \{(1,2),(1,4),(2,3),(3,4)\}$



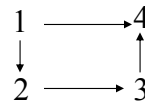
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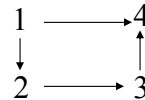
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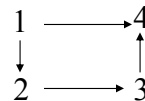
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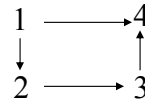
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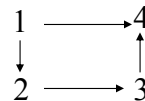
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Connectivity

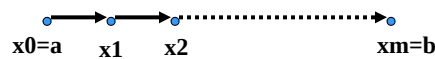
Lemma 1: Let A be a set with n elements, and R a relation on A .

If there is a path from a to b , then there exists a path of length $< n$ in between (a,b) . Consequently:

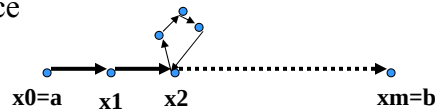
$$R^* = \bigcup_{k=1}^n R^k$$

Proof (intuition):

- There are at most n different elements we can visit on a path if the path does not have loops



- Loops may increase the length but the same node is visited more than once



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Connectivity

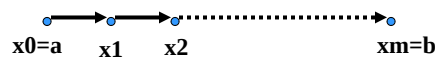
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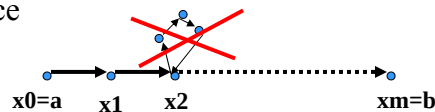
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Transitivity closure and connectivity relation

Theorem: The transitive closure of a relation R **equals** the connectivity relation R^* .

Based on the **Lemma 1**.

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Equivalence relation

Definition: A relation R on a set A is called an **equivalence relation** if it is reflexive, symmetric and transitive.

Example: Let $A = \{0,1,2,3,4,5,6\}$ and

- $R = \{(a,b) \mid a,b \in A, a \equiv b \pmod{3}\}$ (a is congruent to b modulo 3)

Congruencies:

- $0 \pmod{3} = ?$

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- $4 \pmod{3} = ?$

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Relation R has the following pairs:

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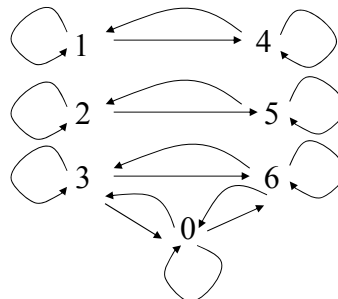
- $(0,0) \quad (0,3), (3,0), (0,6), (6,0)$
- $(3,3), (3,6), (6,3), (6,6) \quad (1,1), (1,4), (4,1), (4,4)$
- $(2,2), (2,5), (5,2), (5,5)$

Equivalence relation

- Relation R on $A = \{0,1,2,3,4,5,6\}$ has the following pairs:**

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- Is R reflexive?

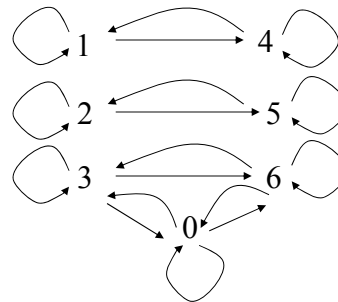


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- Is R reflexive? **Yes.**
- Is R symmetric?



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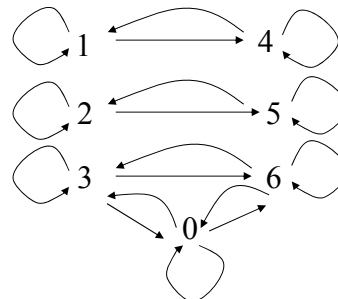
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- Is R reflexive? **Yes.**
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- Is R transitive?



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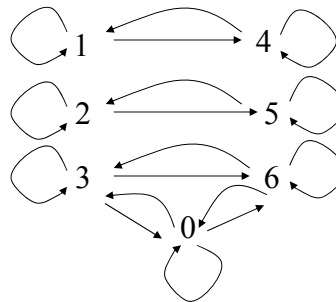
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- Is R reflexive? **Yes.**
- Is R symmetric? **Yes.**
- Is R transitive. **Yes.**

Then

- **R is an equivalence relation.**



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Equivalence class

Theorem: Let R be an **equivalence relation** on a set A. The following statements are equivalent:

- i) $a R b$
- ii) $[a] = [b]$
- iii) $[a] \cap [b] \neq \emptyset$.

Proof: (iii) $\rightarrow$ (i)

- Suppose $[a] \cap [b] \neq \emptyset$, want to show $a R b$.
- $[a] \cap [b] \neq \emptyset \rightarrow x \in [a] \cap [b] \rightarrow x \in [a]$ and $x \in [b] \rightarrow (a,x)$ and $(b,x) \in R$.
- Since R is symmetric $(x,b) \in R$. By the transitivity of R $(a,x) \in R$ and $(x,b) \in R$ implies $(a,b) \in R \rightarrow a R b$.

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Lecture 24

Relations IV

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Equivalence relation

Definition: A relation R on a set A is called an **equivalence relation** if it is reflexive, symmetric and transitive.

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Congruencies:

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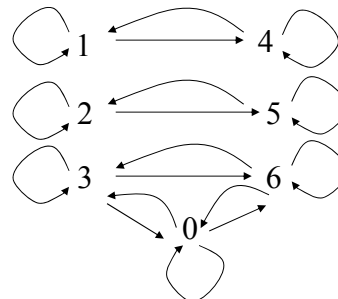
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(0,0)	(0,3), (3,0), (0,6), (6,0)
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- Is R reflexive? **Yes.**
- Is R symmetric? **Yes.**
- Is R transitive. **Yes.**

Then

- **R is an equivalence relation.**



Equivalence class

Definition: Let R be an equivalence relation on a set A . The set $\{x \in A \mid a R x\}$ is called **the equivalence class of a** , denoted by $[a]_R$ or simply $[a]$. If $b \in [a]$ then b is called **a representative of this equivalence class**.

Example:

- Assume $R = \{(a,b) \mid a \equiv b \pmod{3}\}$ for $A = \{0,1,2,3,4,5,6\}$
 $R = \{(0,0), (0,3), (3,0), (0,6), (6,0), (3,3), (3,6), (6,3), (6,6), (1,1), (1,4), (4,1), (4,4), (2,2), (2,5), (5,2), (5,5)\}$
- **Pick an element $a = 0$.**
- $[0]_R = \{0,3,6\}$
- Element 1: $[1]_R = \{1,4\}$
- Element 2: $[2]_R = \{2,5\}$
- Element 3: $[3]_R = \{0,3,6\} = [0]_R = [6]_R$
- Element 4: $[4]_R = \{1,4\} = [1]_R$ Element 5: $[5]_R = \{2,5\} = [2]_R$

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Equivalence class

Example:

- Assume $R = \{(a,b) \mid a \equiv b \pmod{3}\}$ for $A = \{0,1,2,3,4,5,6\}$
- $R = \{(0,0), (0,3), (3,0), (0,6), (6,0), (3,3), (3,6), (6,3), (6,6), (1,1), (1,4), (4,1), (4,4), (2,2), (2,5), (5,2), (5,5)\}$

Three different equivalence classes all together:

- $[0]_R = [3]_R = [6]_R = \{0,3,6\}$
- $[1]_R = [4]_R = \{1,4\}$
- $[2]_R = [5]_R = \{2,5\}$

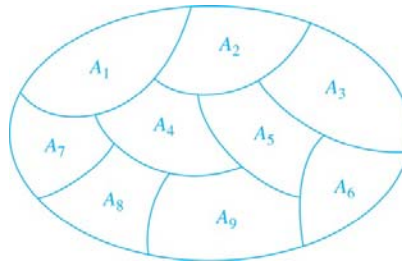
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Partition of a set S

Definition: Let S be a set. A collection of nonempty subsets of S $A_1, A_2, \dots, A_k$ is called **a partition of S** if:

- $A_i \cap A_j = \emptyset, i \neq j$ and $S = \bigcup_{i=1}^k A_i$



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Example: Let $S = \{1, 2, 3, 4, 5, 6\}$ and

- $A_1 = \{0, 3, 6\}$ $A_2 = \{1, 4\}$ $A_3 = \{2, 5\}$
- Is A_1, A_2, A_3 a partition of S ? **Yes.**
- Give a partition of S ?
- $\{0, 2, 4, 6\}$ $\{1, 3, 5\}$
- $\{0\}$ $\{1, 2\}$ $\{3, 4, 5\}$ $\{6\}$

Equivalence classes and partitions

Theorem: Let R be an equivalence relation on a set A . Then the union of all the equivalence classes of R is A :

$$A = \bigcup_{a \in A} [a]_R$$

Proof: an element a of A is in its own equivalence class $[a]_R$ so union cover A .

Theorem: The equivalence classes form a partition of A .

Proof: The equivalence classes split A into disjoint subsets.

Theorem : Let $\{A_1, A_2, \dots, A_i, \dots\}$ be a partitioning of S . Then there is an equivalence relation R on S , that has the sets A_i as its equivalence classes.

Partial orderings

Definition: A relation R on a set S is called a *partial ordering*, or *partial order*, if it is reflexive, antisymmetric, and transitive. A set together with a partial ordering R is called a *partially ordered set*, or *poset*, and is denoted by (S, R) . Members of S are called *elements* of the poset.

Example: Assume R denotes the “greater than or equal” relation $(\geq)$ on the set $S = \{1, 2, 3, 4, 5\}$.

- Is the relation reflexive? Yes
- Is it antisymmetric? Yes
- Is it transitive? Yes
- **Conclusion:** R is a partial ordering.

Partial orderings

Example: Assume R is the divisibility relation ($|$) on the set of integers $S = \{1, 2, 3, 4, 5, 6\}$

- Is the relation reflexive? Yes
- Is it antisymmetric? Yes
- Is it transitive? Yes
- **Conclusion:** R is a partial ordering.

Comparability

Definition 1: The elements a and b of a poset $(S, \preceq)$ are *comparable* if either $a \preceq b$ or $b \preceq a$. When a and b are elements of S so that neither $a \preceq b$ nor $b \preceq a$ holds, then a and b are called *incomparable*.

Definition 2: If $(S, \preceq)$ is a poset and every two elements of S are comparable, S is called a *totally ordered* or *linearly ordered set*, and $\preceq$ is called a *total order* or a *linear order*. A totally ordered set is also called a *chain*.

Definition 3: $(S, \preceq)$ is a well-ordered set if it is a poset such that $\preceq$ is a total ordering and every nonempty subset of S has a least element.

Lexicographical ordering

Definition: Given two posets $(A_1, \preceq_1)$ and $(A_2, \preceq_2)$, the *lexicographic ordering* on $A_1 \times A_2$ is defined by specifying that (a_1, a_2) is less than (b_1, b_2) , that is, $(a_1, a_2) < (b_1, b_2)$, either if $a_1 <_1 b_1$ or if $a_1 = b_1$ then $a_2 <_2 b_2$.

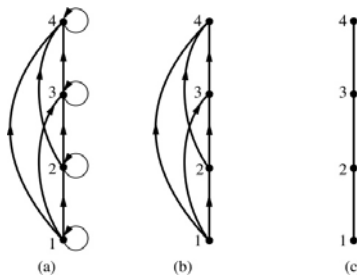
The definition can be extended to a lexicographic ordering on strings

Example: Consider strings of lowercase English letters. A lexicographic ordering can be defined using the ordering of the letters in the alphabet. This is the same ordering as that used in dictionaries.

- *discreet* < *discrete*, because these strings differ in the seventh position and $e < t$.
- *discreet* < *discreetness*, because the first eight letters agree, but the second string is longer.

Hasse diagram

Definition: A *Hasse diagram* is a visual representation of a partial ordering that leaves out edges that must be present because of the reflexive and transitive properties.



- (a) A partial ordering. The loops are due to the reflexive property
- (b) The edges that must be present due to the transitive property are deleted
- (c) The Hasse diagram for the partial ordering (a).

Procedure for constructing Hasse diagram

- To represent a finite poset $(S, \preceq)$ using a Hasse diagram, start with the directed graph of the relation:
 - Remove the loops (a, a) present at every vertex due to the reflexive property.
 - Remove all edges (x, y) for which there is an element $z \in S$ such that $x \prec z$ and $z \prec y$. These are the edges that must be present due to the transitive property.
 - Arrange each edge so that its initial vertex is below the terminal vertex. Remove all the arrows, because all edges point upwards toward their terminal vertex.

CS 441 Discrete Mathematics for CS Lecture 24b

Graphs (chapter 10)

Milos Hauskrecht

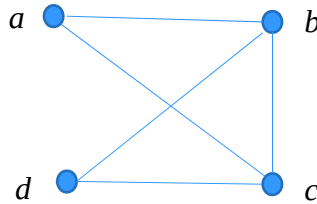
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Definition of a graph

- **Definition:** A graph $G = (V, E)$ consists of a nonempty set V of vertices (or nodes) and a set E of edges. Each edge has either one or two vertices associated with it, called its *endpoints*. An edge is said to *connect* its endpoints.

- **Example:**

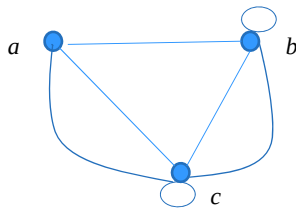


Terminology

- In a **simple graph** each edge connects two different vertices and no two edges connect the same pair of vertices.
- **Multigraphs** may have multiple edges connecting the same two vertices. When m different edges connect the vertices u and v , we say that $\{u,v\}$ is an edge of *multiplicity* m .
- An edge that connects a vertex to itself is called a **loop**.
- A **pseudograph** may include loops, as well as multiple edges connecting the same pair of vertices.

Terminology

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Directed graph

Definition: An *directed graph* (or *digraph*) $G = (V, E)$ consists of a nonempty set V of vertices (or *nodes*) and a set E of *directed edges* (or *arcs*). Each edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u,v) is said to *start at* u and *end at* v .

Remark:

- Graphs where the end points of an edge are not ordered are said to be *undirected graphs*.

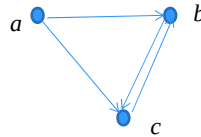
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Directed graph

- A *simple directed graph* has no loops and no multiple edges.

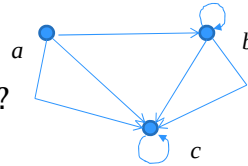
Example:



- A *directed multigraph* may have multiple directed edges. When there are m directed edges from the vertex u to the vertex v , we say that (u,v) is an edge of *multiplicity* m .

Example:

- multiplicity of (a,b) is ?
- and the multiplicity of (b,c) is ?



CS 441 Discrete Mathematics for CS
Lecture 25

Graphs

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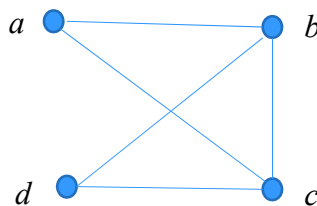
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- **Example:**



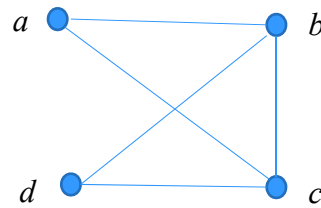
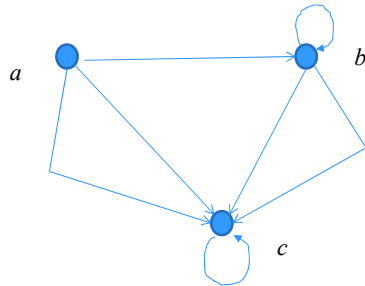
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Graphs: basics

Basic types of graphs:

- **Directed graphs**
- **Undirected graphs**



Terminology

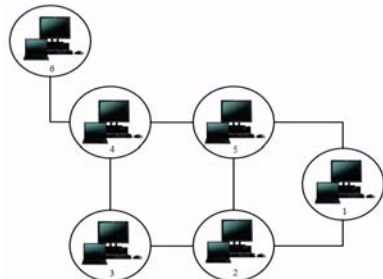
- In a *simple graph* each edge connects two different vertices and no two edges connect the same pair of vertices.
- *Multigraphs* may have multiple edges connecting the same two vertices. When m different edges connect the vertices u and v , we say that $\{u,v\}$ is an edge of *multiplicity* m .
- An edge that connects a vertex to itself is called a *loop*.
- A *pseudograph* may include loops, as well as multiple edges connecting the same pair of vertices.

Graphs

- **Graphs and graph theory can be used to model:**
 - Computer networks
 - Social networks
 - Communications networks
 - Information networks
 - Software design
 - Transportation networks
 - Biological networks

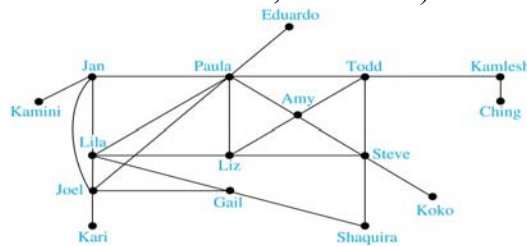
Graphs

- **Computer networks:**
 - **Nodes** – computers
 - **Edges** - connections



Graph models

- Graphs can be used to model social structures based on different kinds of relationships between people or groups.
- **Social network**, vertices represent individuals or organizations and edges represent relationships between them.
- Useful graph models of social networks include:
 - **friendship graphs** - undirected graphs where two people are connected if they are friends (in the real world, on Facebook, or in a particular virtual world, and so on.)

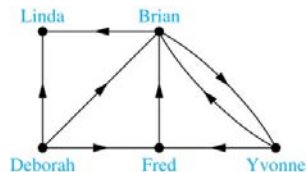


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Graph models

- Useful graph models of social networks include:
 - **influence graphs** - directed graphs where there is an edge from one person to another if the first person can influence the second person



- **collaboration graphs** - undirected graphs where two people are connected if they collaborate in a specific way

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Collaboration graphs

- The **Hollywood graph** models the collaboration of actors in films.
 - We represent actors by vertices and we connect two vertices if the actors they represent have appeared in the same movie.
 - Kevin Bacon numbers.
- An **academic collaboration graph** models the collaboration of researchers who have jointly written a paper in a particular subject.
 - We represent researchers in a particular academic discipline using vertices.
 - We connect the vertices representing two researchers in this discipline if they are coauthors of a paper.
 - *Erdős number*.

Information graphs

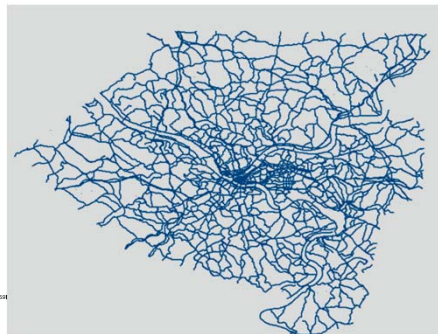
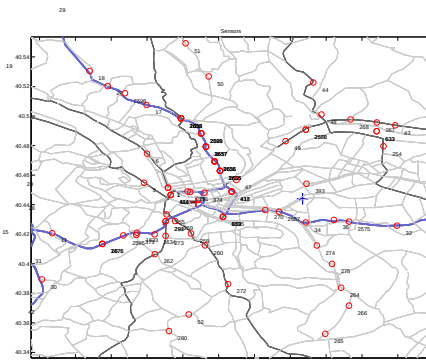
- Graphs can be used to model different types of networks that link different types of information.
- In a **web graph**, web pages are represented by vertices and links are represented by directed edges.
 - A web graph models the web at a particular time.
- In a **citation network**:
 - Research papers in a particular discipline are represented by vertices.
 - When a paper cites a second paper as a reference, there is an edge from the vertex representing this paper to the vertex representing the second paper.

Transportation graphs

- Graph models are extensively used in the study of transportation networks.
- **Airline networks** modeled using directed multigraphs:
 - airports are represented by vertices
 - each flight is represented by a directed edge from the vertex representing the departure airport to the vertex representing the destination airport
- **Road networks** can be modeled using graphs where
 - vertices represent intersections and edges represent roads.
 - undirected edges represent two-way roads and directed edges represent one-way roads.

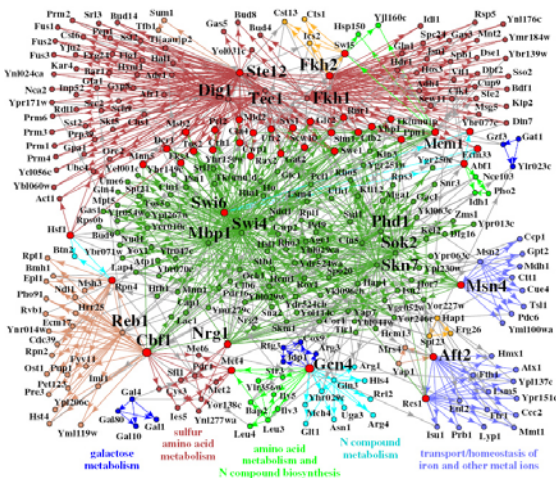
Transportation graphs

- Graph models are extensively used in the study of transportation networks.



Graphs

- **Biological networks:**



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Graph characteristics: Undirected graphs

Definition 1. Two vertices u, v in an undirected graph G are called **adjacent (or neighbors)** in G if there is an edge e between u and v . Such an edge e is called *incident with* the vertices u and v and e is said to *connect* u and v .

Definition 2. The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called **the neighborhood of v** . If A is a subset of V , we denote by $N(A)$ the set of all vertices in G that are adjacent to at least one vertex in A . So,

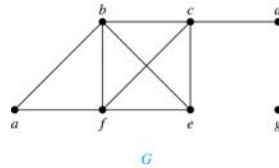
Definition 3. The **degree of a vertex in a undirected graph** is the number of edges incident with it, except that a loop at a vertex contributes two to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

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Undirected graphs

Example: What are the degrees and neighborhoods of the vertices in the graphs G ?



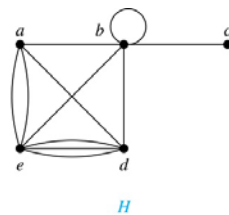
Solution:

G : $\deg(a) = 2$, $\deg(b) = \deg(c) = \deg(f) = 4$, $\deg(d) = 1$,
 $\deg(e) = 3$, $\deg(g) = 0$.

$N(a) = \{b, f\}$, $N(b) = \{a, c, e, f\}$, $N(c) = \{b, d, e, f\}$,
 $N(d) = \{c\}$, $N(e) = \{b, c, f\}$, $N(f) = \{a, b, c, e\}$, $N(g) = \emptyset$.

Undirected graphs

Example: What are the degrees and neighborhoods of the vertices in the graphs H ?



Solution:

H : $\deg(a) = 4$, $\deg(b) = \deg(e) = 6$, $\deg(c) = 1$, $\deg(d) = 5$.

$N(a) = \{b, d, e\}$, $N(b) = \{a, b, c, d, e\}$, $N(c) = \{b\}$,
 $N(d) = \{a, b, e\}$, $N(e) = \{a, b, d\}$.

Undirected graphs

Theorem 1 (Handshaking Theorem): If $G = (V, E)$ is an undirected graph with m edges, then

$$2m = \sum_{v \in V} \deg(v)$$

Proof:

Each edge contributes twice to the degree count of all vertices. Hence, both the left-hand and right-hand sides of this equation equal twice the number of edges.

Think about the graph where vertices represent the people at a party and an edge connects two people who have shaken hands.

Undirected graphs

Theorem 2: An undirected graph has an even number of vertices of odd degree.

Proof: Let V_1 be the vertices of even degree and V_2 be the vertices of odd degree in an undirected graph $G = (V, E)$ with m edges. Then

$$2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

must be even since $\deg(v)$ is even for each $v \in V_1$

This sum must be even because $2m$ is even and the sum of the degrees of the vertices of even degrees is also even. Because this is the sum of the degrees of all vertices of odd degree in the graph, there must be an even number of such vertices.

Directed graphs

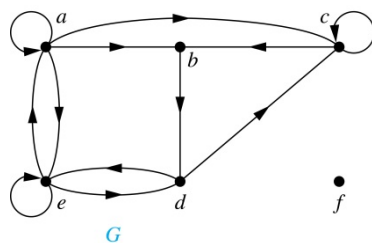
Definition: An **directed graph** $G = (V, E)$ consists of V , a nonempty set of *vertices* (or *nodes*), and E , a set of *directed edges* or *arcs*. Each edge is an ordered pair of vertices. The directed edge (u, v) is said to start at u and end at v .

Definition: Let (u, v) be an edge in G . Then u is the *initial vertex* of this edge and is *adjacent to* v and v is the *terminal* (or *end*) *vertex* of this edge and is *adjacent from* u . The initial and terminal vertices of a loop are the same.

Directed graphs

Definition: The *in-degree* of a vertex v , denoted $\deg^-(v)$, is the number of edges which terminate at v . The *out-degree* of v , denoted $\deg^+(v)$, is the number of edges with v as their initial vertex. Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of the vertex.

Example: Assume graph G :



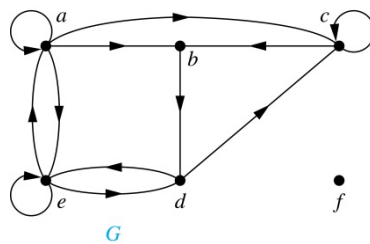
What are in-degrees of vertices: ?

$\deg^-(a) = 2, \deg^-(b) = 2, \deg^-(c) = 3,$
 $\deg^-(d) = 2, \deg^-(e) = 3, \deg^-(f) = 0.$

Graphs: basics

Definition: The *in-degree* of a vertex v , denoted $\deg^-(v)$, is the number of edges which terminate at v . The *out-degree* of v , denoted $\deg^+(v)$, is the number of edges with v as their initial vertex. Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of the vertex.

Example: Assume graph G :



What are out-degrees of vertices: ?

$\deg^+(a) = 4, \deg^+(b) = 1, \deg^+(c) = 2,$
 $\deg^+(d) = 2, \deg^+(e) = 3, \deg^+(f) = 0.$

Directed graphs

Theorem: Let $G = (V, E)$ be a graph with directed edges. Then:

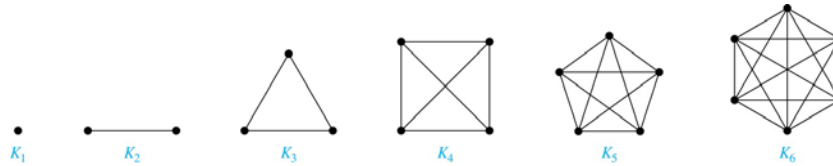
$$|E| = \sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v).$$

Proof:

The first sum counts the number of outgoing edges over all vertices and the second sum counts the number of incoming edges over all vertices. It follows that both sums equal the number of edges in the graph.

Complete graphs

A *complete graph* on n vertices, denoted by K_n , is the simple graph that contains exactly one edge between each pair of distinct vertices.

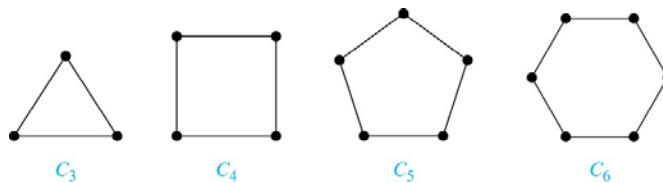


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A cycle

A *cycle* C_n for $n \geq 3$ consists of n vertices $v_1, v_2, \dots, v_n$, and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.

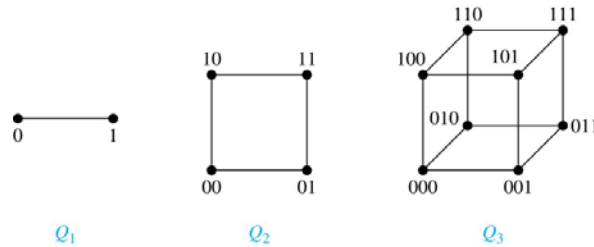


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N-dimensional hypercube

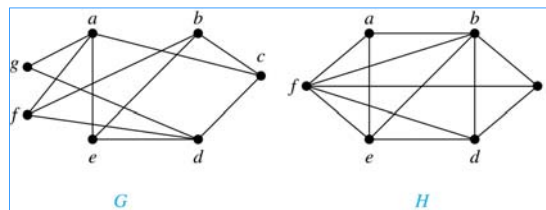
An n -dimensional hypercube, or n -cube, Q_n , is a graph with 2^n vertices representing all bit strings of length n , where there is an edge between two vertices that differ in exactly one bit position.



Bipartite graphs

Definition: A simple graph G is **bipartite** if V can be partitioned into two disjoint subsets V_1 and V_2 such that every edge connects a vertex in V_1 and a vertex in V_2 . In other words, there are no edges which connect two vertices in V_1 or in V_2 .

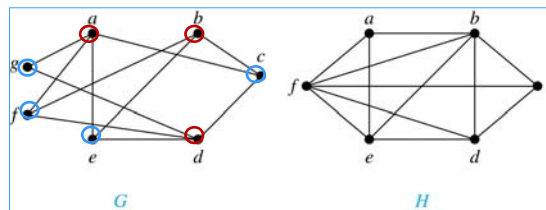
Note: An equivalent definition of a bipartite graph is a graph where it is possible to color the vertices red or blue so that no two adjacent vertices are the same color.



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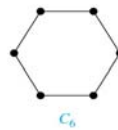


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Bipartite graphs

Example: Show that C_6 is bipartite.



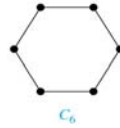
Solution:

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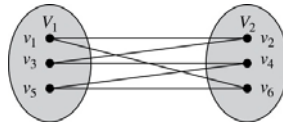
Bipartite graphs

Example: Show that C_6 is bipartite.



Solution:

- We can partition the vertex set into $V_1 = \{v_1, v_3, v_5\}$ and $V_2 = \{v_2, v_4, v_6\}$ so that every edge of C_6 connects a vertex in V_1 and V_2 .



Bipartite graphs

Example: Show that C_3 is not bipartite.



Solution:

Bipartite graphs

Example: Show that C_3 is not bipartite.



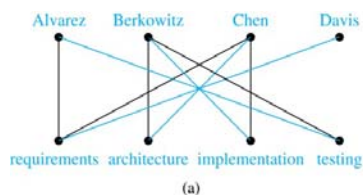
Solution:

If we divide the vertex set of C_3 into two nonempty sets, one of the two must contain two vertices. But in C_3 every vertex is connected to every other vertex. Therefore, the two vertices in the same partition are connected. Hence, C_3 is not bipartite.

Bipartite graphs and matching

Bipartite graphs are used to model applications that involve **matching** the elements of one set to elements in another, for example:

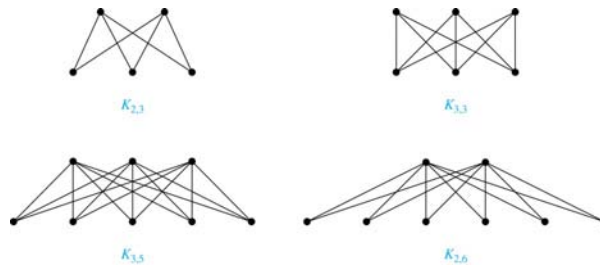
Example: *Job assignments* - vertices represent the jobs and the employees, edges link employees with those jobs they have been trained to do. A common goal is to match jobs to employees so that the most jobs are done.



Complete bipartite graphs

Definition: A *complete bipartite graph* $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets V_1 of size m and V_2 of size n such that there is an edge from every vertex in V_1 to every vertex in V_2 .

Example: We display four complete bipartite graphs here.



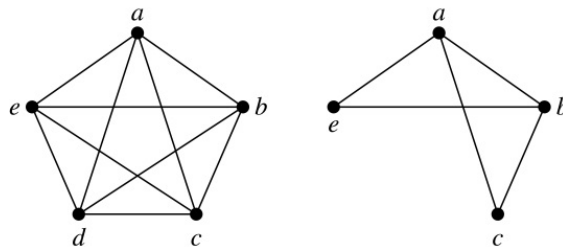
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Subgraphs

Definition: A *subgraph of a graph* $G = (V, E)$ is a graph (W, F) , where $W \subset V$ and $F \subset E$. A subgraph H of G is a proper subgraph of G if $H \neq G$.

Example: K_5 and one of its subgraphs.



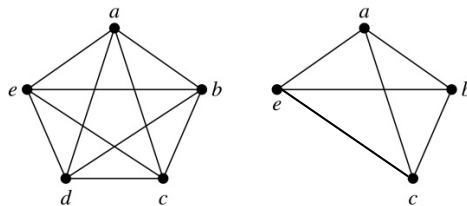
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Subgraphs

Definition: Let $G = (V, E)$ be a simple graph. The *subgraph induced* by a subset W of the vertex set V is the graph (W, F) , where the edge set F contains an edge in E if and only if both endpoints are in W .

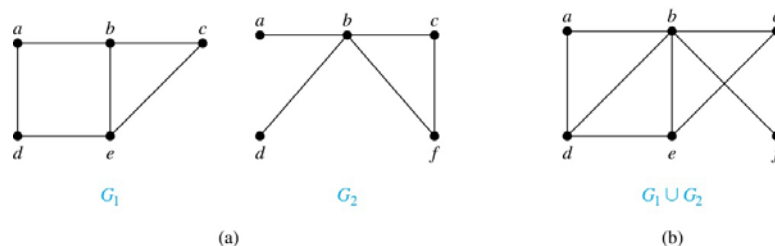
Example: K_5 and the subgraph induced by $W = \{a, b, c, e\}$.



Union of the graphs

Definition: The *union* of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with vertex set $V_1 \cup V_2$ and edge set $E_1 \cup E_2$. The union of G_1 and G_2 is denoted by $G_1 \cup G_2$.

Example:



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Lecture 26

Graphs

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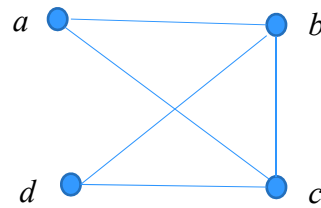
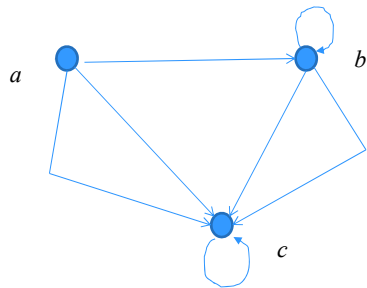
Final exam

- **Saturday, April 26, 2014 at 10:00-11:50am**
- **The same classroom as lectures**
- **The exam is:**
 - **Closed book**
 - **cumulative**
- **Please bring your own calculators**

Graphs: basics

Basic types of graphs:

- Directed graphs
- Undirected graphs

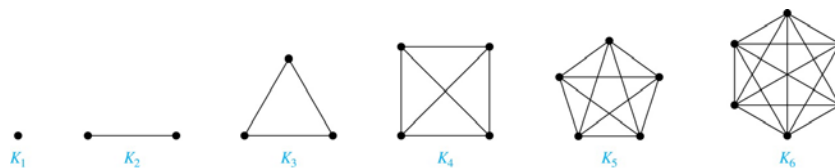


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Complete graphs

A *complete graph on n vertices*, denoted by K_n , is the simple graph that contains exactly one edge between each pair of distinct vertices.

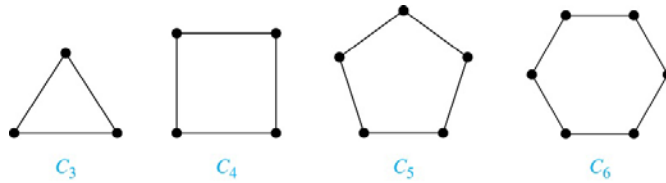


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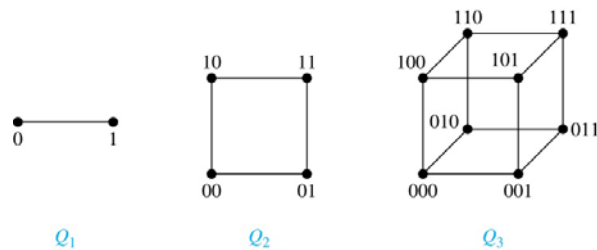
A cycle

A cycle C_n for $n \geq 3$ consists of n vertices $v_1, v_2, \dots, v_n$, and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.



N-dimensional hypercube

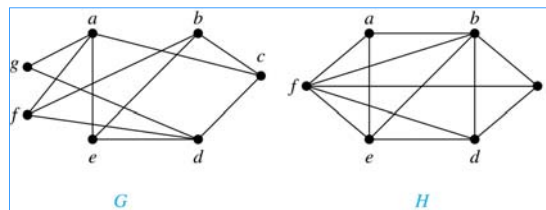
An n -dimensional hypercube, or n -cube, Q_n , is a graph with 2^n vertices representing all bit strings of length n , where there is an edge between two vertices that differ in exactly one bit position.



Bipartite graphs

Definition: A simple graph G is **bipartite** if V can be partitioned into two disjoint subsets V_1 and V_2 such that every edge connects a vertex in V_1 and a vertex in V_2 . In other words, there are no edges which connect two vertices in V_1 or in V_2 .

Note: An equivalent definition of a bipartite graph is a graph where it is possible to color the vertices red or blue so that no two adjacent vertices are the same color.



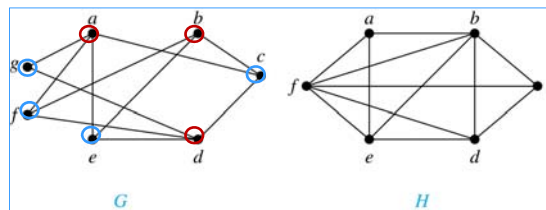
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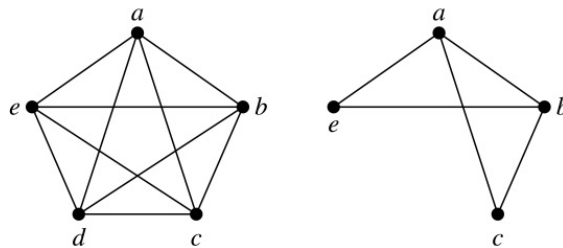
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Subgraphs

Definition: A *subgraph of a graph* $G = (V, E)$ is a graph (W, F) , where $W \subset V$ and $F \subset E$. A subgraph H of G is a proper subgraph of G if $H \neq G$.

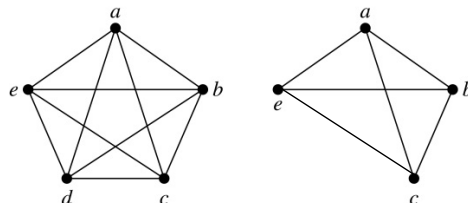
Example: K_5 and one of its subgraphs.



Subgraphs

Definition: Let $G = (V, E)$ be a simple graph. The *subgraph induced* by a subset W of the vertex set V is the graph (W, F) , where the edge set F contains an edge in E if and only if both endpoints are in W .

Example: K_5 and the subgraph induced by $W = \{a, b, c, e\}$.



Representation of graphs

Definition: An *adjacency list* can be used to represent a graph with no multiple edges by specifying the vertices that are adjacent to each vertex of the graph.

Example:

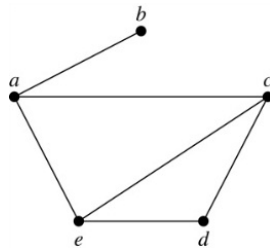


TABLE 1 An Adjacency List for a Simple Graph.

Vertex	Adjacent Vertices
a	b, c, e
b	a
c	a, d, e
d	c, e
e	a, c, d

Representation of graphs

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Example:

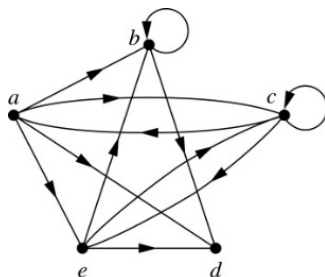


TABLE 2 An Adjacency List for a Directed Graph.

Initial Vertex	Terminal Vertices
a	b, c, d, e
b	b, d
c	a, c, e
d	c, e
e	b, c, d

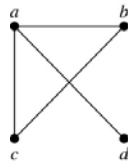
Adjacency matrices

Definition: Suppose that $G = (V, E)$ is a simple graph where $|V| = n$. Arbitrarily list the vertices of G as $v_1, v_2, \dots, v_n$. The *adjacency matrix* $\mathbf{A}_G$ of G , with respect to the listing of vertices, is the $n \times n$ zero-one matrix with 1 as its (i, j) th entry when v_i and v_j are adjacent, and 0 as its (i, j) th entry when they are not adjacent.

– In other words, if the graph's adjacency matrix is $\mathbf{A}_G = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

Example:



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

The ordering of vertices is a, b, c, d .

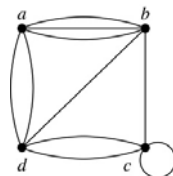
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Adjacency matrices

- Adjacency matrices can also be used to represent graphs with loops and multiple edges.
- A loop at the vertex v_i is represented by a 1 at the (i, i) th position of the matrix.
- When multiple edges connect the same pair of vertices v_i and v_j , (or if multiple loops are present at the same vertex), the (i, j) th entry equals the number of edges connecting the pair of vertices.

Example: The adjacency matrix of the pseudograph shown here using the ordering of vertices a, b, c, d .



$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}$$

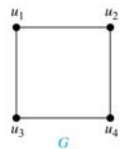
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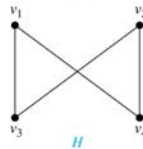
Graph isomorphism

Definition: The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are *isomorphic* if there is a **one-to-one and onto function** f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all a and b in V_1 . Such a function f is called an *isomorphism*. Two simple graphs that are not isomorphic are called *nonisomorphic*.

Example:



Are the two graphs isomorphic?



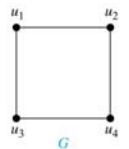
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Example:



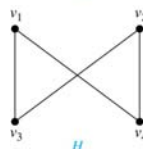
Are the two graphs isomorphic?

$u1 \rightarrow v1$

$u2 \rightarrow v4$

$u3 \rightarrow v2$

$u4 \rightarrow v3$



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Connectivity in the graphs, paths

Informal Definition: A *path* is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of the graph. As the path travels along its edges, it visits the vertices along this path, that is, the endpoints of these.

Applications: Numerous problems can be modeled with paths formed by traveling along edges of graphs such as:

- determining whether a message can be sent between two computers.
- efficiently planning routes for mail/message delivery.

Connectivity in the graphs

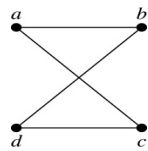
- We can use the adjacency matrix of a graph to find the **number of the different paths between two vertices in the graph.**

Theorem: Let G be a graph with adjacency matrix $\mathbf{A}$ with respect to the ordering $v_1, \dots, v_n$ of vertices (with directed or undirected edges, multiple edges and loops allowed). The number of different paths of length r from v_i to v_j , where $r > 0$ is a positive integer, equals the (i,j) th entry of $\mathbf{A}^r$.

Connectivity in the graphs

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Example:



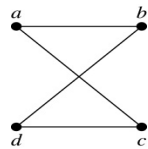
$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Paths of length 4.

Connectivity in the graphs

Theorem: Let G be a graph with adjacency matrix $\mathbf{A}$ with respect to the ordering $v_1, \dots, v_n$ of vertices (with directed or undirected edges, multiple edges and loops allowed). The number of different paths of length r from v_i to v_j , where $r > 0$ is a positive integer, equals the (i,j) th entry of $\mathbf{A}^r$.

Example:



$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

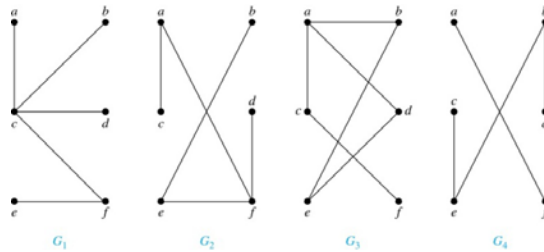
Paths of length 4: The adjacency matrix of G (ordering the vertices as a, b, c, d) is given above. Hence the number of paths of length four from a to d is the $(1, 4)$ th entry of $\mathbf{A}^4$

$$\mathbf{A}^4 = \begin{bmatrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{bmatrix}$$

Trees

Definition: A *tree* is a **connected undirected** graph with **no simple circuits**.

Examples:



Tree: yes

Tree: yes

Tree: no

Tree: no

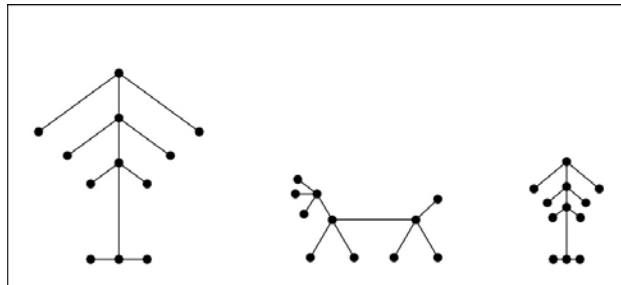
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Connectivity in the graphs

Definition: A *forest* is a graph that has no simple circuit, but is not connected. Each of the connected components in a forest is a tree.

Example:

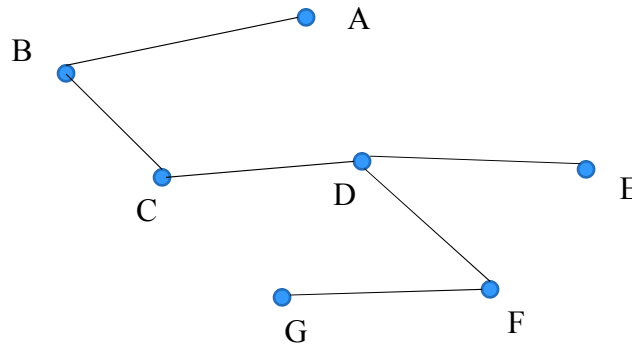


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Trees

Theorem: An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.



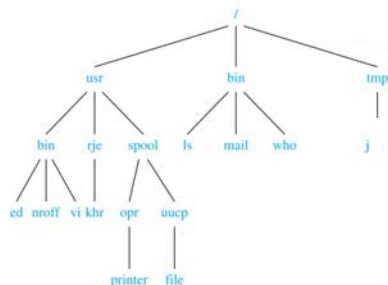
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Application of trees

Examples:

- The organization of a computer file system into directories, subdirectories, and files is naturally represented as a tree.
- structure of organizations.



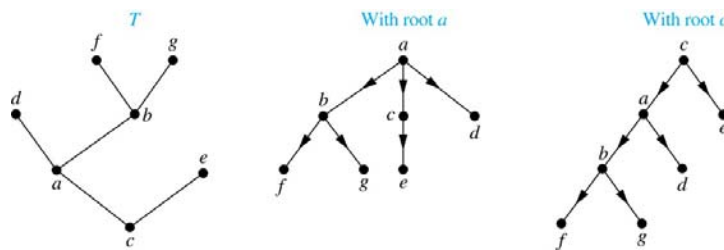
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Rooted trees

Definition: A *rooted tree* is a tree in which one vertex has been designated as the *root* and every edge is directed away from the root.

Note: An *unrooted tree* can be converted into different rooted trees when one of the vertices is chosen as the root.

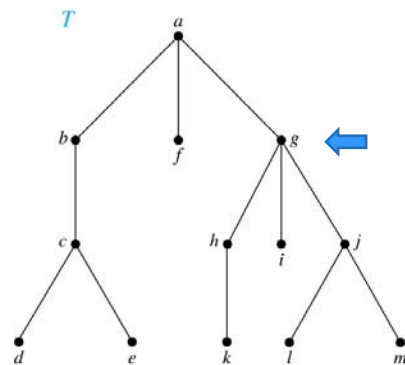


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Rooted trees - terminology

- If v is a vertex of a rooted tree other than the root, the *parent* of v is the unique vertex u such that there is a directed edge from u to v . When u is a parent of v , v is called a *child* of u . Vertices with the same parent are called *siblings*.



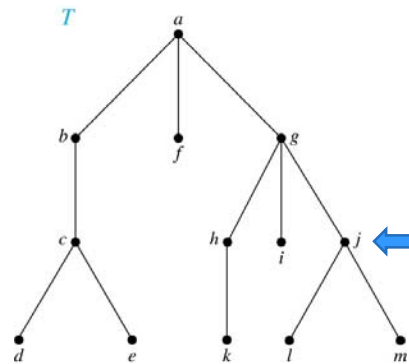
Parent of g : a
 Children of g : h, j, k
 Siblings of g : b, f

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Rooted trees - terminology

- The *ancestors* of a vertex are the vertices on the path from the root to this vertex, excluding the vertex itself and including the root. The *descendants* of a vertex v are those vertices that have v as an ancestor.



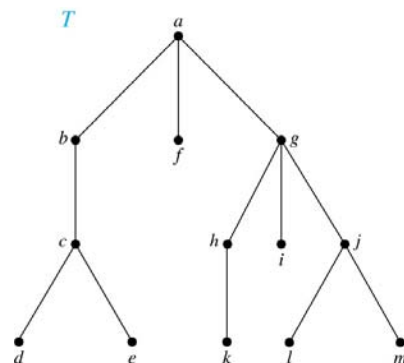
Ancestors of j : g, a
Descendants of j : l, m

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Rooted trees - terminology

- A vertex of a rooted tree with no children is called a *leaf*. Vertices that have children are called *internal vertices*.



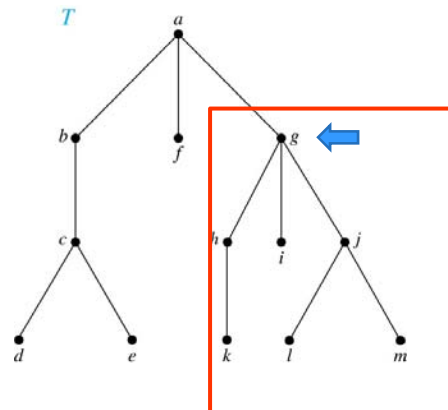
Leafs: d, e, k, l, m
Examples of internal nodes:
 b, g, h

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Rooted trees - terminology

- If a is a vertex in a tree, the *subtree* with a as its root is the subgraph of the tree consisting of a and its descendants and all edges incident to these descendants.



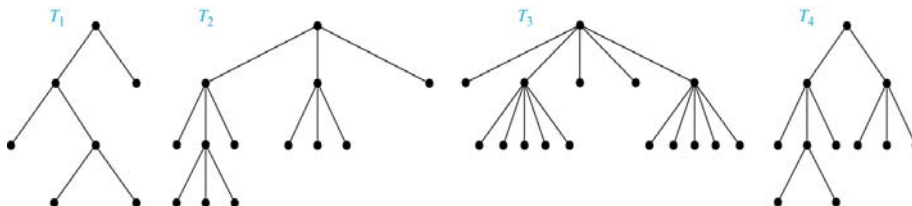
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M-ary tree

Definition: A rooted tree is called an m -ary tree if every internal vertex has no more than m children. The tree is called a *full m -ary tree* if every internal vertex has exactly m children. An m -ary tree with $m = 2$ is called a *binary tree*.

Example: Are the following rooted trees full m -ary trees for some positive integer m ?



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Binary trees

Definition: A *binary tree* is an ordered rooted tree where each internal vertex has at most two children. If an internal vertex of a binary tree has two children, the first is called the *left child* and the second the *right child*. The tree rooted at the left child of a vertex is called the *left subtree* of this vertex, and the tree rooted at the right child of a vertex is called the *right subtree* of this vertex.

